ZIMBABWE

MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT

HIGHER EDUCATION EXAMINATIONS COUNCIL

2025

REGULATIONS AND MODULES FOR THE

NATIONAL CERTIFICATE

IN

ELECTRICAL POWER ENGINEERING

Course Code: 321/25/CR/0

Implementation date: January 2026

HERITAGE-BASED EDUCATION 5.0

PREAMBLE

The course is designed to develop an artisan with knowledge, skills and attitudes to competently practice Electrical Power Engineering in the Electrical industry. The total duration of the course is **1 820** notional hours spread over a period of two (2) semesters and one (1) year On- the- Job Education and Training (OJET). The minimum entry requirements into this course are English Language, relevant Science and Mathematics passed at Ordinary Level with grade C or better and any other two subjects or relevant National Foundation Certificate (NFC) subjects or equivalent. The course is offered on a full time, part time, Block release or Open Distance e-Learning (ODL) or Part Qualification on a Single Modular basis (The single modular basis is exempted from 5 O' levels requirement). Assessment is through field based assignments, continuous assessment, written examination and On- the- Job Education and Training (OJET). The course will consider gender mainstreaming, sustainable development, physical challenges, health dispositions, and the intersections between race, class and culture. It shall embrace innovative heritage-based education and training philosophy to solve national problems and to produce goods and services for industrialization and modernization.

CONSULTATION		**YEAR**
1.	Tendo Electronics and Power Engineering (pvt) ltd	2025
2.	Dunlop Zimbabwe (pvt) ltd	2025
3.	Zimbabwe Electricity Distribution Power Company	2025
4.	National Railways of Zimbabwe	2025
5.	Zimbabwe Institution of Engineers	2025
6.	Trojan Nickel Mine	2025
7.	National Breweries	2025
8.	Mimosa mining company (pvt) ltd	2025
9.	Zimbabwe Electricity Supply Authority	2025
10.	Freda Rebecca Gold Mine	2025

PART 1: COURSE REGULATIONS

1.0 TITLE AND LEVEL OF AWARD

National Certificate in **Electrical Power Engineering**

2.0 AIM

The aim of the course is to produce an Electrical Engineering Mechanic with knowledge, skills and attitudes to satisfy the needs of the Electrical Engineering Industry.

3.0 LEARNING OUTCOMES

By the end of the course the students should be able to:

3.1 demonstrate knowledge of safety, health, environmental and quality management in the Electrical Power Engineering field

3.2 apply electrical and electronic concepts in Power generation, transmission, distribution and line construction

3.3 apply electrical and electronic basics to design and solve electrical problems

3.4 develop electrical and electronic skills through designed projects

3.5 employ electrical and electronics principles to produce given project

3.6 draw engineering drawings and design electrical circuits

3.7 apply electrical principles in electrical Plant & Equipment Installation. and equipment repair

3.8 communicate with other personnel in the organization using standard communication symbols and information technologies.

3.9 acquire business concepts to apply in the electrical power engineering field.

3.10 relate with the nation by being patriotic and participating in national development.

3.11 demonstrate the practical application of electrical power engineering field concepts in a working environment.

3.12 Manipulate automated systems through programmable Logic controllers

3.13 Construct power lines for 11kV lines and distribution lines 415 V

3.14 Develop electrical trade skills through workshop based tasks

4.0 STRUCTURE

	MODULE TITLE	CODE	NOTIONAL HOURS
	SEMESTER 1		
1	Safety ,Health Environment Quality Management	321/25/M01	80
2	National Studies	401/24/M01	50
3	Power generation, transmission , distribution, & Utilization	321/25/M02	200
4	Engineering Mathematics 1	321/25/M06/1	100
5	Electrical Engineering Technology 1	321/25/M05/1	150
6	Workshop Practice (Skills Proficiency)*	321/25/M13	
	SEMESTER 2		
7	Engineering Mathematics 2	321/25/M06/2	100
8	Engineering Drawing, Electrical Circuit Designing	321/25/M03	150
9	Electronics 1	321/25/M13/1	100
10	Communication and Computer Skills	321/25/M04	150
11	Entrepreneurial Skills Development	402/25/M01	50
12	Workshop Practice (Skills Proficiency)*	321/25/M12	
	SEMESTER 3 & 4		
13	On The Job Education and Training	321/25/M11	1 year
	SEMESTER 5		
14	Electronics 2	321/25/M13/2	100
15	Electrical Engineering Technology 2	321/25/M05/2	150
16	Electrical Plant & Equipment Installation, Maintenance & Repair	321/25/M07	200
17	Workshop Practice (Skills Proficiency)*	321/25/M12	
	SEMESTER 6		
18	Programmable Logic Controllers	321/25/M08	150
19	Overhead &Underground Power Line Construction	321/25/M09	150
20	Project	321/25/M10	40(supervised)
21	Workshop Practice (Skills Proficiency)*	321/25/M12	
	TOTAL		**1 820 hours**

Workshop Practice (Skills Proficiency) Marks will be submitted in the semester 6

5.0 DURATION

The course duration is 1820 notional hours spread over two (4) semesters and one (1) year On -the Job Education and Training (OJET).

6.0 ENTRY REQUIREMENTS

6.1 English Language, relevant Science and Mathematics passed at Ordinary Level with grade C or better and any other two Ordinary Level subjects or relevant National Foundation Certificate subjects.

6.2 The single module part Qualification pathway is exempted from 5 O' levels requirement

6.3 The single module part qualification should be taken one (1) module at a time by those without 5 Ordinary levels

7.0 MODE OF STUDY

Full time : 1820 notional hours

Part time : 1820 notional hours

Block Release : 1820 notional hours

ODeL : 18200 notional hours

8.0 ASSESSMENT

8.1 ASSESSMENT SCHEME

SUBJECT TITLE AND CODE	MODE OF ASSESSMENT		WEIGHTING
	WRITTEN EXAMINATION 40%	CONTINUOUS ASSESSMENT 60%	
Safety ,Health Environment Quality Management 321/25/M01	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%
Power generation, transmission , distribution and Utilization 321/25/M02	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%

Electronics (1) 321/25/M13/1/2	3 – hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100 %
Engineering Mathematics 321/25/M06/1/2	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%
Communication and Computer Skills 321/25/M04	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%
Electrical Engineering Technology 321/25/M05/1/2	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%
Engineering Drawing Electrical Circuit Designing 321/25/M07	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%
Electrical Plant & Equipment Installation Maintenance & Repair 321/25/M07	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%
Programmable Logic Controllers 321/25/M08	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%
Underground & Overheard Power Line Construction 321/25/M09		A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	
Entrepreneurial Skills Development 402/25/M01	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%

National Studies 401/24/M01	3 hour written paper	A minimum of • 1 Research based Assignments 20% • 1 Innovation based case study 20% • 1 Tests 20%	100%
Project* 321/25/M10	Submit Written project.	Submit Marks	100%
On the job Education and Training 321/25/M11	LOGBOOK	Submit Marks	100%
Workshop Practice (Skills Proficiency) 321/25/M12	Continuous Assessment	Submit Marks	100%

9.0 CONDITIONS OF GRADING

0% to 49% - Fail
50% to 59% - Pass
60% to 79% - Credit
80% and above - Distinction

10.0 CONDITIONS OF AWARD

10.1 A candidate should attend at least 100% of learning sessions to qualify for examinations. (An approved absent is shall be considered as a present)

10.2 Approved absenteeism shall not exceed 15% learning sessions

10.3 The final mark should be obtained through aggregation provided the candidate scores at least 50 % in each of continuous assessment, Skills Proficiency and examinations.

10.4 The pass mark shall be 50 %.

10.5 Candidates should pass all modules to be awarded a National Certificate.

10. 6 Single module candidates will be awarded part certificates in single module

11.0 RE-WRITES

11.1 Re-write(s) should conform to current course structure.

11.2 Candidates should pass at least two thirds of the course to qualify for a referral.

11.3 Any candidate who fails to pass at least two thirds of the course should repeat the whole course, including the subjects they would have passed.

11.4 There is no time limit for which to re-write a failed examination.

11.5 There is no aggregation for re-writes.

11.6 All re-writes should pass on performance in the examination.

11.7 If a candidate fails continuous assessment he/she repeats the subject

12.0 EXEMPTIONS AND TRANSFER OF CREDITS

12.1 Exemptions are only granted in modules already attained from a completed accredited qualification provided an exemption certificate specifying modules of exemption is produced.

12.2 Transfer of credits are ONLY granted in modules passed from accredited course programmes

12.3 Exemption or Transfer of credits Certificate should be applied for at enrolment and produced before registration of examinations.

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13.0 IRREGULAR PRACTICES

13.1 Cheating in examinations will result in disqualification from the whole course and all other HEXCO courses. The candidate will be suspended for two years.

13.2 Plagiarisms in any of the assessments will result in automatic disqualification in the course and any other HEXCO courses and the penalty as in 13.1 will apply.

14.0. **RESOURCES**

14.1 **Lecturer's Qualifications**

The minimum qualification for a lecturer is at least a National Diploma in a relevant area or an equivalent.

14.2 **Tools and Equipment**

The following are the minimum equipment requirements for the course per class.

Benchfitting

Minimum of 6 x 2 viced workbenches

Hacksaws
Drilling machines
Center punches
Dies and Tapes
Ladder.
Bearing puller.
Chain block.
Megger

Wiring (Panel)

Side cutters
Pliers
Multimeters
Screw drivers set
Spanners set
Tool boxes
Knives

Wiring House

Installation boards
Conduit benders
Hammers
Spirit levels
Plum bobs
Installation accessories

Motors and Transformers

Wiring boards
Wiring accessories
Contractors
Ammeters
Voltmeters
Tachometers
Frequency meters
Multimeter
3 phase induction motor 3 terminals
3 phase induction motor 6 terminals
1 phase electric motor capacitor start
1 phase electric motor capacitor start capacitor run
3 phase induction motor rotor slip ring

Kits

Basic Electrician Training tool kits.
Laptop

14.3 SUGGESTED REFERENCES

Theraja B.L.(2017) *Electrical Technology* Amazon London

Petruzella F.D.(2017) *Programmable Logic Controllers* McGraw-Hill New York

Jurek, S. F. (2015) Electrical *Machines for Engineer* Longman New York

Gonen, J. (2006) Electronic *Power Distribution* McGraw-Hill New York

Bhattacharya S.K. (2015) *Electrical Power* Bookboon

Bird, J. (2010) *Basic Engineering Mathematics 5th Ed* Elsevier Ltd

Floyd, T. L. (2015) Digital *Fundamentals* Prentice Hall

Boylestad R.L. (2016) *Introductory Circuit Analysis* Prentice Hall

Ulaby, L (2016) *Circuits* NTSP

Stoner, J.A.F. (2018) *Managemen*t Prentice Hall U.S.A

Other sources e.g. *internet, magazines, and journals*.

Module Code:	**321/25/M01**
Module Title:	**SAFETY, HEALTH, ENVIRONMENTAL AND QUALITY MANAGEMENT**
ZNQF Level:	4
Credits:	**8**
Duration:	**80** hours
Relationship with Qualification Standards:	Based on Unit Standard SAFETY, HEALTH, ENVIRONMENTAL AND QUALITY MANAGEMENT of Qualification Standard for an Artisan
Pre-requisite modules:	NONE
Purpose of Module:	The purpose of this module is to equip Electricians, Instrumentation Technicians, Communication Artisans, and Computer Systems Technicians with essential skills, knowledge, and attitudes necessary to maintain a safe working environment and implement effective safety precautions. It emphasizes adherence to Occupational and Environmental Health and Safety regulations, highlighting the responsibilities of both employers and employees. Key components include understanding workplace safety responsibilities, selecting and using appropriate personal protective equipment (PPE), following emergency procedures, maintaining good housekeeping standards, interpreting safety signs, and managing waste effectively. The module also addresses techniques for preventing electric shock, ensuring circuit isolation, and promoting safe practices when working at heights and with electrical equipment. Ultimately, it aims to foster awareness of environmental impacts and encourage continuous

	improvement in safety and health performance within the electrical and electronic engineering sector.
List of Learning Outcomes:	1.1 Framework of Occupational Safety and Health in Zimbabwe 1.2 Workplace Hazards and Safe Practices 1.3 Managing Workplace Risks and Hazards 1.4 Workplace Hazard Controls 1.5 Fire Safety and Emergency Preparedness Response Planning 1.6 Environmental and Waste Management
Learning Outcome 01	**FRAMEWORK OF OCCUPATIONAL SAFETY AND HEALTH (OSH) IN ZIMBABWE**
Assessment Criteria:	1.1.1 OSH Structure and Roles 1.1.2 Legal and Regulatory Framework (Domestic laws): 1.1.3 The Labour Act and Constitution of Zimbabwe
Content:	**1.1.1 OSH Structure and Roles** • Describe the structure of Occupational Safety and Health (OSH) in Zimbabwe. • Identify and state the roles of: • the Ministry of Public Service, Labour and Social Welfare • National Social Security Authority (NASA) • the Employers • the Employees. • Explain the functions of OSH committees and representatives **1.1.2 Legal and Regulatory Framework (Domestic laws):** **a. Factories and Works Act (Chapter 14:08):**

	(i) Understand and apply safety requirements for electrical installations, machinery, and maintenance:
	• Live parts must be insulated or barricaded
	• Non-circuit conductive parts must be earthed.
	• Use of fuses, circuit breakers, and thermal overload relays (IEE Regulations)
	• Lockout-Tagout (LOTO) during maintenance (IEE Regulations)
	• Worksite Electrical Safety (Construction Industry)
	(ii) Ensure safe operation, guarding, and maintenance of electrical machinery:
	• Machine Guarding
	• Personal Safety when working on machinery
	• Responsible Person's Role
	(iii) Accident Reporting:
	1. Fatal accidents:
	2. Non-fatal injuries/electrical shocks
	(iv) Fire & Emergency Preparedness:
	a) Know Emergency exits, Fire extinguisher locations, and Evacuation procedures.
	b) Follow Safety Signs (e.g., "High Voltage," "Danger: Electrical Hazard").
	b. The National Occupational Safety and Health Policy of Zimbabwe:
	• Analyse the objectives of the National OSH Policy and its alignment with:
	  • ILO Conventions C155 and C187
	  • Sustainable Development Goals (SDGs)
	• Explain:

	a) Employer and employee duties b) Risk assessments and reporting requirements • Appreciate the relevance of international standards (e.g., ISO 45001, ILO conventions) to Zimbabwe's OSH compliance. **c. Environmental Management Act:** a) State the role of the Environmental Management Act in waste disposal and pollution control. b) Describe methods for effective waste management and environmental impact assessments (EIAs). **1.1.3 The Labour Act and the Constitution of Zimbabwe** • Define the terms "employer obligations" and "employee rights" under the Labour Act and the Constitution. • Illustrate how company SHEQ policies and procedures align with national regulations. • State the advantages of adhering to SHEQ frameworks for workplace safety.
Assessment Tasks:	a) Written assessment on the identification of hazards and risks associated with tasks. b) Oral assessment on adherence to relevant regulations and permits. c) Practical assessment demonstrating the selection, usage, and maintenance of personal protective equipment (PPE).
Conditions/Context of assessment	• Assessment can be conducted in a classroom environment. • Practical assessments will take place in a simulated work environment or training institution. • Use facilities and materials, including safety manuals and relevant legislation
Learning Outcome 02	**WORKPLACE HAZARDS AND SAFE PRACTICES**
Assessment Criteria:	**1.2.1 Walking and Working Surfaces**

	1.2.2 Manual Handling and Ergonomics **1.2.3 Hazardous Substances** **1.2.4 Personal Protective Equipment (PPE)** **1.2.5 Hand Tools Safety**
Content:	**1.2.1 Walking and Working Surfaces** • Define Tripping hazard and Identify hazards associated with Tripping • Define Slipping hazard and Identify hazards associated with Slipping • Define Fall protection (guardrails, fall arrest systems) and Identify hazards associated with Falls • Apply control measures using the hierarchy of controls: ✓ **Elimination:** Remove obstructions (e.g., tape down cords). ✓ **Engineering:** Install guardrails (42" height), abrasive strips. ✓ **Administrative:** Post warning signs, implement 5S housekeeping. ✓ **PPE:** Slip-resistant footwear, safety harnesses. **Demonstrate safe practices for:** • Ladders: 3-point contact, 75° angle (4:1 ratio), **at least 1 meter above the landing surface** for safe access. • Scaffolds: Load capacity checks, guardrail installation. **1.2.2 Manual Handling and Ergonomics** • Explain the 5 P's of Manual Handling: Plan, Position, Posture, Pick, Proceed. • Illustrate correct lifting techniques • Analyse common manual handling injuries • Recommend controls for manual handling risks

	1.2.3 Hazardous Substances • Identify materials commonly used in the electrical and electronic industry that can be hazardous to workers' health, including; ❖ Solvents ❖ Hydrogen and other gases ❖ Paints. Lead, isocyanates, polymers ❖ Coolants, PCBs. etc. ❖ Thermal laggings such as Asbestos ❖ Mercury • Discuss health risks associated with hazardous substances, including chemicals that can cause respiratory issues, skin irritation, and long-term health effects. • Outline specific safety procedures for the use, handling, and storage of hazardous substances. • Apply the risk assessment process specifically to hazardous substances, including identification, evaluation, and control measures. • Explain the hierarchy of controls for managing hazardous substances and the relevant regulations governing their use. • Describe appropriate emergency response measures for hazardous substance incidents. • Discuss safe disposal methods for hazardous substances. **1.2.4 Personal Protective Equipment (PPE)** • Define Personal Protective Equipment and its significance in minimizing workplace hazards. • Statutory Requirements: Outline employer obligations regarding PPE provision and maintenance as per relevant regulations. • Identify different types of PPE and their specific uses: ❖ Head protection

	❖ Eye and face protection ❖ Hearing protection ❖ Respiratory protection ❖ Hand protection ❖ Foot protection ❖ Protective clothing ❖ Safety Belt a) Discuss considerations for selecting appropriate PPE. b) Explain proper cleaning and maintenance practices for PPE. **c)** Highlight the importance of training workers on the use of PPE and compliance with safety standards**.** **1.2.5 Hand Tools Safety** • Carry out safe practices when using hand tools. • Compare and contrast proper practices and safeguards. • Distinguish between various types of hand tools and their specific safety measures. • Describe how to implement effective controls for portable power tools. • State and explain the importance of regular maintenance and inspection • Conduct a Job Hazard Analysis (JHA) for a task (e.g., operating a grinder): a) Step 1: Identify hazards (e.g., flying debris, vibration). **b)** Step 2: Recommend controls (face shield, anti-vibration gloves
Assessment Tasks:	a) Written assessment on safe manual handling procedures and hazard identification. b) Practical assessment demonstrating the correct usage of PPE and adherence to safety signs.
Conditions/Context of	a) Written assessments may occur in a classroom setting.

assessment	b) Practical assessments will be conducted in a simulated work environment. c) Include relevant tools and personal protective equipment
Learning Outcome 03	**MANAGING WORKPLACE RISKS AND HAZARDS**
Assessment Criteria:	**1.3.1 Principles of Risk Management** **1.3.2 Hazard Identification and Risk Assessment (HIRA** **1.3.3 Accident Prevention and Investigation**
Content:	**1.3.1 Principles of Risk Management** **a) Define** the terms *risk*, *hazard*, *risk management*, and *risk tolerance*. **b) Explain** the objectives of risk management in preserving organizational assets. **c) Describe** the five-step risk management process: (1) Risk Identification (2) Risk Analysis (3) Risk Control (Elimination/Reduction) (4) Risk financing (5) Risk administration. **Apply** the **hierarchy of controls** to mitigate workplace risks, providing examples for each level: (1) Elimination (2) Substitution (3) Engineering controls (4) Administrative controls (5) **Personal Protective Equipment (PPE)**

	1.3.2 Hazard Identification and Risk Assessment (HIRA)
	• Compare proactive (e.g., inspections, audits) and reactive (e.g., incident analysis) approaches to hazard identification.
	• Use hazard identification tools, including:
	    • Workplace inspections
	    • Job Safety Analysis (JSA)
	    • Accident/incident data analysis
	    • Safety audits.
	• Classify hazards into categories: physical, chemical, mechanical, electrical, biological, ergonomic, and psychosocial.
	• Conduct a HIRA for a given scenario by:
	    • Identifying hazards
	    • Recommending controls based on the hierarchy of controls.
	3. Accident Prevention and Investigation
	a) Distinguish between *accidents*, *incidents*, and *near misses*, providing examples.
	b) Determine which incidents warrant investigation based on severity and potential consequences.
	c) Outline tools for accident investigations, such as interviews, photographic evidence, and checklists.
	d) Identify reportable incidents (e.g., fatalities, major injuries, dangerous occurrences).
	e) Describe the steps in accident investigation:
	    • Scene management
	    • Evidence collection
	    • Witness interviews
	    • Root cause analysis
	    • Reporting findings.
	a) Apply root cause analysis techniques:
	    a) *5 Whys*

	b) *Fault tree analysis*. b) Investigate a simulated accident by compiling evidence, identifying causes, and proposing corrective actions. c) State Zimbabwe's statutory reporting requirements, including forms like FG2 and WCIF 14.14.
Assessment Tasks:	• Written assessment on risk management principles and hazard identification. • Practical assessment involving a Job Hazard Analysis (JHA) for a specific task.
Conditions/Context of assessment	• Assessments can be conducted in a classroom environment. • Practical assessments will take place in a training institution with relevant tools and safety equipment. • Use case studies and simulated scenarios for analysis
Leaning Outcome 04	**WORKPLACE HAZARD CONTROLS**
Assessment Criteria:	**1.4.1 Electrical Safety** **1.4.2 Machinery Safety** **1.4.3 Safety Signs**
Content:	**1.4.1 Electrical Safety** • Identify the four main hazards of electricity: (1) Electric shock (e.g., ventricular fibrillation). (2) Arc flash/arching (UV radiation, thermal burns). (3) Electrical fires (causes: short circuits, overloads). (4) Static electricity (ignition risks in flammable environments). • Explain the causes of electrical conductor overheating (e.g., overcurrent, poor connections).

	• Describe how electricity induces fibrillation or fires/explosions (e.g., spark ignition).
	• State the primary controls for electrical hazards:
	    • Engineering (e.g., insulation, grounding).
	    • Administrative (e.g., LOTO procedures).
	    • PPE (e.g., insulated gloves).
	• Explain the function of a Ground Fault Circuit Interrupter (GFCI) in preventing shocks.
	• Apply Lockout-Tagout (LOTO) procedures during equipment maintenance.
	• Demonstrate safe practices near overhead power lines (e.g., crane operations).
	• Describe electrical regulations for:
	    ✓ Overload protection (fuses, circuit breakers).
	    ✓ Equipment labelling and inspection.
	• Carry out first aid for electric shock victims, emphasizing:
	    ✓ Scene safety (non-conductive tools).
	    ✓ CPR and recovery position.
	• Analyse hazards in Lead-Acid Battery charging (e.g., hydrogen gas) and state controls.
	• Explain lightning safety protocols, including external (lightning rods) and internal (surge protectors) protection.
	
	1.4.2 Machinery Safety
	
	a) Define "machinery" per regulatory standards and explain its workplace significance.
	b) Identify machinery hazards
	c) Compare and contrast safeguarding methods:
	    • Guards (fixed, interlocked e.t.c).

	• Devices (light curtains, pressure mats e.t.c). • Distance/location (remote operation). d) Explain the hierarchy of controls for machinery risks, illustrating guard types with advantages/limitations. e) Describe the roles of a "responsible person" in machinery safety maintenance. f) Illustrate the LOTO process for controlling energy during maintenance. **1.4.3 Safety Signs** ❖ Classify safety signs into four categories: (1) Prohibitive (e.g., "No Smoking"). (2) Warning (e.g., "High Voltage"). (3) Mandatory (e.g., "Wear PPE"). (4) Safe condition (e.g., "Emergency Exit"). ❖ Interpret sign features: a. Shape/colour b. Graphics ❖ Discuss the purpose of safety signs in injury prevention. ❖ Apply knowledge to identify signs for: a) Hazard locations. b) Directional/traffic guidance.
Assessment Tasks:	• Written assessment on electrical safety measures and machinery safeguarding. • Practical assessment demonstrating lockout/tagout (LOTO) procedures and the use of safety signs.

Conditions/Context of assessment	a) Assessments may be conducted in both classroom and practical settings. b) Practical assessments will utilize relevant equipment and safety devices. c) Ensure compliance with workplace safety regulations during assessments.
Learning Outcome 05	**FIRE SAFETY AND EMERGENCY PREPAREDNESS RESPONSE PLANNING**
Assessment Criteria:	**1.5.1 Fire Safety** **1.5.2 Emergency Response Planning** **1.5.3 First Aid and Incident Documentation**

Content:	**1.5.1 Fire Safety**
	
	1.5.1.1 Fundamentals of Fire
	
	• Define "fire" and explain the fire triangle (fuel, heat, oxygen)
	• Describe common causes of workplace fires
	1.5.1.2 Fire Classification and Control
	
	a) Classify fires into Classes A, B, C, D, and K (per Zimbabwean/ISO standards).
	b) Explain methods of extinguishing fires
	c) Describe fire extinguishing media and their suitability for fire classes
	
	1.5.1.3 Fire Prevention and Mitigation;
	
	• Identify sources of ignition
	• Apply fire prevention strategies:
	    • **Safe Chemical Storage & Ventilation**
	    • **Effective Housekeeping**

	• **Planned Equipment Maintenance** • **Controlled Ignition Sources** • **Fire Safety Training** 1. Discuss products of combustion (Smoke, Heat, Toxic gases, Flames) and their hazards. **1.5.2 Emergency Response Planning** **1.5.2.1 Key Definitions and Emergency Priorities** 2. Define the following terms: ✓ Emergency ✓ Disaster ✓ Disaster Management 3. Explain the following priorities during emergencies (ranked from highest to lowest importance): • **Safety of People** • **Protection of Property** • **Clean-up and Salvage** • **Restoring Operations** **1.5.2.2 Types of Emergencies and Response Steps** 4. Identify the following Types of Emergencies: • Natural Emergencies: e.g., floods, insect infestations • Fires and Explosions: e.g., structural fires, chemical explosions • System Failures: e.g., boiler overheating, electrical grid collapse • Toxic Gas Releases: e.g., chlorine gas leaks from chemical spills • Civil Disturbances: e.g., workplace violence, public protests • Medical Emergencies: e.g., road traffic accidents, disease outbreaks (cholera, etc.) • Utility Failures: e.g., ventilation system breakdowns, plumbing failures • Weather-Related Events: e.g., lightning strikes

	• Describe Steps during any Emergency (include the following Response Phases): ✓ **Pre-Emergency Preparedness** ✓ **During Emergency** ✓ **Post-Emergency Actions** • Outline key elements of an emergency response plan: ○ Roles and Responsibilities (Fire marshals, first aid teams, evacuation coordinators e.t.c, External agencies-fire brigade, medical services). ○ Communication Protocols (Internal and External). ○ Risk Assessments ○ Implement controls ○ Training Requirements ○ Equipment and Resources a) Differentiate the following phases of disaster management: • **Preparedness** • **Response** • **Recovery** (Short/Long-term). **1.5.3 First Aid and Incident Documentation** **1.5.3.1 Demonstrate First Aid treatment for:** • **Cuts and Abrasions:** ○ Clean with soap and water. ○ Apply antiseptic (e.g., iodine, alcohol-free solution). ○ Cover with sterile bandage. ○ Monitor for infection (redness, swelling, pus). ▪ **Burns** ○ Cool under running water (10–20 mins) ○ Remove jewellery /clothing (if not stuck to skin). ○ Cover with non-stick dressing (cling film or burn gel).

	○ Seek medical help for: Severe burns, Burns on face, hands, or genitals, and Chemical/electrical burn ▪ **Electrical shock** ○ Turn off power source (do not touch victim if current is live). ○ Call emergency services immediately. ○ Check responsiveness & breathing: If unresponsive, start CPR (30 compressions, 2 breaths). ○ Treat burns (if present) ○ Monitor for cardiac arrest (even if conscious). **1.5.3.1 Post-Incident Procedures** • **Documentation:** • Complete FG2 forms/WCIF 14 (Zimbabwean accident reporting). • Record near-misses and analyse root causes (e.g., 5 Whys technique). • **Labour Act** a) Notify supervisors & safety officers. b) Review corrective actions (prevent recurrence)
Assessment Tasks:	• Written assessment on fire classification, prevention strategies, and emergency response procedures. • Practical assessment demonstrating first aid for fire-related injuries and conducting emergency drills
Conditions/Context of assessment	a) Written assessments can occur in a classroom environment. b) Practical assessments will take place in a controlled training environment, simulating emergencies. **c)** Use firefighting equipment and emergency response plans during assessments**.**
Learning Outcome 06	**ENVIRONMENTAL AND WASTE MANAGEMENT**
Assessment Criteria	**1.6.1 Environmental Compliance**

	1.6.2 Waste Management **1.6.3 Housekeeping and Sanitation**
Content:	**1.6.1 Environmental Compliance** • Describe the Basel Convention: It regulates transboundary movements of hazardous waste, requiring prior informed consent. • Explain the Stockholm Convention: It aims to eliminate Persistent Organic Pollutants (POPs) to protect human health and the environment. • Evaluate the Environmental Management Act (EMA): It consolidates environmental laws in Zimbabwe to promote pollution prevention and sustainable development. • Define and appreciate Pollution Prevention and Resource Conservation **1.6.2 Waste Management** a) Define waste and describe the three types of waste: 1. Gaseous (e.g., air emissions) 2. Liquid (e.g., wastewater) 3. Solid (e.g., municipal waste) b) Explain the environmental and health impacts of improper waste disposal. c) Describe the waste hierarchy using a diagram to illustrate prioritization from prevention to disposal, and explain its significance. Highlight its practical application in industries through the 5Rs: Reduce, Reuse, Recycle, Recover, and Redesign—the key principles of effective waste management. d) Explain key waste management principles: • Extended Producer Responsibility (EPR) • Polluter Pays Principle • Cradle-to-Grave Approach

	• Life Cycle Principle • Precautionary Principle • Compare landfill, incineration, recycling, biological reprocessing, and waste reduction techniques. State the advantages and disadvantages of each waste disposal method • Explain key waste management regulations (e.g., hazardous waste disposal laws) • Outline best practices for storing, transporting, and treating waste (e.g., liquid waste containment). • Discuss hazardous waste management (toxicity, reactivity, flammability e.t.c). • Describe accident prevention measures • Identify safety measures for material storage • Explain spill response and emergency protocols for hazardous waste incident **1.6.3 Housekeeping and Sanitation** a) Explain the Importance of Housekeeping and Sanitation b) Describe Best Practices for Effective Housekeeping and Waste Management c) Illustrate how proper housekeeping reduces risks: Provide examples of how effective practices lead to safer environments d) Explain the five steps of 5S (Sort, Set in Order, Shine, Standardize, Sustain) with workplace examples specific to electrical industrial settings.
Assessment Tasks:	Written assessment on environmental compliance and waste management principles. Practical assessment demonstrating proper waste disposal procedures and adherence to environmental regulations.
Conditions/Context of	Assessments can be conducted in a classroom setting.

assessment:	Practical assessments will occur in a simulated environment with waste management tools. Include relevant legislation and safety procedures during assessments

ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOPIC	WEIGHTING %
01	Framework of Occupational Safety and Health in Zimbabwe	15%
02	Workplace Hazards,Safe Practices and Controls	25%
03	Managing Workplace Risks and Hazards	20%
04	Fire Safety and Emergency Preparedness Response Planning	20%
05	Environmental and Waste Management	20%
	TOTAL	**100%**

- **Learning Resources**

  Relevant training manual (learners' guide) and facilitators' guide

- **Reference Materials (recommended textbooks, recommended readings)**

Pedro Arezes et al. (2019). Occupational and Environmental, Safety and Health. Springer Nature, Switzerland: Cham, Switzerland.

Ronald Burke et al. (2011). Occupational Health and Safety, Gower Publishing: Aldershot United Kingdom.

John Wiley (2010). Handbook of International Electrical Safety Practices, Scrivener Publishing: Amazon Princeton Energy Resources International.

Module	**321/25/M02**
Module Title:	**POWER GENERATION, TRANSMISSION, DISTRIBUTION AND UTILISATION**
ZNQF Level:	**4**
Credits:	**20**
Duration:	**200 hours**
Relationship with Qualification Standards:	Based on Unit Standard Power generation, transmission, distribution and utilization of Qualification Standard for an electrician
Pre-requisite modules:	NONE
Purpose of Module:	This unit will enable an individual to exhibit knowledge of, installation and maintenance generation, transmission, distribution equipment in compliance with the IEE, SAZ, CAS wiring Regulations and Electricity Supply Regulations.
Learning Outcomes:	1. understand types, principles, components and economic factors for power generation and transmission 2. understand various methods of distribution systems, consumer distribution wiring systems 3. understand the various distribution substation plant and equipment 4. understand various protection, isolation and switching equipment for distribution systems and consumer installation 5. understand various methods, components, and economic considerations for consumer metering and tariffs 6. understand and apply various methods of protection against electric shock including earthing and earth leakage protection 7. understand and apply relevant IEE and/or SAZ Regulations to various types of lamps and methods of illumination 8. Understand and apply relevant I.E.E. and SAZ wiring regulations to the various special types of electrical installations

	9. Understand the design and installation procedure of a photo-voltaic system,
Learning Outcome 1	**Describe types, principles, components and economic factors for power generation and transmission**
Assessment Criteria:	1.1 Explain the principles of ac power generation 1.2 State and explain the methods of ac power generation 1.3 Explain the principles of ac power transmission 1.4 Design a solar system domestic and commercial application
Content	1.1 explain the principles of ac power generation 1. Single coil generator 2. Three-phase generators; frequency; phase displacement; voltage 1.2 explain the methods of ac power generation 1. Renewable (Hydro), Fossil fuel; Solar and Nuclear; 2. State advantages and disadvantages - economic, social impact, environmental 1.3 explain the principles of ac power transmission 1. explain reasons for stepping up generation voltages, - voltage drop - power loss - conductor and switchgear sizes 2. draw typical line diagram to include voltages; 3. draw national grid diagram; sub-stations; transformers 4. state and evaluate advantages and disadvantages of national grid
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand types, principles, components and economic factors for power generation and transmission as outlined in the assessment criteria and content above.
Conditions/Context of assessment	Written assessment can be conducted in a classroom environment.
Learning Outcome 2	**Outline various methods of distribution systems, consumer distribution wiring systems**
Assessment Criteria:	2.1 describe with line diagrams distribution systems 2.2 describe with line diagrams to show how loads are connected to distribution systems 2.3 state application of the distribution systems in 2.2

Content:	2.1 describe with line diagrams distribution systems 1. Radial distribution system 2. Parallel feeders distribution 3. Ring main distribution system 4. Interconnected distribution Including control and switching systems and compare the systems. 2.2 describe with line diagrams to show how loads are connected to distribution systems 1. Three-phase three wire 2. Three-phase four wire 3. Single-phase two wire
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand various methods of distribution systems, consumer distribution wiring systems as outlined in the assessment criteria and content above.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 3	**Describe the various distribution substation plant and equipment**
Assessment Criteria:	3.1 State various types of substations 3.2 Draw line/block diagrams of an 11kV/380V substation 3.3 Apply relevant IEE, CAS, and SAZ regulations pertaining to substation
Content:	3.1 state various types of substations 3.2 Draw line/block diagrams of an 11kV/380V substation showing the following equipment: - H. V. switchgear, isolator - Transformer - Low voltage switchgear, isolators - Bus-bar chamber - Distribution boards 3.3 Apply relevant regulations (IEE, CAS, and SAZ) pertaining to substation enclosures or buildings.

Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand the various distribution substation plant and equipment as outlined in the assessment criteria and content above.
Conditions/Context of assessment	Written assessment can be conducted in a classroom environment.
Learning Outcome 4	**Describe various protection, isolation and switching equipment for distribution systems and consumer installation**
Assessment Criteria:	4.1 Describe, illustrate and state advantages and disadvantages with the aid of schematic diagrams and, wiring diagrams the operation and construction of various switch gear 4.2 Identify and apply relevant regulations of the above circuit protection devices
Content:	4.1 Describe, illustrate and state advantages and disadvantages with the aid of schematic diagrams and, wiring diagrams the following switchgear: 1. High voltage switchgear: air blast circuit breaker, oil, sulphur hexafluoride (SF6), vacuum. 2. Low voltage switchgear: moulded case circuit breaker, miniature circuit breaker, switch-fuse, fuse-switch, isolator, double pole switch 3. Fuses – Rewirable, Cartridge, High breaking capacity (HBC) 4.2 Identify and apply relevant regulations of the above circuit protection devices
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand types, principles, components and economic factors for power generation and transmission as outlined in the assessment criteria and content above.
Conditions/Context of assessment	Written assessment can be conducted in a classroom environment.
Learning Outcome 5	**Outline various methods, components, and economic considerations for consumer installation including metering and tariffs**
Assessment Criteria:	5.1 Describe and draw line/block diagram of consumer installation 5.2 Describe methods of supplying power to consumer installation 5.3 Identify and apply IEE, SAZ and CAS regulations

	5.4 Explain metering and tariffs for electrical consumers 5.5 Carry out simple consumer installation 5.6 Illustrate and explain the connection of power factor correction devices
Content:	5.1 Describe, draw line/block diagram of consumer installation to include: - Service cable, Service MCB, Meter, Main switch, Distribution board indicating final circuits. 5.2 Describe the methods of supplying power to consumer installation 1. rising main (vertical supply) 2. single floor (horizontal supply) 5.3 Identify and apply IEE, SAZ and CAS regulations. 5.4 Explain metering and tariffs for electrical consumers to include domestic, commercial and industrial purposes: - Load limiter; - Two-part; - Maximum demand; - Block tariff; - Off-peak; - Pre-paid 5.5 Carry out simple consumer installation – to include components in 6.1 5.6 Illustrate and explain the connection of power factor correction devices. - Bank/group capacitors - Synchronous motor - Phase advancers
Assessment Tasks:	1. Written and/or oral assessment on the skills and knowledge required understanding various methods, components, and economic considerations for consumer installation including metering and tariffs as outlined in the assessment criteria and content above. 2. Practical assessment on the carrying out consumer installation
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. 2. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 3. The practical assessment will be conducted in situ or simulated work environment in the training institution. 4. The context of assessment should include the facilities, tools, equipment and materials listed below: Basic Electrician tool kit: Laptop, Ladder, Digital Multimeter, Insulation Resistance tester, Earth leakage

	tester, Materials: distribution board, circuit breakers, various switches, socket outlets, and other accessories
Learning Outcome 6	**Demonstrate and apply various methods of protection against electric shock including earthing and earth leakage protection**
Assessment Criteria:	6.1 Describe methods of protection against direct and indirect contact: 6.2 Define terms associated with earthing and give suitable example; 6.3 Illustrate earthing arrangement and state application: 6.4 Describe with illustration earth electrodes and voltage gradients, 6.5 Draw the earth fault loop path. 6.6 Describe earth leakage protection with aid of diagrams
Content:	6.1 Describe methods of protection against direct and indirect contact: 1. Use of barriers 2. Placing out of reach 3. Insulation, double insulation 4. Use of extra-low voltage 5. Isolation transformers 6. Earthing 6.2 Define terms associated with earthing and give suitable example; 1. Earthing 2. Exposed conductive parts 3. Extraneous conductive part 4. Bonding conductors 5. Circuit protective conductors 6. Earth electrode 7. Earth electrode resistance area 6.3 Illustrate earthing arrangement and state application: 1. TT 2. TN-C-S 3. TN-C 4. TN-S 6.4 Describe with illustration earth electrodes and voltage gradients, 1. As listed in IEE and/or SAZ Regulations (542). 2. Earth electrode resistance test 6.5 Draw the earth fault loop path and determine.

	1. Loop impedance 2. Prospective shock voltage 6.6 Describe earth leakage protection with aid of diagrams 1. Residual current device. 2. Earth monitoring device.
Assessment Tasks:	1. Written and/or oral assessment on the skills and knowledge required understanding various methods of protection against electric shock including earthing and earth leakage protection as outlined in the assessment criteria and content above. 2. Practical assessment on the carrying out of earth electrode resistance test
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. 2. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 3. The practical assessment will be conducted in situ or simulated work environment in the training institution. 4. The context of assessment should include the facilities, tools, equipment and materials listed below: Basic Electrician tool kit: Laptop, Ladder, Digital Multimeter, Insulation Resistance tester, Earth leakage tester, Materials: distribution board, circuit breakers, various switches, socket outlets, and other accessories
Learning outcome 7	**Apply relevant IEE and/or SAZ Regulations to various types of lamps and methods of illumination**
Assessment Criteria:	7.1 Define the following terms related to illumination: 7.2 Use inverse square law and cosine law to calculate illuminance. 7.3 Describe construction and operation of various lamps: 7.4 Identify and rectify faults on the above lamps. 7.5 Define stroboscopic effect and describe the methods of minimizing 7.6 Determine the number and positioning of luminaries for a given installation.

	7.7 Apply relevant IEE and/or SAZ Regulations.
Content:	7.1 Define the following terms related to illumination: 1. Light 2. Luminous flux 3. Luminous intensity 4. Illumination/illuminance 5. Glare 7.2 Use inverse square law and cosine law to calculate illuminance. 7.3 Describe construction and operation of the following lamps: 1. Compact Fluorescent Lamp (CFL) 2. LED's 3. Fluorescent 4. Sodium vapor 5. Mercury vapor 7.4 Identify and rectify faults on the above lamps. 7.5 Define stroboscopic effect and describe the methods of minimizing it: 1. Lead-lag circuit 2. Mixture of discharge incandescent 3. Distribution of lamps over three phase supply. 7.6 Determine the number and positioning of luminaries for a given installation.
Assessment Tasks:	1. Written and/or oral assessment on the skills and knowledge required understanding various methods of protection against electric shock including earthing and earth leakage protection as outlined in the assessment criteria and content above. 2. Practical assessment on the carrying out of earth electrode resistance test
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. 2. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 3. The practical assessment will be conducted in situ or simulated work environment in the training institution. 4. The context of assessment should include the facilities, tools, equipment and materials listed below: Basic Electrician tool kit: Laptop,

	Ladder, Digital Multimeter, Insulation Resistance tester, Earth leakage tester, Materials: distribution board, circuit breakers, various switches, socket outlets, and other accessories
Learning outcome 8	**Understand and apply relevant I.E.E. and SAZ wiring regulations to the various special types of electrical installations**
Assessment Criteria:	8.1 Identify fire hazardous areas. 8.2 Define the following zones in fire hazardous areas: 8.3 Describe basic fire alarm systems stating advantages, disadvantages and applications: 8.4 Describe the operation and application of the following emergency supply systems: 8.5 Describe the operation and application of emergency lighting systems stating advantages and disadvantages of: 8.6 Recognize the risks associated with the wiring of various special installations:
Content:	8.1 Identify fire hazardous areas. 8.2 Define the following zones in fire hazardous areas: 1. ZONE O 2. ZONE 1 3. ZONE 2 8.3 Describe basic fire alarm systems stating advantages, disadvantages and applications: 1. Open circuit 2. Closed circuit 8.4 Describe the operation and application of the following emergency supply systems:

	1. Generator 2. Battery 3. Uninterruptible power supply 8.5 Describe the operation and application of emergency lighting systems stating advantages and disadvantages of: 1. Maintained 2. Non-maintained 8.6 Recognize the risks associated with the wiring of: 1. Agricultural and horticultural installation 2. Temporary installation 3. Hospital installation 4. Electric fence installations Paying attention to and giving remedies of: a. Corrosion b. Erosion c. Fire hazards d. Damage by flora and fauna e. Danger to fauna and workers
Assessment Tasks:	1. Written and/or oral assessment on the skills and knowledge required understanding special types of electrical installations as outlined in the assessment criteria and content above.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. 2. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 3. The practical assessment will be conducted in situ or simulated work environment in the training institution. 4. The context of assessment should include the facilities, tools, equipment and materials listed below: Basic Electrician tool kit: Laptop, Ladder, Digital Multimeter, Insulation Resistance tester, Earth leakage tester, Materials: distribution board, circuit breakers, various switches, socket outlets, and other accessories
Learning outcome 9	**Design the installation procedure of a photo-voltaic system,**

Assessment Criteria	9.1 Describe the construction and connection of the various PV (Photovoltaic) cells, 9.2 Describe the various types of PV systems its components, advantages, and disadvantages 9.3 Describe the various types batteries used for PV system stating advantages and disadvantages, including charging and maintenance 9.4 Perform designing of PV system, 9.5 Describe the various solar wires and cables and the installation process 9.6 Carry out a simple design and installation of solar system
Content	9.1 Describe the construction and connection of the various PV (Photovoltaic) cells, and give advantages and disadvantages of each; 1. Mono-crystalline and poly-crystalline 2. series and parallel 9.2 Describe the various types of PV systems its components, advantages, and disadvantages 1. Grid-Tied system, 2. Off-Grid system 3. Hybrid Solar system 9.3 Describe the various types batteries used for PV system stating advantages and disadvantages, including charging and maintenance - Lead acid, nickel cadmium, lithium iron, flow batteries 9.4 Perform designing of PV system, - Load assessment - Determination of PV modules, inverter, battery, solar charge controller, and MPPT charge controller. 9.5 Describe the various solar wires and cables and the installation process 9.6 Carry out a simple design and installation of - Domestic, Solar water pumping system
Assessment Tasks:	1. Written and/or oral assessment on the skills and knowledge required understanding various design and installation procedure of a photo-voltaic system, as outlined in the assessment criteria and content above.

	2. Practical assessment on the carrying out of installation of Domestic, Solar water pumping system
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. 2. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 3. The practical assessment will be conducted in situ or simulated work environment in the training institution. 4. The context of assessment should include the facilities, tools, equipment and materials listed below: Basic Electrician tool kit: Laptop, Ladder, Digital Multimeter, Insulation Resistance tester, Earth leakage tester, Materials: distribution board, circuit breakers, various switches, socket outlets, and other accessories, PV modules, batteries, invertors, charge controllers, and other accessories

ASSESSMENT SPECIFICATION GRID

LEARNING OUTCOME	**Weighting %**
1. POWER GENERATION AND TRANSMISSION	10
2. DISTRIBUTION SYSTEMS, CONSUMER DISTRIBUTION WIRING SYSTEMS	10
3. DISTRIBUTION SUBSTATION PLANT AND EQUIPMENT	10
4. PROTECTION, ISOLATION AND SWITCHING EQUIPMENT FOR DISTRIBUTION SYSTEMS AND CONSUMER INSTALLATION	10
5., COMPONENTS, AND ECONOMIC CONSIDERATIONS FOR CONSUMER METERING AND TARIFFS	20
6. PROTECTION AGAINST ELECTRIC SHOCK INCLUDING EARTHING AND EARTH LEAKAGE PROTECTION	10
7. IEE AND/OR SAZ REGULATIONS TO VARIOUS TYPES OF LAMPS AND METHODS OF ILLUMINATION	10
8. I.E.E. AND SAZ WIRING REGULATIONS TO THE VARIOUS SPECIAL	10

TYPES OF ELECTRICAL INSTALLATIONS	
9. DESIGNING AND INSTALLATION PROCEDURE OF A PHOTO-VOLTAIC SYSTEM,	10
TOTAL	**100%**

Approach to Teaching and Learning:

1. Observation of adult learning principles.
2. Both institution-based and work-based learning to facilitate the integration of theory and practice.
3. Face-to-face education and learning.
4. Problem-based learning.
5. Online/distance education and learning.
6. Blended/hybrid education and learning.
7. Use of social media.

Approach to Assessment:

1. Weighting of practical and theory assessment: 70% practical and 30% theory.
2. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50% work-based assessment.
3. Oral assessment to be conducted by a panel of two or more assessors.
4. RPL assessment.*
5. Portfolio of evidence.
6. Assessment of work conducted by both individual learners and team of learners.

Resources:

3. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA). The lecturer should have a minimum of National Diploma in Electrical Power Engineering

4. Facilities, Tools, Equipment and Materials

The following are the minimum equipment requirements for the course per class.

Workshop and class room

Filling

Hacksaws

Drilling machines Center

Punches

Dies and Tapes

Ladder.

Bearing puller.

Chain block.

Megger
Wiring (Panel)
Side cutters
Pliers
multimeters
Screw drivers
Spanners
Tool boxes
Knives Wiring
House Conduit
benders
Hammers
Spirit levels
Motors and
Transformers
Contractors
Ammeters
Voltmeters
Tachometers
Trequenas meters
Multimeter
Basic electrician tool kits.
Lapto**p**

5. Learning Resources
Relevant training manual (learner's guide) and facilitator's guide

6. Reference Materials (recommended textbooks, recommended readings)
Lewis ML (2014) Electrical Installation Technology Hutchinson & Co. Ltd. London
Trevor L (2005) Advanced Electrical Installation Work 4th Edition
Mehta VK (2008) Principles of Power Systems 4th Edition
Lewis ML (2016) Electrical Installation Technology Hutchinson & Co. Ltd.
Mark H. (1995) Solar Electric System for Africa Common Wealth Science Council
Bimbhra PS (2008) Electrical Machinery Khanna
IEE Regulations 15 or 16th Edition
Z Wiring Regulations and code of Practice (latest edition)

Module Code:	**321/25/M13/1**
Module Title:	**ELECTRONIC 1**
ZNQF Level:	**4**
Credits:	**10**
Duration:	**100 hours**
Relationship with Qualification Standards:	**Based on Unit Standard of Qualification Standard for electrical artisans**
Pre-requisite modules:	**none**
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an Instrumentation and Control Mechanic, electrical power artisan, computer services artisan and electronic communication artisan to understand electronic principles. This includes resistors, capacitors, inductors, semi-conductor material, pn diodes and applications, zener diode and applications, BJT and applications and FETs and application The advantages of basic electrical principles includes providing a foundation for the study of the different electrical courses. Access to this module is open to all target groups including the unemployed youths, women and men wishing to establish or improve SMEs in electrical engineering.
List of Learning Outcomes:	**LO1: Introduction to Electronics and Resistors** **LO2: Capacitors and inductors** **LO3: Basic electronic materials** **LO4 : The PN junction diode** **LO5 : Applications of pn diode** **LO6: Zener diode and its applications** **LO7: Bipolar Junction Transistors (BJTs)** **LO8: field Effect transistors (FETs)** **LO9: OPTOELECTRONIC DEVICES**
Learning Outcome 01	**Introduction to Electronics and Resistors**

Assessment Criteria:	**1.1Demonstrate knowledge resistors.** **1.2 Tabulate the resistor colour and printed code (BS1852).** **1.3 Summary of terms, units and symbols**
Content:	Introduction to Electronics and Resistors 1.1 State that Electronics is that branch of Electrical Engineering which deals with design, manufacture, operation and application of devices that handle electrons 1.1.1 Give examples of application of electronic devices in Power Engineering, Instrumentation and Control, Communications and Electronic Servicing. 1.2 Resistors Define resistance and resistors 1.2.1State and explain resistor specifications. • Nominal value (Use the E12 series). • Power rating. • Stability • Tolerance 1.2.2 State the different types of resistors used in electronic circuits and the draw their symbols: a) Fixed Resistors - Carbon composition - Wire-wound - Metal oxide Tabulate the resistor color and printed code (BS1852). b) Variable Resistor c) Preset Resistors d). Non-linear resistors: Draw circuit symbols and explain their principle of operation , draw the characteristics curves and applications of the following - NTC and PTC thermistors. - Voltage dependent resistors. - Light dependent resistors. e) surface mount resistors - to decode their values from the printed code inscribed 1.3 State faults on resistors and how to test resistors
Assessment Tasks:	2. Written assessment on the skills and knowledge required to understand the resistors

Conditions/Context of assessment	4. Written assessment can be conducted in a classroom environment.
Learning Outcome 02	**CAPACITORS AND INDUCTORS**
Assessment Criteria	2.1 Demonstrate knowledge of capacitors and inductors and their types.
	2.2 state their specifications
	2.3 how to decode their values from the printed code inscribed of surface mounted
	Capacitors and inductors 2.1 Capacitors • State the different types of capacitors (polarized and non-polarised), draw their symbols and state their application 1.Fixed Capacitors e) Paper f) Mica g) Ceramic h) Polyester i) Electrolytic 2. Preset capacitors. • Explain capacitor specifications: d) Nominal values e) Working voltage f) Tolerance g) Types of dielectrics • Describe the capacitor colour code and use it to determine the value of a capacitor. • Describe the operation of the basic capacitor in DC and AC circuits • state applications of capacitors • Surface mount capacitors: how to decode their values from the printed code inscribed • State faults on capacitors and how to test capacitors 2.2 Inductors • Define inductance and inductors • State the different types of inductors, draw their symbols and

	state their applications c) Air-cored. d) Iron-cored. e) Dust/ferrite cored. f) Laminated-core inductors. • Explain inductor specifications. c) Nominal value d) Frequency range e) Working current • Surface mount inductor :how to decode their values from the printed code inscribed • State faults on inductors and how to test inductors
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand capacitors and inductors
Conditions/Context of assessment	5. Written assessment can be conducted in a classroom environment.
Learning Outcome 03	**BASIC ELECTRONIC MATERIALS**
Assessment Criteria:	3.1Demonstrate knowledge of n type and p type material 3.2 Differentiate between conductors, insulators and semi- conductors 3.3 Demonstrate some knowledge associated with some terms ie doping, donor, acceptor, intrinsic and extrinsic
Content	LO3: BASIC ELECTRONIC MATERIALS • State the basic materials used in Electronics, for example, conductors, semiconductors and insulators. Give examples in each case. • Using Bohr's model of the atom with its constituent particles (electrons, protons and neutrons), explain the energy band theory. c) Valence band d) Conduction band e) Forbidden gap • Doping. d) Doping materials used and how to create P-type and N-type semi-conductor materials. e) Majority and minority charge carriers. • Describe conduction in electronic materials energy band theory.

	• Intrinsic and extrinsic conduction
Assessment Tasks	**1**. Written and/or oral assessment on the skills and knowledge required on BASIC ELECTRONIC MATERIALS
Conditions/Context of assessment	**1.** Written and/or oral assessment can be conducted in a classroom environment**.**
Learning Outcome 04	**PN JUNCTION DIODE AND APPLICATIONS**
Assessment Criteria	4.1 Demonstrate knowledge of pn diode
	4.2 Demonstrate knowledge on different types of rectifiers, clippers and clampers
	4.3 Differentiate between ideal diode and practical diode
	4.4 Demonstrate knowledge on voltage doubles and trippler
Content	PN JUNCTION DIODE AND APPLICATIONS THE PN JUNCTION • Explain the formation of a PN junction. • Describe the formation of potential barrier and the depletion layer in an unbiased junction. • Explain the effects of forward/reverse biasing a PN junction (using block symbol) on: c) Currents d) Depletion region e) Barrier potential • Describe the effect of minority charge carriers on blocking current (reverse bias • Explain the Zener and avalanche effects. • Sketch typical forward and reverse-bias static characteristics of a PN diode and draw the test circuit to obtain them. • Explain: c) Power dissipation in a diode, Pd. d) Maximum forward voltage, VF max. e) Maximum forward current, IF max. f) Peak inverse voltage, PIV. **APPLICATION OF THE RECTIFIER DIODE**

	• Describe, with the aid of sketches, the operation of diode AC circuits: d) DC power supply block diagram. e) The half wave rectifier. f) The centre-tap full wave rectifier. g) The full-wave bridge rectifier - • Draw the input and output waveforms for the rectifier circuits above, assuming sinusoidal inputs and a purely resistive load. • Calculate, for the rectifier circuits above: b) The average voltage Vdc and average current Idc. c) The efficiency • Define peak-to-peak ripple voltage $Vrpp$ and the ripple factor. • Clipping and clamping circuits: a) Simple series and parallel diode clippers. b) Positive and negative clampers. • Voltage Multipliers. d) Doubler e) Voltage Tripler.
Assessment Tasks:	**1. Written assessment on the skills and knowledge required to understand PN JUNCTION DIODE AND APPLICATIONS**
Conditions/Context of assessment	**1. Written assessment can be conducted in a classroom environment.**
Learning Outcome 05	**THE ZENER DIODE**
Assessment criteria	Demonstrate the knowledge on construction, V/I characteristics Perform calculation on zener as a regulator
Content	LO05 :THE ZENER DIODE • Explain the operation of the zener diode. • Draw the Zener diode characteristics to indicate the following parameters: e) IZK, IZT, IZM f) VZ g) Z (or ZZT) • Draw the equivalent circuit and characteristics of the Zener diode assuming a constant Zener voltage and a constant reverse resistance RZ. • Explain the operation of the Zener diode as a voltage stabilizer. • Perform simple calculations on zener diode voltage stabilizer circuits. d) Varying load, fixed supply voltage. e) Varying supply voltage, fixed load.

	f) Zener voltage is temperature dependent. • Describe the application of the Zener diode as a clipper and perform simple calculations. • State faults found on a diodes and how to test a diode Surface mount diodes • how to decode them from the printed code inscribed
Learning Outcome 06	**THE BIPOLAR JUNCTION TRANSISTOR**
Assessment Criteria:	5.1Demonstrate knowledge of npn and pnp transistors 5.2 Calculate different voltages on different biasing methods 5.3 state the difference 3 configurations
Content:	LO 05 :THE BIPOLAR JUNCTION TRANSISTOR • The basic principles of operation of PNP and NPN transistors. • Understand the construction and operation of the PNP and NPN transistors (correct bias voltages). • Describe transistor configurations – CBC, CEO and CCC. • Define important transistor parameters. c) Voltage gain (amplification). d) Current gain. e) Power gain. f) Input and output impedance. g) Phase relation. • Define the DC amplification factors hFE (â), and hFb (β • Derive the relationship between α and β • Draw and explain the static characteristics of an NPN transistor in the common emitter mode and the circuit to obtain them. • Indicate and explain cut off, saturation and active regions. • Determine the DC gain, hFE (â) from the output characteristics of an NPN transistor. • Explain the concept of thermal runaway in transistors and its effect on â hfE . **THE TRANSISTOR AS AN AMPLIFIER** • Explain the need for biasing transistor amplifier circuits; the quiescent point, Q. • Draw and explain the circuit diagram of a single-stage amplifier having a resistance load, RL (CEC). • Describe transistor biasing methods and explain component functions: d) Fixed bias e) Self bias

	f) Potential-divider bias. • Determine bias conditions mathematically for a, b and c. • Derive the DC load line and draw it on to the output characteristics of an amplifier (CEC). • Find the major BJT parameters from both characteristics and datasheet. c) $PDmax$. d) $VCEmax$ e) $Icmax$ f) hFE • Explain the function of transistor amplifier components. Surface mount BJTs • how to decode them from the printed code inscribed • State faults on BJTs and how to test
Assessment Tasks:	**1. Written assessment on the skills and knowledge required to understand the BIPOLAR JUNCTION TRANSISTOR**
Conditions/Context of assessment	**1. Written assessment can be conducted in a classroom environment.**
Learning Outcome 07	**THE FIELD EFFECT TRANSISTOR**
Assessment Criteria:	**6.1Demonstrate knowledge of JFET,DEMOSFET and EMOSFET** **6.2 Demonstrate knowledge of FETs as a switch**
Content:	THE FIELD EFFECT TRANSISTOR **BASIC JFET OPERATION** • State that the JFET is basically a voltage operated device. • State the devices' advantages and disadvantages as compared to the BJT. • Describe, with the aid of a construction diagram, the operation of the JFET (N channel and P channel devices). • Explain why thermal runaway is not a problem with FETs. • Sketch typical output and transfer characteristics of a JFET. Indicate the cut off, ohmic and pinch off regions. **BASIC MOSFET OPERATION**

	• Understand the construction and operation of both the depletion mode and enhancement – mode MOSFETs (P channel and N channel). . • State that enhancement-mode MOSFETs are normally off devices whereas depletion-mode devices are normally on devices. • Draw the circuit symbols for the four types of MOSFETs to show the correct bias voltages. • Describe biasing methods for both JFETs and MOSFETs and simple calculations a) Simple bias. b) Fixed bias. c) Source-self bias. d) Drain-self bias. e) Potential divider bias. **THE FET AS A SWITCH** • Draw the circuit diagram to show how the FET can be used as a switch (CSC). Explain the operation. • Explain possible applications of the FET switch Surface mount FETs • how to decode their values from the printed code inscribed • State faults on FETs and how to test FETs •
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand THE FIELD EFFECT TRANSISTOR
Conditions/Context of assessment	**1. Written assessment can be conducted in a classroom environment.**
Learning Outcome 08	**OPTOELECTRONIC DEVICES**
Assessment Criteria:	8.1 describe construction, principle of operation and application of opto devices 7.2 State application of opto devices
Content:	**OPTOELECTRONIC DEVICES** • Describe the basic construction and operation of optoelectronic devices; - Photoconductive cell (Light Dependent Resistor) - Solar cell.

	- Light-emitting diodes (LEDs). - Photo transistor • Give applications of the optoelectronic devices above. • Describe operation and applications of the opto- coupler; the need for isolating electronic circuits **THE TRANSISTOR AS A SWITCH** • State the bias conditions for switching the transistor on and off. • Describe the principle of operation of transistor switching circuits incorporating the following components: a) Phototransistor (include construction and principles of operation). b) Photodiode. c) Photoconductive cell. d) Solar cell. e) Thermistors.
Assessment Tasks:	**1. Written assessment on the skills and knowledge required to understand.** **OPTOELECTRONIC DEVICES**
Conditions/Context of assessment	**1. Written assessment can be conducted in a classroom environment.**

ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOPIC	WEIGHTING %
1	INTRODUCTION TO ELECTRONICS AND RESISTORS	10
2	CAPACITORS AND INDUCTORS	10
3	BASIC ELECTRONIC MATERIALS	10
4	THE PN JUNCTION DIODE AND ITS APPLICATIONS	20
5	THE ZENER DIODE	10
6	BIPOLAR TRANSISTOR	15

7	**FIELD EFFECT TRANSISTOR**	**15**
8	**OPTOELECTRONIC DEVICES**	**10**
	TOTAL	**100**

Approach to Teaching and Learning:

5. Observation of adult learning principles.
6. Both institution-based and work-based learning to facilitate the integration of theory and practice.
7. Face-to-face education and learning.
8. Problem-based learning.
9. Online/distance education and learning.
10. Blended/hybrid education and learning.
11. Use of social media.

Approach to Assessment:

1. Weighting of practical and theory assessment: 70% theory and 30% practical.
2. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50%.
3. Oral assessment to be conducted by a panel of two or more assessors.
4. RPL assessment.
5. Portfolio of evidence.
6. Assessment of work conducted by both individual learners and teams of learners.

Resources:

3. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

4. Facilities, Tools, Equipment and Materials
   - Electrical tool box
   - Digital multi meter
   - Ladder
   - Laptop
   - Soldering kit
   - PPE
   - Drilling machine
   - Insulation tape
   - Mutton cloth
   - Detergents
   - Cables
   - Cable enclosures

- Electrical accessories

5. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

6. Reference Materials (recommended textbooks, recommended readings)

1 Floyd T.L. (2014) Electronic Devices and Circuit Prentice Hall New Jersey

2 Floyd T.L. (2015) Digital fundamentals Prentice Hall New Jersey

3 Horowitz, P. (2015) The Art of Electronics 3rd Ed Cambridge University Press

4 Bimbhra P.S. (2012) Power Electronics Khanna

5 Lang J.(2016) Foundations of Analog and Digital Electronic Circuits Morgan Kaufmann Publishers

6 Forrest M (2017) Getting Started in Electronics Book Renter, Inc.

7 Scherz P (2016) Practical Electronics for Inventors McGraw-Hill Education

8 Platt C. (2015) Encyclopedia of Electronic Components Volume 2: LEDs, LCDs, Audio, Thyristors, Digital Logic, and Amplification Maker Media, Inc

9 Platt C. (2017) Easy Electronics (Make: Handbook) Maker Media, Inc;

10 Cohen S. (2018) Make It, Wear It: Wearable Electronics for Makers, Crafters, and

Cosplayers McGraw-Hill Education TAB

Module Code:	**321/25/M13/2**
Module Title:	**ELECTRONICS 2**
ZNQF Level:	**4**
Credits:	**10**
Duration:	**100 hours**
Relationship with Qualification Standards:	**Based on Unit Standard of Qualification Standard for electrical artisans**
Pre-requisite modules:	**ELECTRONICS 2**
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an Instrumentation and Control Mechanic, electrical power artisan, computer services artisan and electronic communication artisan to understand electrical and electronic principles. This includes an understanding OF operational amplifiers, digital electronics fundamentals, power electronics transducers and sensors. The advantages of ELECTRONICS 2 includes providing a foundation for the study of the different electrical courses. Access to this module is open to all target groups including the unemployed youths, women and men wishing to establish or improve SMEs in electrical engineering.
List of Learning Outcomes:	**LO1: THE OPERATIONAL AMPLIFIER** **LO2: POWER ELECTRONICS** **LO3: DIGITAL FUNDAMENTALS** **LO4: SENSORS AND TRANSDUCERS**
Learning Outcome 01	**THE OPERATIONAL AMPLIFIER**
Assessment Criteria:	**1.1Demonstrate knowledge the virtual earth concept.** **1.2 State the ideal and practical properties of an Operational Amplifier** **1.3 Perform simple calculations**
Content:	**THE OPERATIONAL AMPLIFIER** **INTRODUCTION TO THE IC OPRATIONAL AMPLIFIER** .

	• Define the term Operational Amplifier. • Draw the block symbol of an Operational Amplifier. • Explain that the operational amplifier is used with feedback. • State that the operational amplifier basically amplifies the difference between its input signals, that is, VO = A (V2 – V1). • State the ideal and practical properties of an Operational Amplifier: d) Input Resistance e) Output Resistance f) Open-Loop gain g) Bandwidth h) Offset voltage • Draw and explain the transfer characteristic of an Operational Amplifier. • Describe single-rail and dual-rail biasing techniques for the Operational Amplifier • Define the following terms g) Bias current. h) Differential Mode gain Ad. i) Common-Mode gain,. j) Common Mode Rejection Ratio, CMRR. k) Input offset voltage. l) Output offset current. • Perform simple calculations on items b, c, d, e and f above • Use the 741 IC OP AMP as an example. **OPERATIONAL AMPLIFIER APPLICATIONS** • Understand the virtual earth concept. • Derive the expressions and perform simple calculations for the voltage gain of an OP-AMP when connected as a: a) Voltage follower b) Inverting amplifier c) Non-inverting amplifier d) Summing amplifier e) Differential amplifier • State the practical applications of the amplifier circuits above

Assessment Tasks:	3. Written assessment on the skills and knowledge required to understand OPERATIONAL AMPLIFIER
Conditions/Context of assessment	6. Written assessment can be conducted in a classroom environment.
Learning Outcome 02	**POWER ELECTRONICS**
Assessment Criteria	2.1 Demonstrate knowledge of SCR
	2.2 Demonstrate an understanding of DAIC AND TRIAC
	2.3 Draw waveforms of rectification.
	POWER ELECTRONICS **THE SILICON-CONTROLLED RECTIFIER (SCR)** • Describe the construction and operation of the SCR: • draw the voltage/ current characteristics • Use transistor- pair analogy. • State the methods for triggering the SCR: f) Using forward voltage g) Gate triggering • Thyristor ratings: a) Voltage ratings **b) Current ratings** **APPLICATIONS OF THE SCR** • DC/AC motor speed control • DC over voltage protection • Controlled rectification **THE DIAC AND TRIAC** • Describe construction and principle of operation. • Draw and explain conduction characteristics. State the Applications of DAIC and TRIAC THE UNIJUNCTION TRANSISTOR (UJT) • Describe the construction and operation of the UJT • Draw the typical characteristic for the UJT.

	• Applications of the UJT: c) SCR triggering d) Relaxation oscillator: sketch waveforms to show the exponential charge and discharge of the capacitor
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand POWER ELECTRONICS
Conditions/Context of assessment	**7. Written assessment can be conducted in a classroom environment.**
Learning Outcome 03	**DIGITAL ELECTRONICS FUNDAMENTALS**
Assessment Criteria:	3.1Demonstrate knowledge of logic gate 3.2 Demonstrate knowledge of universal gate, Boolean laws. 3.3 Demonstrate some knowledge of K-maps and combinational logic.
Content	**DIGITAL ELECTRONICS FUNDAMENTALS** INTRODUCTION • Understand the difference between analogue and digital signals. • Give examples of digital systems. **LOGIC ELEMENTS** • Explain, using symbols and truth tables, the operation of logic elements. d) AND e) OR f) NOT g) NAND h) NOR i) EX-OR j) EX-NOR **BOOLEAN** • Introduce Boolean variables, operations and expressions. • State Boolean laws and theorems: • Boolean identities • De Morgan's theorems • Simplify Boolean expressions using the identities and theorem above • Derive Boolean expressions from truth tables (Mini terms only) and logic circuits.

	• Draw the logic circuits to generate Boolean expression. **THE UNIVERSAL GATES** • Demonstrate that NAND or NOR gates may be used to replace any other gate. • Transform a logic network to use only one type of logic gate (NAND or NOR). • Graphical method: replaces each gate by its NAND/NOR equivalent. • Analytical method using De Morgen's Theorems. **KARNAUPH MAPS** • Construct K-maps of up to four variables from truth tables and Boolean expressions. • Simplify Boolean expressions of up to four variables using the K-map Method **COMBINATIONAL LOGIC** • Define the term Combinational Logic. • Design simple combinational logic circuits. • Draw the timing diagram for simple logic circuits. • Appreciate Combinational logic application: a) Half and full decoder b) Encoders and decoders • Multiplexers and demultiplexers (Qualitative treatment using block diagrams).
Assessment Tasks	**1. Written and/or oral assessment on the skills and knowledge required on digital electronics fundamentals.**
Conditions/Context of assessment	**1. Written and/or oral assessment can be conducted in a classroom environment.**
Learning Outcome 04	**SENSORS AND TRANSDUCERS**
Assessment Criteria	**4.1 Demonstrate knowledge on sensors**
	4.2 Demonstrate knowledge on transducers
	4.3 Demonstrate knowledge on their applications
	4.3 State the difference between sensors and transducers
	SENSORS AND TRANSDUCERS • Define sensor and transducer

	• Explain construction, application and operation of the following transducers Types of Transducers: b) Temperature transduces ➢ Thermocouple ➢ Resistance Temperature Detectors (RTDs) ➢ Thermistors c) Pressure Transducers ♦ Piezoelectric sensors d) Light Transducers: ♦ Photodiodes ♦ Phototransistors ♦ Light-Dependent Resistors (LDRs) e) Sound Transducers ♦ Microphones ♦ Piezoelectric sensors e) Position and Displacement Transducers ➢ Potentiometers ➢ Linear Variable Differential Transformers (LVDTs) ➢ Rotary Variable Differential Transformers (RVDTs) • State the application of following sensors above • State the difference between sensors and transducers
Assessment Tasks:	**1. Written assessment on the skills and knowledge required to understand Transducers and Sensors**
Conditions/Context of assessment	**1. Written assessment can be conducted in a classroom environment.**

ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOPIC	WEIGHTING %
1	**THE OPERATIONAL AMPLIFIER**	**20**
2	**POWER ELECTRONICS**	**20**
3	**INTRODUCTION TO DIGITAL, LOGIC GATES, BOOLAN**	**15**
4	**UNIVERSAL GATES AND K-MAPS**	**15**

5	**COMBINATIONAL LOGIC**	**10**
6	**SENSORS AND TRANSDUCERS**	**20**
	TOTAL	**100**

Approach to Teaching and Learning:

12. Observation of adult learning principles. Both institution-based and work-based learning to facilitate the integration of theory and practice.
13. Face-to-face education and learning.
14. Problem-based learning.
15. Online/distance education and learning.
16. Blended/hybrid education and learning.
17. Use of social media.

Approach to Assessment:

7. Weighting of practical and theory assessment: 70% theory and 30% practical.
8. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50%.
9. Oral assessment to be conducted by a panel of two or more assessors.
10. RPL assessment.
11. Portfolio of evidence.
12. **Assessment of work conducted by both individual learners and teams of learners.**

Resources:

7. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

8. Facilities, Tools, Equipment and Materials
    - Electrical tool box
    - Digital multi meter
    - Ladder
    - Laptop
    - Soldering kit
    - PPE
    - Drilling machine
    - Insulation tape
    - Mutton cloth
    - Detergents
    - Cables
    - Cable enclosures
    - Electrical accessories

9. **Learning Resources**

Relevant training manual (learners' guide) and facilitators' guide

10. Reference Materials (recommended textbooks, recommended readings)

J.O Bird (1997) Electrical circuit theory and technology Butterworth, Heinnman, Oxford
Boylestad Introduction to circuit analysis
S.A Knight (1994) Electrical and Electronic Principles 2
Ashfaq Hussan Electrical Engineering principles Butterworth, Heinnman, Oxford

Module Code:	**321/25/M03**
Module Title:	**ENGINEERING DRAWING, ELECTRICAL CIRCUIT DESIGNING**
ZNQF Level:	**4**
Credits:	**15**
Duration:	**150** hours
Relationship with Qualification Standards:	Based on Unit Standard of Qualification Standard for electrical artisans
Pre-requisite modules:	none
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an Instrumentation and Control Mechanic, electrical power artisan, computer services artisan and electronic communication artisan to understand engineering drawing and electrical circuit design principles. This unit will enable an individual to design electrical circuit that meets the IEE, SAZ, and CAS, wiring Regulations, BS3939 Standards and Electricity Supply Regulations.
List of Learning Outcomes:	6.1 explain and apply the basics of engineering drawing equipment, types of lines, lettering, the relevant conventional standards, etc. 6.2 convert orthographic projection drawings to pictorial views and vice-versa. 6.3 Apply dimensions to orthographic sketches drawings 6.4 produce and interpret section drawings. 6.5 design and interpret Electrical circuit in electrical power, computer, electronic and Instrumentation and control engineering. 6.6 use AutoCAD concepts to draw all diagram done on the drawing board. 6.7 use standard drawing software/CAD to design and interpret electrical circuits.

Learning Outcome 01	Explain and apply the basics of engineering drawing equipment, types of lines, lettering, the relevant conventional standards, etc.
Assessment Criteria:	
Content:	1.1 Explain and apply the basics of engineering: - drawing equipment - types of lines - lettering - the relevant conventional standards - numbering and dimensioning 1.2 State the importance of presenting drawings to standard. 1.3 State the advantages of using drawings as means of technical communication. 1.4 Identify and state the different pencil grades used for drawing particular lines, arcs and circles. 1.5 Identify and state the standard paper sizes used in Engineering Drawing. 1.6 Copy to a given scale. 1.7 Identify, use and list drawing instruments and equipment as e.g. compass, set square, T rule, drawing board. 1.8 Identify the different types of lines. 1.9 State the use of the different types of lines. 1.10 Draw the different types of lines using the appropriate pencil grades. 1.11 Produce clear and uniform freehand letters and numerals in accordance with BS308. 1.12 Draw border lines. 1.13 Draw title, blocks and print required information. 1.14 Show a working knowledge of BS308: 1972 to Publication PD 7308:

	1978. 1.15 Bisect a line. - Construct a perpendicular from a given point to a line. - Divide a line into proportional parts. 1.16 Define an angle. - Identify the different types of angle. - Bisect and angle. - Construct specific angles without a protractor. - Copy an angle. 1.17 Define a circle. - Identify parts of a circle. - Find the centre of a given arc or circle. - Draw a circle through any three point. - Join straight lines with arcs. - Join two or more arcs and circles. 1.18 Identify types of triangles. - Construct a triangle given - The sides. - Two angles and sides. - Perpendicular height and the base. - Construct the following triangle: - Inscribed - Circumscribed - Escribed 1.19 Define a tangent.

	- Construct internal and external tangents to any two circles. 1.19 Name the regular polygons up to eight sided. - Construct the above polygons using common and particular methods 1.2 Describe an ellipse and state its parts. -Construct an ellipse given major and minor axes using: - Trammel methods - Auxiliary circle method - Rectangular method - Approximate method - Focal point method
Assessment Tasks:	1. Written assessment on the skills and knowledge required the of engineering drawing equipment.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 02	**Convert orthographic projection drawings to pictorial views and vice-versa.**
Assessment Criteria	2.1 Identify types of pictorial drawings.
	2.2 Identify types of orthographic drawings.
	2.3 Sketch isometric and oblique drawings.
	2.4 Draw to a given scale isometric drawing.
	2.5 Draw to a given scale oblique drawing.
	2.6 Draw to a given scale 1st angle orthographic drawing.
	2.7 Draw to a given scale 3st angle orthographic drawing.
	2.8 Convert pictorial diagram to orthographic diagram.
	2.9 Convert orthographic to pictorial diagram.

	2.10 Draw isometric and oblique drawings using AutoCAD.
Content	2.1 **PICTORAL DRAWING** - State the advantages of pictorial drawings. - State the two types of pictorial views i.e. isometric and oblique. - Identify isometric and oblique drawings and sketches. - Sketch and draw simple engineering components in oblique and isometric views. - use AutoCAD concepts to draw pictorial drawings. 2.2 **ORTHOGRAPHIC PROJECTION** - Define projection. - Describe how to obtain views in orthographic projection, i.e. first angle and third angle. - Explain the difference between first and third angle orthographic projection. - Select the side most suitable to be the front view. - Appreciate the possibilities of various numbers of views to completely describe objects. - Illustrate the positioning (layout) of the views on paper. - Sketch and draw objects in first and third angle projection, including hidden details . - Sketch and draw objects in first and third angle projection using AutoCAD. - Convert or orthographic draws (first and third angle) to pictorial views (isometric and oblique) and vice versa.
Assessment Tasks	Written and/or practical assessment on the skills and knowledge required draw pictorial and orthographic drawings.

Conditions/Context of assessment	1. Written and/or practical assessment can be conducted in a classroom or computer laboratory environment.
Learning Outcome 03	**Apply dimensions to orthographic sketches drawings**
Assessment Criteria	
Content	- State and illustrate the different methods of dimensioning. - State and illustrate the basic rules of dimensioning. - Illustrate the acceptable layout of dimensions. - Interpret all relevant information from given first and third angle orthographic views. - Dimension the sketches and drawings according to standard conventions. - Dimension the sketches and drawings using AutoCAD
Assessment Tasks	Written and/or practical assessment on the skills and knowledge required for dimensioning orthographic drawings and sketches.
Conditions/Context of assessment	Written and/or practical assessment can be conducted in a classroom environment.
Learning Outcome 04	**Produce and interpret section drawings.**
Assessment Criteria:	
Content:	- State the reasons for sectioning. - State the general rules to be observed when sectioning. - Identify and state the different types of sections. - List features which are not sectioned. - Illustrate the correct methods of hatching. - Interpret relevant information from given section drawings. - Sketch and draw sectional views of simple engineering components.
Assessment Tasks:	Written and/or practical assessment on the skills and knowledge required

	for sectioning orthographic drawings and sketches.
Conditions/Context of assessment	Written and/or practical assessment can be conducted in a classroom environment.
Learning Outcome 05	**Produce and interpret block and schematic diagrams of electronic, electrical and instrumentation drawings.**
Assessment Criteria:	
Content:	- Identify BS3939 symbols and state the meaning of each symbol. - Draw the relevant symbols for all relevant components. - Identify block and schematic diagrams. - Interpret all relevant information from given block and schematic diagrams.
Assessment Tasks:	1. Written assessment on the skills and knowledge required to Identify BS3939 symbols 2. Written and/or practical assessment on the skills and knowledge required to drawn electric circuit using BS 3939 symbols. 3. Written and/or oral assessment on the skills and knowledge required identify and interpret all relevant information from block and schematic diagrams.
Conditions/Context of assessment	2. Written and/or oral assessment can be conducted in a classroom environment.
Learning Outcome 06	**Design and interpret Electrical circuit in electrical power, computer, electronic and instrumentation and control engineering.**
Assessment Criteria:	**6.1 Obtain circuit application** **6.2 Implement the designed circuit** **6.3 Modify circuit diagram**

Content:	**6.6.1 Obtain circuit application** ○ Obtain oral, written and circuit application ○ Draft control/power circuits ○ Simulate designed draft circuits **6.6.2 Implement the designed circuit** ○ Apply standard drawing software/ CAD. ○ Gather drawing instruments/tools/paper. ○ Draw drafted circuits to required standards ○ Get approval for drawn circuits ○ File/Shelf approved drawn circuits **6.6.3 Modify circuit diagram** ○ Retrieve existing copy of circuit diagram ○ Analyse copy of circuit diagram ○ Incorporate changes ○ Simulate modified circuit diagram ○ Adopt modified circuit diagram
Assessment Tasks:	1. Written and/or oral assessment on the skills and knowledge required to conduct appropriate Electrical Circuit Designing as outlined in the assessment criteria and content above. 2. Practical assessment on the conducting the following of SHEQ requirements, the consideration of environmental factors which affect the plant.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 3. The context of assessment should be as follows for different courses. a) **Electrical Power Engineering**: - Design and draw AC and DC motor control diagrams *(face-plate starter, thyristor control for DC motors only, DOL and Star-Delta starters including sequential starting, auto-transformer and rotor resistance starter)*

	- Design and draw, using given data and floor plan, simple electrical installation plans. - Identify, read, design and interpret power distribution diagrams of one line and three lines. b) **Electronic Engineering**: **Instrumentation and Control Systems**: - Identify BS3939 symbols on given circuit diagrams/process loops and interpret all relevant information from circuit diagrams. - Design and draw, from given data, circuit diagrams/process loops using BS3939 symbols (temperature, pressure, level, pH and flow rate). - Interpret and draw block diagrams of PLC monitoring systems. - Design and draw ladder diagrams of a PLC monitoring system using given data. c) **Electronic Engineering: Communication Systems** - Interpret and draw circuit symbols for input and output transducers, for example, microphones and loudspeakers. - Interpret and draw circuit symbols for filters, amplifiers, Op-amps. - Interpret symbols for power and signal lines and ground. - Interpret and draw functional and block diagrams of receivers and transmitters, for example, radio receiver and T.V. - Interpret and draw digital symbols and circuits (British and American), for example, multiplexers, coders, decoders and demultiplexers. d) **Electronic Engineering: Computer Systems:** - Draw the symbols of various electronic components - Detailed block diagram of a computer - Drawing and component layout of motherboard, display card, Ethernet card, etc

	e) **Electronic Engineering: Telecommunication Systems** - Interpret and draw circuit symbols for input and output transducers, for example, microphones and loudspeakers. - Interpret and draw circuit symbols for filters, amplifiers, Op-amps. - Interpret symbols for power and signal lines and ground. - Interpret and draw functional and block diagrams of receivers and transmitters, for example, radio receiver and T.V. - Interpret and draw digital symbols and circuits (British and American), for example, multiplexers, coders, decoders and demultiplexers.

ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOPIC	WEIGHTING %
1	BASICS OF ENGINEERING DRAWING & AUTOCAD	20
2	PICTORIAL DRAWINGS	15
3	ORTHOGRAPHIC	15
4	SECTIONING	10
5	CIRCUIT DESIGN & DIAGRAMS	40
	TOTAL	**100**

Approach to Teaching and Learning:

8. Observation of adult learning principles.
9. Both institution-based and work-based learning to facilitate the integration of theory and practice.
10. Face-to-face education and learning.
11. Problem-based learning.
12. Online/distance education and learning.
13. Blended/hybrid education and learning.

14. Use of social media.

Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

The lecturer should have a minimum of National Diploma in Electrical Power Engineering

8. Facilities, Tools, Equipment and Materials

- Drawing sets
- Computer
- Printer
- AutoCAD
- Multism/Eagle etc

9. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

10. Reference Materials (recommended textbooks, recommended readings)

1. Yarwood (2013) Geometrical & Technical Drawing Book 1 & 2, Thomas Nelson Son Hong Kong

2. Blackie, B. (2018) Technical Drawing Blakie & Son Harare

3. French, Svensen, Helsen & Urbanick (2016) Mechanical Drawing, 8th ed McGrawHill

4. Michael N. (2012) Basic Electrical Installation, 3rd ed. Macmillan Press

5. Morling, K. (2013) Geometric & Engineering Drawing Macmillan Press

6. Panchal V.M. (2014) Engineering Drawing 53rd Ed Chorotar Publishing House

7. Bhatt N.D. (2014) Machine Drawing Chorotar Publishing House

8. Bland, S. (2016) Graded Exercises in Technical Drawing. Macmillan Press

9. Verma, G. (2016) AutoCAD Electrical 2016 Black Book

Module Code	**321//25/M07**
Module Title:	**ELECTRICAL PLANT & EQUIPMENT INSTALLATION, AND MAINTENANCE**
ZNQF Level:	4
Credits:	**20**
Duration:	**200** hours
Relationship with Qualification Standards:	Based on Unit Standard **PLANT AND EQUIPMENT INSTALLATION** of Qualification Standard for an electrician
Pre-requisite modules:	1. **PLANT AND EQUIPMENT** 2. **MANTAIN ELECTRICAL EQUIPMENT** 3. **SAFETY, HEALTH, ENVIRONMENTAL AND QUALITY MANAGEMENT**
Purpose of Module:	This unit will enable an individual to install plant and equipment. After learning this module, one will be able to install plant and equipment following IEE, SAZ & CAS wiring Regulations. This module is intended to develop individuals who would be employed as artisans in Electrical Installation & Maintenance. Or individuals wishing to establish enterprises in Electrical Installation & Maintenance.
List of Learning Outcomes:	**LO 1 Conduct Installation survey** **LO 2 Design installation plan** **LO 3 Produce quantities of required resources** **LO 4 Conduct installation** **LO 5 Inspect Installation** **LO 6 Test and Re-commission plant and equipment** **LO 7: Monitor running Plant** **LO 8: Design maintenance plan** **LO 9: Request required resource** **LO 10: Carry out planned maintenance** **LO 11: Carry out breakdown maintenance**

	LO 12: Test and re-commission plant and equipment
Learning Outcome 01	**Conduct Installation survey**
Assessment Criteria:	2.1.1 Access power supply adequacy 2.1.2 Establish environmental factors that affect installation 2.1.3 Establish installation requirements 2.1.4 Produce survey report
Content:	2.1.1 Access power supply adequacy 1. Single Phase AC Theory 2. Three Phase Theory 3. Distribution Wiring Systems 4. Distribution Plant and Equipment 2.1.2 Establish Environmental factors that affect installation 1. Cables and Enclosures 2. Earthing and Earth Leakage Protection 3. Earth monitoring unit 4. Special Installations 2.1.3 Establish Installation requirements 1. Consumer Installations 2. Tools and equipment 3. Bill of quantities 4. Quotations 2.1.4 Survey report is produced 1. Data Processing (Microsoft Word) 2. Data Processing (Spreadsheet) 3. Writing technical reports 4. Writing recommendatory reports
Assessment Tasks:	2. Written and/or oral assessment on the skills and knowledge required to conduct appropriate installation survey as outlined in the assessment criteria and content above. 3. Practical assessment on conducting the SHEQ requirements, the consideration of environmental factors which affect the plant.
Conditions/Context of assessment	3. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 4. The practical assessment will be conducted in the plant or simulated work environment in the training institution.

Learning Outcome 02	Design installation plan
Assessment Criteria:	2.1.1 Establish standards and regulations 2.1.2 Obtain site plan/floor plan 2.1.3 Indicate on the plan installation positions/cable routes/distribution boards /panels/equipment/power permits/luminaries 2.1.4 Produce installation wiring diagrams 2.1.5 Produce according to wiring standards and regulations the installation overlays
Content:)	2.1.1 Establish standards and regulations 1. Outline the importance of application of the regulations 1. Electricity supply regulations 2. Electricity at work regulations 3. Health and safety at work 4.IEE regulations regarding design, selection and installation of electrical equipment 1. Cable selection procedure 2. Cable enclosure selection 3. Switch gear selection 2.1.2 Produce site plan/floor plan 1. Define site plan 2.Wiring systems for domestic and industrial installations 3.Illumination - types of lamps - design of lighting schemes 4 Power -Socket outlets -cooker circuits - water heating -special installation 2.1.3 Indicate on the plan installation positions/cable routes/DBs/panels/equipment/power permits/luminaries 1. Cable management systems -conduit -trunking/Bus Bar/Rising mains -ducting -cable ladders -cable trays 2.Selection and positioning of Distribution Boards 2.1.4 Interpret installation wiring diagrams circuit diagrams 1.Use circuit diagrams in electrical engineering

	2. Identify electrical symbols to BS3939 3.Draw relevant symbols for all relevant components 4.Identfy block and schematic diagrams 5.Interpret all relevant information from given block and schematic diagrams 2.1.5 Produce according to wiring standards and regulations the installation overlays 1 Circuit Diagrams 2.Interpretation of circuit diagrams 3.Presentation of circuit on overlays 4. Use of overlays in conducting installation work
Assessment Tasks:	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or simulated work environment in the training institution.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 3. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 4. The practical assessment will be conducted in the plant or simulated work environment in the training institution.
Learning Outcome 03	**Produce quantities of required resources**
Assessment Criteria:	3.1 Determine installation materials 3.2 Establish human resources requirements 3.3 Establish project critical path
Content:	4.1 Determine installation materials • How to determine installation materials • Use of wiring diagrams • Floor plans • Types enclosures • Types of cables • Types of accessories • Types of switchgear • Environmental conditions 3.2 Establish human resources requirements

	1 identify scope of works 2 establish time frame for the project 3 identify personnel for various tasks 2.2 Establish project critical path 1 definition of critical path 2 importance of critical path 3 drawing up of critical path 4 application of critical path
Assessment Tasks:	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or simulated work environment in the training institution.
Conditions/Context of assessment	3 Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 4 The practical assessment will be conducted in the plant or simulated work environment in the training institution. 5 Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 6 The practical assessment will be conducted in the plant or simulated work environment in the training institution.
Learning Outcome 04	**Conduct installation**
Assessment Criteria:	2.4.1 Observe SHEQ 2.4.2 Prepare installations 2.4.3 Mark installation positions according to standard 2.4.4 Install cable enclosure 2.4.5 Mount distribution boards/panels 2.4.6 Draw in cables 2.4.7 Connect switch gear/fitting/equipment/accessories 2.4.8 Adhere to relevant standards and regulations
Content:	2.4.1 Observe SHEQ Regulations governing electrical installations • IEE Wiring regulations • Electricity supply regulations • Low voltage electrical equipment(safety) regulations • Health and safety at work 2.4.2 Prepare installations • Type of building construction

	• Wiring systems • Mounting arrangements 2.4.3 Mark installation positions according to standard • Measurements and use of measuring equipment • Adherence to standards and regulation 2.4.4 Install cable enclosure • Space factor for various cable enclosures • -importance of observing space factor requirements • Procedure for drawing in and laying of cables 2.4.5 Mount distribution boards/panels • Factors influencing the positioning of distribution boards/panels • Proximity to supply authority points • Central point of load • Accessibility to users • security 2.4.6 Draw in cables • Types of cables • Tools and equipment • Environmental conditions 2.4.7 Connect switch gear/fitting/equipment/accessories • Types off switch gear • Termination requirements for switch gear/fitting/equipment/accessories 2.4.8 Adhere to relevant standards and regulations
Assessment Tasks:	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or simulated work environment in the training institution.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 3. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 4. The practical assessment will be conducted in the plant or simulated work environment in the training institution.

Learning Outcome 05	**Inspect Installation**
Assessment Criteria:	2.1 Conduct visual inspection 2.2 Verify terminations/connections 2.3 Check fire labels and correct colour coding 2.4 Verify circuit segregation 2.5 Complete installation check list
Content:	2.1 Conduct visual inspection 1 What is visual inspection regarding electrical installations 2 Inspection check list 2.2 Verify terminations/connections 1 Types of joints 2 Effects of loose connections 3 Corrosion 4 Continuity tests 2.3 Check fire labels and correct colour coding 1 Types of fire equipment 2 Types of fire extinguishers • Correct colour coding • Combustion 2.4 Verify circuit segregation 1 Definition of a circuit 2 Define segregation 3 Zoning 4 Categories 2.5 Complete installation check list 1 Visual inspection 2 Tests that should be carried out after completion of installation
Assessment Tasks:	1. Written and/or oral assessment on the skills and knowledge required to conduct appropriate inspection of installation as outlined in the assessment criteria and content above. 2. Practical assessment on conducting SHEQ requirements, in consideration of environmental factors which affect the installation.
Conditions/Context of assessment	6 Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 7 The practical assessment will be conducted in the plant or simulated work environment in the training institution.
Learning Outcome 06	**Test and Re-commission plant and equipment**

Assessment Criteria:	**2.1** Test continuity 2.2 Test polarity 2.3 Carry out insulation resistance test 2.4 Carry out earth loop impendence test 2.5 Verify supply voltage 2.6 Verify supply phase rotation 2.7 Complete installation test check list 2.8 Conduct installation test run 2.9 Do pre-commission test 2.10 Perform production test runs 2.11 Do installation documentation and handover
Content:	7.1 Test continuity 1) continuity tests 2.2 Test polarity 1) polarity test 2) continuity of ring circuit 2.3 Carry out insulation resistance test 1) insulation resistance tests 2.4 Carry out earth loop impendence test 1) earth loop impendence tests 2.5 Verify supply voltage 1) supply voltage verification 2.6 Verify supply phase rotation 1) supply phase rotation verification 2.7 Complete installation test check list 1) installation test check listing 2.8 Conduct installation test run 1) installation test runs 2.9 Do pre-commission test 1) pre-commission testing 2.10 Perform production test runs 1) production test runs 2.11 Do installation documentation and handover 1) installation documentation and handover excise
Assessment Tasks:	1. Written and/or oral assessment on the skills and knowledge required to conduct appropriate inspection, testing and commissioning as outlined in the assessment criteria and content above. 2. Practical assessment on conducting SHEQ requirements, in consideration of environmental factors which affect the plant.

Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the installation or simulated work environment in the training institution. 3. The context of assessment should include the facilities, tools, equipment and materials listed below: Basic Electrician tools and equipment kit: Laptop, Ladder, Megger, Lux meter, Earth leakage tester, digital multimeter, tong tester, and other useful tools . Materials: Cables ,Insulation tape ,conduit and electrical accessories etc.
Learning Outcome 07	**Monitor running Plant**
Assessment Criteria:	**2.1 Adhere to safety, health, environmental and quality procedures.** **2.2 Complete operation checklist** **2.3 Attend to operation breakdowns** **2.4 Document frequency and nature of operation breakdowns**
Content:	**2.1 Adhere to safety, health, environmental and quality procedures** • Switching and isolation of electrical equipment • Definition of Isolation & Switching • Purpose of Isolation & Switching of Electrical Installations • Distinguish between Isolation & Switching • Devices used in Isolation & Switching • Identify High Voltage & Low Voltage switchgear. • Positioning of Isolating & switching devices in an installation • Install switching and isolating devices and systems • Issue of safety documents • Risk and hazard identification **2.2 Complete operation checklist** • Operation procedures • Definition of PPE • Purpose of PPE • Identification of appropriate PPE • Selection of proper tools & equipment • Carryout hot checks on equipment • Carryout cold checks on equipment • Isolate ,barricade ,Lock out equipment **2.3 Attend to operation breakdowns** • Equipment and machinery user manuals • Procedure for fault finding • Fault identification • Fault rectification

	• Fault rectification reports **2.4 Document frequency and nature of operation breakdowns** • Record keeping • Purpose of records in maintenance work • Tools used to create records for machines • Plant Maintenance • Types of Maintenance • Maintenance schedules • Preparation of maintenance schedules • Procedures for carrying out maintenance • Maintenance & breakdown reports
Assessment Tasks:	4. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 5. Practical assessment on the conducting the following of SHEQ requirements, the consideration of environmental factors which affect the plant.
Conditions/Context of assessment	5. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 6. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 7. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, **Materials**: Cables Insulation tape PVC conduit Electrical accessories
Learning Outcome 08	**Design maintenance plan**
Assessment Criteria:	**2.1 Categorise Plant equipment and machinery.** **2.2 Note Different maintenance schedules** **2.3 Collate Breakdown history** **2.4 Draft Maintenance plan** **2.5 Schedule Maintenance activities**
Content:	**2.1 Categorise Plant equipment and machinery** - Identify and classify types of machinery in the plant - **Transformers:** - Essential features of Transformer construction - Principle of operation of an ideal transformer

	- Use the relationship Ep/Es=Np/Ns=Is/Ip=Vp/Vs - Identify transformer losses & ways to minimize them - **Draw & explain the following types of transformer windings & core construction**: - Core type - Shell type - Concentric winding - Sandwich winding - Toroidal core - Describe principle of operation - Auto Transformer, - Current Transformers - Compare Double-Wound & Auto transformers - **Perform Transformer Tests:** - Open Circuit - Short Circuit - **Transformer Protection:** - Types of Transformer protection devices - Winding Temperature - Oil temperature - Buchholtz relay - Induction over current relay - Lightning protection - **Transformer Cooling Methods:** - Natural air - Forced air - Water - **DC Machines:** - Describe the construction, operation & application of DC Generators - Types of DC Generators: - Shunt - Series - Compound - Application of the above DC Generators - Installation of DC Generators - Alignment of shafts & couplings - Fitment of Bearings - Identify & rectify faults in DC Generators. - Describe the construction, operation & application of DC Motors - Series - Shunt - Compound - Application of the above DC Motors - Methods of Starting & speed control of DC Motors - Field rheostat - Faceplate

	- Electronic Drives - Identify & rectify faults in DC Motors - **A.C Machines** - Describe construction & operation of single phase machines - Universal motor/ single phase series motor - Shaded pole motor - Single phase synchronous motor - Split phase motor - Capacitor start - Repulsion Motor - Servo motor - Multispeed motor - Describe construction & operation of Three phase machines - Define Synchronous speed, Rotor speed & Slip. - Identify differences between three phase cage rotor Synchronous & three phase wound rotor induction motors - Starting Methods for A.C Induction Motors - Direct online - Star delta - Auto transformer - Rotor Resistance - Soft Starters including programmable logic controllers - Reactance starter ✓ Starting Methods for A.C Synchronous Motors - Pony Motor - Induction Starting ✓ Speed Control for three phase induction motors - Pole Changing - Rotor Resistance - Electronic Drive **2.2 Note different maintenance schedules** - **Types of maintenance schedules** ○ **Routine, predictive, prescriptive and preventive** - Draw up a planned maintenance scheme for the various types of motors & switch gear & Transformers. - Fault identification **2.3 Collate Breakdown history** - Analysis of machinery breakdown history - Types of Electrical machines - Common faults associated with A.C Machines - Repairs & to Electrical machines - Keeping records of repairs of electrical machines **2.4 Draft Maintenance plan** - define maintenance plan -outline the purpose of a maintenance plan

	- Outline how to execute maintenance plan - Planning - Importance of Keeping Plant breakdown history - Importance of maintenance plan in equipment repairs. - Decision making **2.5 Schedule Maintenance activities** - Identify maintenance activities - Establish maintenance timelines - Importance of establishing timelines in carrying out maintenance
Assessment Tasks:	3. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 4. Practical assessment on the conducting the following of SHEQ requirements, the consideration of environmental factors which affect the plant.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 3. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Tongtester Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, **Materials**: Cables Insulation tape PVC conduit Electrical accessories
Learning Outcome 09	**Request Required Resources**
Assessment Criteria:	**3.1 Identify resource parameters** **3.2 Quantify relevant resources** **3.3 Verify requested resources** **3.4 Complete requisition form**
Content:	**3.1 Identify resource parameters** ✓ Determine resource specifications ✓ Electrical Measuring Instruments ✓ Moving coil ✓ Moving Iron ✓ Thermocouple ✓ Watt meter ✓ Energy meter ✓ Digital voltmeter ✓ Define sensitivity of an instrument

	✓ Compare the digital & Analogue instruments **so** **3.2 Quantify relevant resources** ✓ **Produce Bill of quantities** ✓ **Drawings & Diagrams** ✓ Types of Drawings & Diagrams ✓ Block diagrams ✓ Circuit diagrams ✓ Schematic diagrams ✓ Assembly Diagrams ✓ Layout/ location diagrams ✓ Site plans ✓ Application of drawings & diagrams to produce bill of quantities ✓ **Cost required resources** ✓ Sourcing for quotations from suppliers ✓ Production of Comparative schedule for components/ materials ✓ Request for procurement. **3.3 Complete requisition form** ✓ Observe procurement procedures ✓ Compilation of required materials on requisition forms ✓ Submit requisition form to procurement office **3.4 Verify requested resources** ✓ Receive materials. - What aspects are checked upon receipt of materials - Technical specifications - Quantities - Checking for physical damages ✓ Tools used to verify delivered materials - Using source documents to verify specification from received materials - ✓ Importance of verification of materials - Facilitation for payment of the suppliers - Ascertaining correctness of received materials -
Assessment Tasks: (	1. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 2. Practical assessment on the conducting the following of SHEQ requirements, the consideration of environmental factors which affect the plant.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or

	simulated work environment in the training institution. 3. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, **Materials**: Cables Insulation tape PVC conduit Electrical accessories
Learning Outcome 10	**Carry out planned maintenance**
Assessment Criteria:	**4.1 Obtain and analyse maintenance work order and permit** **4.2 Observe SHEQ** **4.3 Carry out tasks according to the maintenance work order** **4.4 Test and commission operation and performance** **4.5 Close maintenance work order and work permit**

Content:	**4.1 Obtain and analyse maintenance work order and permit**
	• Isolation and switching
	• Conduct tests to ensure circuits are dead
	✓ voltage test
	✓ Use of grounding wires
	✓ Lock out isolating switchgear.
	• barricading work area
	✓ Importance of barricading work area
	✓ Types of barricading materials
	✓ warning signs
	
	4.2 Observe SHEQ
	• specifying and adhering to Safety, Health and environmental standards
	4.3 Carry out tasks according to the maintenance work order
	• Analysing Operator's job card
	• Repairing of machinery conducted
	
	4.4 Test and commission operation and performance
	• Verification of terminations
	- Carrying out Commissioning tests
	  - Visuals Inspection
	  - Test Continuity of Protective conductors
	  - Test for continuity of ring final circuit
	  - Test for Earth Electrode resistance
	  - Test for Earth fault loop impedance
	  - Test for polarity
	  - Test for operation of RCD
	- Conduct test run on the Plant/equipment
	
	Commission the plant

	4.5 Close maintenance work order and work permit • Work permit/ order closed
Assessment Tasks:	3. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 4. Practical assessment on the conducting the following of SHEQ requirements, the consideration of environmental factors which affect the plant.
Conditions/Context of assessment	5. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 6. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 7. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, **Materials**: Cables Insulation tape PVC conduit Electrical accessories
Learning Outcome 11	**Carry out breakdown maintenance**
	5.1 Obtain and analyse maintenance work order and permit **5.2 Observe SHEQ** **5.3 Analyse machine history** **5.4 Diagnose breakdown machine according to laid down procedure** **5.5 Inspect and test repaired machine** **5.6 Close maintenance work order and work permit**
Content: (	**5.1 Obtain and analyse maintenance work order and permit** - Maintenance work Orders - Define Maintenance work Order - Types of Maintenance work orders - Repairs - Service - Preventive Maintenance order - Predictive Maintenance order - Daily work orders - Industries where maintenance orders are used - Manufacturing - Facilities management - Hospital and agriculture installations - Power utilities

	- Oil & Gas plants - **Permit to Work** - Define work permit - Contents of permit to work - Work done, the equipment to be used& the personnel involved - Precautions to be taken when performing the task - Other groups to be informed of work being performed in their area - Authorization for work to commence - Duration that the permit is valid - Wit ness mechanism that all work has been complete and the work site restored to a clean ,safe condition - Actions to be taken in an emergency. **5.3 Analyse machine history** - Analysis of machinery breakdown history - What is machinery History - Recorded outline on all actions taken on the machinery - Why keep maintenance record - Prevent expensive repair works - Helps in creating specialised maintenance programs - Prevents problems regarding warranty claims - Increases the safety of operators - Helps in tracking who is accountable for the Machinery - Helps to decide on replacement of machinery to reduce maintenance costs or to continue to repair & maintain. - It increases resale value. - What are the tools for gathering machine History - Using Machine history cards - Information on machine history cards - Serial number - Date of maintenance - Break down problem - Maintenance activity - Person in charge - Remarks **5.4 Diagnose breakdown machine according to laid down procedure** - **Categorize Causes of Equipment breakdown** - Technical - Thermally induced - Mechanically - Erratic - Human - Untrained personnel operating equipment - Ignoring warning signals - Failure to read owner's manual - Overrunning machine capability

	- Improper maintenance - Replacement of parts when required - Poor electrical connections - Improper storage - **Troubleshoot Using recommended procedure** - Verifying existence of the problem - Involvement of the operator - Narrowing down the problem's root cause - Using Test equipment - Isolating supply - Disassembling where necessary - identifying low maintenance items first - Compliance with safety protocols - Correcting the cause of the problem - Correcting the Problem **5.5 Inspect and test repaired machine** - Verifying the rectified problem - Checking all the gauges for correct operation - Performing manufacturer's recommended procedure - Performing Electrical tests - Continuity test - Insulation resistance test - Test run the equipment with operator **5.6 Close maintenance work order and work permit** - Work permit/ order closing Compiling of commissioning reports
Assessment Tasks:	3. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 4. Practical assessment on the conducting the following of SHEQ requirements, the consideration of environmental factors which affect the plant.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 3. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, **Materials**: Cables Insulation tape PVC conduit Electrical

	accessories
Learning Outcome 12	**Test and Re-commission plant and equipment**
Assessment Criteria:	**6.1 Adhere to manufacturer's operating procedures** **6.2 Carry out operation inspection according to checklist** **6.3 Carry out visual inspection** **6.4 Carry out corrective action where necessary** **6.5 Carry out proper handover procedure**
Content:	**7.2 Adhere to manufacturer's operating procedures** • Testing & Commissioning in Electrical work ✓ Outline reasons for Testing& commissioning Electrical Installations - Compliance with The IEE Regulations - Compliance with Manufacturer's instructions - Enable the supply authority to supply electricity to the installation ✓ Outline the kinds of electrical installation tests carried out - Continuity Testing - Insulation resistance Testing - Earth Continuity Testing - Earth Electrode resistance Testing - Polarity Testing - Electrical performance Testing. **7.3 Carry out operation inspection according to checklist** • **Electrical Drawings** ✓ Identify Types of Drawings - Circuit - Schematic - Layout - Site ✓ Outline the application of Electrical Drawings - Operation of the electrical installation/equipment - Location of electrical components within the installation - Installer's /Operator's guide in wiring the system - Repairer's guide in fault finding - In creating bills of quantities for materials to be used in an installation - Guide for inspection & testing - Show routes for cabling. **7.4 Carry out visual inspection** • Define visual inspection ✓ Outline purpose for visual inspection - Checking for correctness of installed equipment with reference to

	the manufacturers' specifications. - Checking for correctness of sizes of cables for given circuits - Checking for proper connections to equipment - Checking for deviations in the specifications - Verifying colour codes for cables used - Verifying Earth arrangements - Verifying the ratings for protective devices **7.5 Carry out corrective action where necessary** ✓ Implementation of inspection & test results - Correcting all the issues on the installation found not meeting the standards. - Re – inspection for failed tests ✓ Completion of inspection reports - Purpose of completion reports - Contents of reports - Signatories to the reports **6.5 Carry out proper handover procedure** • **Compiling of commissioning reports** ✓ Define Commissioning - Purpose for commissioning report - Items included in the commissioning report. - Signatories to the report. ✓ Handover of commissioned Installation/Equipment.
Assessment Tasks:	3. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 4. Practical assessment on the conducting the following of SHEQ requirements, the consideration of environmental factors which affect the plant.
Conditions/Context of assessment	4. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 5. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 6. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, **Materials**: Cables Insulation tape PVC conduit Electrical accessories

Learning Outcome 13	**Gather operational history**
Assessment Criteria:	**1.1 Obtain work order/ job card** **1.2 Observe SHEQ** **1.3 Analyze operation /operator`s report** **1.4 Establish machine / equipment history**
Content:	**1.1 Obtain work order/ job card** • Communication skills • Knowledge of MS office ,Excel and Spreadsheet **1.2 Observe SHEQ** • Hazards in electrical work environment • Dangers in electrical work environment. • Precautions in handling electrical tools and materials. • First-aid for electrical accidents. • Statutory regulations. • House keeping **1.3 Analyze operation /operator`s report** • Engineering analytical techniques • Knowledge of equipment manufacturer`s manual **1.4 Establish machine / equipment history** • Inspection of domestic installation. • Inspection of electrical machines o AC machines o DC machines o Portable electrical equipment
Assessment Tasks:	6. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 7. Practical assessment in the conducting of SHEQ requirements and consideration of environmental factors which affect the plant.
Conditions/Context of assessment	7 Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 8 The practical assessment will be conducted in the plant or simulated work environment in the training institution. 9 The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop ,Ladder, Bearing puller, Tong tester, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, insulation resistance tester, electric blower, crowbar **Materials**: Cables ,Insulation tape, PVC conduits and Electrical accessories, winding vanish, stestoflex sleeve tags
Learning Outcome 14	**Diagnose faults**

Assessment Criteria:	**2.1 Carry out visual inspection** **2.2 Carry out necessary tests using appropriate tools and equipment** **2.3 Identify faulty components/ devices/ equipment**

Content:	**2.1 Carry out visual inspection**
	• Identifying faults
	• Causes of faults and breakdown.
	• Interpretation of circuit diagrams.
	• Trouble shooting techniques
	2.2 Carry out necessary tests using appropriate tools and equipment
	• Inspection and testing of machines (e.g. performance test).
	• Domestic installations
	  ○ Types and sequence of test; e.g. polarity, insulation resistance, earthing and continuity of ring circuit.
	  ○ Importance of testing and inspection.
	2.3 Identify faulty components/ devices/ equipment
	• Fault identification
	• Electrical and electronic circuit analysis
	  ○ Knowledge of basic components of circuits (switches, circuit breakers, transistors,
	• Diagnosing techniques

Assessment Tasks:	5. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 6. Practical assessment in conducting the following SHEQ requirements in consideration of environmental factors which affect the plant.
Conditions/Context of assessment	4. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 5. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 6. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, Tong tester, multimeter, crowbar. **Materials**: Cables Insulation tape PVC conduit Electrical accessories
Learning Outcome 15	**Rectify faults**

Assessment Criteria:	**3.1 Carry out Root Cause analysis** **3.2 Carry out appropriate remedial action if necessary** **3.3 Request for required material if necessary following laid down procedure** **3.4 Fix or replace faulty components where possible as per diagnosis**
Content:	**3.1 Carry out Root Cause analysis** • Root cause analysis methods **3.2 Carry out appropriate remedial action if necessary** • Disassembling and assembling of components and equipment • Preparing electrical wire joints, carrying out soldering, crimping and measuring insulation resistance of cables. **3.3 Request for required material if necessary following laid down procedure** • Requisition of materials • Bill of quantities preparation **3.4 Fix or replace faulty components where possible as per diagnosis** • Introduction to electrical tools • Electrical precautions
Assessment Tasks:	3. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 4. Practical assessment in conducting the following SHEQ requirements in consideration of environmental factors which affect the plant.
Conditions/Context of assessment	4. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 5. The practical assessment will be conducted in the plant or simulated work environment in the training institution. 6. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, multimeter ,tong tester, crowbar **Materials**: Cables Insulation tape PVC conduit Electrical accessories
Learning Outcome 16	**Test equipment**
Assessment Criteria:	**4.1 Perform visual inspection** **4.2 Carry out appropriate tests following laid down procedure** **4.3 Carry out bench testing where necessary** **4.4 Carry out in situ testing**

	4.5 Handover is done following laid down procedure **4.6 Close work order**

Content:	**4.1 Perform visual inspection**
	• Labelling and colour coding
	• Equipment inspection procedure
	  ○ Electrical machines
	  ○ Portable electrical equipment
	  ○ Domestic wiring
	  ○ Industrial and commercial wiring
	
	4.2 Carry out appropriate tests following laid down procedure
	• Electrical tests
	  ○ Continuity test
	  ○ Insulation resistance test
	  ○ Test types of circuit breaker (low and high voltage)
	  ○ Test different types of earthing.
	  ○ Oil tests
	
	4.3 Carry out bench work where necessary
	• Workshop practice
	4.4 Carry out in situ testing
	• Electrical tests
	  ○ Continuity test
	  ○ Insulation resistance test
	  ○ Test types of circuit breaker(low and high voltage)
	  ○ Test different types of earthing systems.
	• Test report writing
	4.5 Handover is done following laid down procedure
	• Commissioning procedures
	
	4.6 Work order is closed
	• Hand-over procedures

Assessment Tasks:	5. Written and/or oral assessment on the skills and knowledge required to conduct appropriate plant maintenance as outlined in the assessment criteria and content above. 6. Practical assessment in conducting the following SHEQ requirements in consideration of environmental factors which affect the plant.
Conditions/Context of assessment	8. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 9. The practical assessment will be conducted in the plant or simulated work environment in the training institution.

	10. The context of assessment should include the facilities, tools, equipment and materials listed below: **Basic Electrician tool kit**: Tachometer, Stethoscope, Laptop Ladder, Bearing puller, Chain block, Megger, Infrared camera, Lux meter, Earth leakage tester, **Materials**: Cables Insulation tape PVC conduit Electrical accessories ,mild steel bar ,bench vice ,power saw and pillar drilling machine ,flat files, drill bits, chuck key, round file, apes, stock and dice pipe vice

5.0 SPECIFICATION GRID

TOPIC NO.	TOPIC	WEIGHTING %
1	**Conduct Installation survey**	5
2	**Design installation plan**	5
3	**Produce quantities of required resources**	5
4	**Conduct installation**	10
5	**Inspect Installation**	5
6	**Test and Re-commission plant and equipment**	5
7	**Monitor running Plant**	10
8	**Design maintenance plan**	5
9	**Request Required Resources**	5
10	**Carry out planned maintenance**	10
11	**Carry out breakdown maintenance**	10
12	**Gather operational history**	5
13	**Diagnose faults**	5
14	**Rectify faults**	10
15	**Test equipment**	5
	TOTAL	**100**

Approach to Teaching and Learning:

1. Observation of adult learning principles.
2. Both institution-based and work-based learning to facilitate the integration of theory and practice.
3. Face-to-face education and learning.
4. Problem-based learning.
5. Online/distance education and learning.
6. Blended/hybrid education and learning.
7. Use of social media.

Approach to Assessment:

1. Weighting of practical and theory assessment: 70% practical and 30% theory.
2. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50% work-based assessment.
3. Oral assessment to be conducted by a panel of two or more assessors.
4. Portfolio of evidence.
5. Assessment of work conducted by both individual learners and team of learners.

Resources:

1. **Qualifications and experience of Trainers, Assessors and Moderators**

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

The lecturer should have a minimum of National Diploma in Electrical Power Engineering

2. **Facilities, Tools, Equipment and Materials**
   - Electrical tool box
   - Fish tape
   - Spring bender
   - Clamping tool
   - Chasing hammer
   - heat gun
   - Drilling machine
   - Ladder
   - PPE
   - Chain block
   - Grinder
   - Masonry discs and bits
   - Scaffolding
   - Megger
   - Insulation tape
   - Mutton cloth

- Detergents
- PPE
- Cables
- Cable enclosures
- Electrical accessories

3. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

4. Reference Materials (recommended textbooks, recommended readings)

1. Petruzella F.D. (2017) *Programmable Logic Controllers* McGraw Hill Publishing Rockis G. (2014)
2. Electrical Motor Controls for Integrated Systems American Technical Alexander C.K. (2017)
3. Fundamentals of Electrical Circuits McGraw Hill Publishing Tyler, D. W. (2012)
4. Electrical & Electronic Applications 2 Butterworth & Co. Ltd. Lewis, M. L. (2014)
5. Electrical Installation Technology Hutchinson & Co. Ltd. London W. Bolton (2015)
6. Programmable Logic Controllers, 4th Edition McGraw Hill Publishing Lewis M.L. (2016)
7. Electrical Installation Technology Hutchinson & Co. Ltd. Mark H. (1995)
8. Solar Electric System for Africa Common Wealth Science Council Bimbhra P.S. (2008)
9. Electrical Machinery Khanna IEE Regulations 15 or 16th Edition Z Wiring Regulations and code of Practice (latest edition)

- PPE
- Drilling machine
- Welding machine
- Megger
- Insulation tape
- Mutton cloth
- Detergents
- PPC
- Cables
- Cable enclosures
- Electrical accessories
- Crimping Tool
- Pliers
- Digital Multi Meter
- Tong Tester / Clamp Meter
- Phase Sequence Indicator
- Soldering Iron
- Temperature controlled Soldering Iron
- Stripper
- Hacksaw
- Bending spring (20mm)

- Bending machine
- Workbench
- Bench Vice
- Handrilling machine
- Single/three phase electric motor
- Power saw
- Pedestal grinder
- Stock and dice
- Pipe vice
- Welding machine
- Files
- Scribbers
- Centre punch

5. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

6. Reference Materials (recommended textbooks, recommended readings)

10. Electrical Motor Controls for Integrated Systems American Technical Alexander C.K. (2017)
11. Fundamentals of Electrical Circuits McGraw Hill Publishing Tyler, D. W. (2012)
12. Electrical Installation Technology Hutchinson & Co. Ltd. London W. Bolton (2015)
13. Electrical Installation Technology Hutchinson & Co. Ltd. Mark H. (1995)

Electrical Machinery Khanna IEE Regulations 15 or 16th Edition Z Wiring Regulations and code of practice

Module Code:	**321/25/M04**
Module Title:	**COMMUNICATION AND COMPUTER SKILLS**
ZNQF Level:	**4**
Credits:	**15**
Duration:	**150** hours
Relationship with Qualification Standards:	Based on Unit Standard **321/25/M10 Communication and Computer Skills** of Qualification Standard for Electrical Engineering Discipline
Pre-requisite modules:	N/A
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an Electrical Engineering Artisan to Communicate and use Computer skills. The advantages of Communication and Computer skills include. The ability of students to communicate in different forms of communication, increased computer literacy and ability to execute roles as competent employees with basic managerial skills. Access to this module is open to all target groups including unemployed youths, women and men wishing to establish or improve SMEs in Maintaining Engines.
List of Learning Outcomes:	**L01: COMPUTER SKILLS** **L02: COMMUNICATION**
Learning Outcome 01	**COMPUTER SKILLS**
Assessment Criteria:	**1.1** INTRODUCTION TO COMPUTERS. 1.2 COMPUTER EQUIPMENT AND COMPUTER SYSTEM ARCHITECTURE 1.3.COMPUTER SECURITY 1.4. COMPUTER SOFTWARE 1.5: APPLICATION PACKAGES SOFTWARE 1.6: COMPUTER NETWORKING

<table>
<tr>
<td>Content:</td>
<td>
1.1 INTRODUCTION TO COMPUTERS.

define a computer and draw the basic structure of a computer systems
explain the functions of a computer in terms of the following:

- Receiving (and storing) data
- Processing data
- Outputting data

list and describe input, output and storage devices of a computer
describe and differentiate Random Access Memory and Read Only Memory
Describe the evolution of computers from first generation to date
Describe the following types of computers

Mini computers
Micro computers
Mainframe computers
Super computers

Describe the impact of computers in our everyday life.

1.2 COMPUTER EQUIPMENT AND COMPUTER SYSTEM ARCHITECTURE

Define computer systems architecture
state the main functions of each of the under-listed internal components of a computer system unit:

- Motherboard (ROM, RAM, Processor, Chipsets, Expansion slots)
- Power supply unit (AT & ATX)
- hard disk drive
- CD/DVD -ROM drive
- Processor

Distinguish between internal and external data storage media used with computers.
Care and handling of computer equipment and components
State the necessary conditions for an ideal computer room environment.

1.3.COMPUTER SECURITY

Define the following terms

computer security
Data integrity

Describe the measures to ensure data security
Describe the major threats to Information Systems/ Computers
Define computer crime / fraud and describe the different types of computer crime/fraud
Describe the measures to compact computer crime
Describe a Disaster Recovery Plan (DRP)
Define computer virus and give examples
Describe the types of computer viruses
Describe how computers are infected by computer viruses
State and describe the types of anti-virus packages

</td>
</tr>
</table>

	1.4. COMPUTER SOFTWARE
	
	• define software.
	• explain the following types of software:
	  • system software
	  • application software
	• define and explain operating system software:
	• explain the functions of an operating system
	• State and explain the main differences between DOS and Windows operating system.
	• Describe the following user interfaces
	  • Command Line Interface
	  • Graphical User Interface
	• Perform basic DOS operations
	• Navigate a Windows User interface
	• Describe the following
	  - multiprogramming
	  - Distributed systems
	  - real time systems
	  - Online systems
	  - batch processing systems
	• Install an operating system
	
	1.5: APPLICATION PACKAGES SOFTWARE
	
	• Define and explain the functions of an application software
	• Describe the following sources of application software giving advantages and disadvantages
	  - Customised/In house software
	  - Off the shelf software
	  - Open source software
	  - Shareware
	  - Freeware
	  • Describe software piracy and its implications
	  • Describe software copyright and software licenses
	• State and describe the following application packages giving examples
	  • Word processing software
	  • Spreadsheet software
	  • Database software
	  • Presentation software
	• Navigate and perform basic operations of the above listed application packages
	• Install application packages
	
	1.6: COMPUTER NETWORKING
	• Define computer network
	• State the advantages and disadvantages of networking
	• Describe the following networks
	  • LAN (Local Area Network)
	  • MAN(Metropolitan Area Network)

	• WAN (Wide Area Network)
	• Describe the following types of network topologies
	  • Bus
	  • Star
	  • Ring
	  • Mesh
	  • Hybrid
	• Describe Network Operating System (NOS) and state its features
	• Describe the following transmission media
	• Guided/wired
	  • Twisted pair
	  • Coaxial cable
	  • Fibre optic
	• Unguided/Wireless media
	  • WIFI
	  • Bluetooth
	  • Infrared
	  • Microwave
	• Describe the Internet and state advantages and disadvantages of the internet

	8. Written and/or oral assessment on the skills and knowledge required for Communication and Computer skills as outlined in the assessment criteria and content above. 9. Practical assessment on Communication and Computer skills include installation of operating systems, application packages, antivirus software software including SHEQ requirements based on the performance criteria of the Communication and Computer skills.
Conditions/Context of assessment	8. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 9. The practical assessment will be conducted in the workplace or simulated work environment in the training institution. 10. The context of assessment should include the facilities, tools, equipment and materials listed below.
Learning Outcome 02	**COMMUNICATION**
Assessment Criteria	**2.1 INTRODUCTION TO COMMUNICATION** **2.2 LANGUAGE** **2.3 SPOKEN COMMUNICATION** **2.4 MEETINGS** **2.5 BUSINESS LETTERS**

	2.6 REPORTS **2.7 BUSINESS ORGANISATIONS** **2.8 LEGISLATION AFFECTING EMPLOYERS AND EMPLOYEES.** **2.9 BASIC STORES MANAGEMENT**

Content	**2.1 INTRODUCTION TO COMMUNICATION**
	
	• Define and explain the importance of communication
	• Describe the different forms of communication
	• Describe the following types of communication
	    • Inter personal communication
	    • Intra personal communication
	    • Extra personal communication
	• Describe mass communication
	• Explain communication theories
	• Explain the models of communication
	    • Linear model
	    • Interactive model
	    • Transactional model
	• Define the following
	  - transmitter
	  - receiver
	  - decoder
	  -feedback
	• Explain communication breakdown
	• Define communication barrier
	• Describe the common causes to effective communication
	• Describe the different methods of overcoming barriers to communication
	
	2.2 LANGUAGE
	
	• Define linguistic language
	• Identify and describe style and tone
	• Construct sentences and paragraphs
	• Use appropriate business terms
	2.3 SPOKEN COMMUNICATION
	
	• Describe verbal and non-verbal communication
	• Appreciate use of verbal communication in
	    - interviews
	    - appraisal
	    - reward
	    - counselling
	    - grievances
	    -reprimands
	    -dismissal
	    - termination
	• Describe the advantages and disadvantages of verbal communication

	• Describe oral and written communication stating advantages and disadvantages of each • Describe the following - horizontal communication - vertical communication • Describe non-verbal communication stating advantages and disadvantages • Describe the different ways/methods of non-verbal communication **2.4 MEETINGS** • Describe and state the advantages and disadvantages of meetings • Explain and describe the following: - Formal (private/public) meetings. - Committee meetings - Command meetings - Annual general meetings - Extra ordinary meetings - Board meetings - Shareholders meetings • Practice to convene the meetings stated above • Explain and describe the following terms - Notice - Agenda - minute • Write notices, agenda and minutes. • Explain the procedures of meetings. • define the role of the following in a meeting: - Chairperson - Secretary - Treasurer **2.5 BUSINESS LETTERS** • Define a business letter • State and classify business letters. • Describe, write and outline the format of the following: • Inquiry letter -describe the following different types of inquiry letters i. trade inquiry ii. status inquiry iii. general purpose inquiry • Quotation letter • Order letter • Delivery letter • Collection letter - Define a collection letter - Describe features of a collection letter • Memorandum

	• Complaint and adjustment letter.
	• Curriculum vitae
	
	2.6 REPORTS
	
	• Define a report
	• State the purpose for writing a report
	• Describe and write the following different types of reports
	  - Progress/routine reports.
	  - Technical reports
	  - Recommendatory reports.
	  - Accident reports**.**
	• Outline the format of a report
	
	2.7 BUSINESS ORGANISATIONS
	
	• Describe the contribution of the departments to the enterprise as a whole.
	• Describe the role of the division of labour.
	• Appreciate the need for co-operation flow of information, communication and feedback between departments.
	• Describe the following leadership styles and give advantages and disadvantages of each
	  - Autocratic
	  - Democratic
	  - Bureaucratic
	• Appreciate the role of leadership styles, work ethics and human relations to the success of the enterprise.
	2.8 LEGISLATION AFFECTING EMPLOYERS AND EMPLOYEES.
	
	Appreciate and show understanding of the following :
	• Health and Safety Act
	• Factories Act
	• Workman's Compensation Act
	
	2.8 BASIC STORES MANAGEMENT
	
	• Differentiate between equipment, tools, consumables and working capital.
	• Explain the purpose of requisition and ordering procedures.
	• Describe the need for material, handling and storage specifications.
	• Understand the effect of depreciation on equipment, capital equipment and the flow of production.
	• Prepare workshop and log book records.
	Interpretation of electrical circuit symbols and diagrams

Assessment Tasks	2.1 Written and/or oral assessment on the skills and knowledge required for Communication and Computer skills as outlined in the assessment criteria and content above. 2.2 Practical assessment on Communication and Computer skills include

	installation of operating systems, application packages, antivirus software software including SHEQ requirements based on the performance criteria of the Communication and Computer skills.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the workplace or simulated work environment in the training institution. 3. The context of assessment should include the facilities, tools, equipment and materials listed below.

ASSESSMENT SPECIFICATION GRID

TOPIC NO	TOPIC	WEIGHTING %
1	INTRODUCTION TO COMPUTERS	10
2	COMPUTER EQUIPMENT AND COMPUTER SYSTEM ARCHITECTURE	10
3	COMPUTER SECURITY	5
4	COMPUTER SOFTWARE	10
5	APPLICATION PACKAGES SOFTWARE	5
6	COMPUTER NETWORKING	10
7	INTRODUCTION TO COMMUNICATION	10
8	LANGUAGE	5
9	SPOKEN COMMUNICATION	5
10	MEETINGS	5
11	BUSINESS LETTERS	10
12	REPORTS	5
12	BUSINESS ORGANISATIONS	5
14	LEGISLATION AFFECTING EMPLOYERS AND EMPLOYEES.	3
15	BASIC STORES MANAGEMENT	2
TOTAL		100

Approach to Teaching and Learning:

8. Observation of adult learning principles.
9. Both institution-based and work-based learning to facilitate the integration of theory and practice.
10. Face-to-face education and learning.
11. Problem-based learning.
12. Online/distance education and learning.
13. Blended/hybrid education and learning.
14. Use of social media.

Approach to Assessment:

6. Weighting of practical and theory assessment: 70% theory and 30% practical.
7. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50%.
8. Oral assessment to be conducted by a panel of two or more assessors.
9. RPL assessment.
10. Portfolio of evidence.
11. Assessment of work conducted by both individual learners and teams of learners.

Resources:

7. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

8. Facilities, Tools, Equipment and Materials

- Electrical tool box
- Computer unit
- Application packages software
- Windows operating system
- Computer network
- Mutton cloth
- Computer cleaning solutions

9. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

10. Reference Materials (recommended textbooks, recommended readings)

1. Bishop, P.(2010) Computer Science Nelson United Kingdom
2 Bradley, R. (2014) Understanding Computer Science Stanley Thornes United Kingdom
3 Bryant R.E. (2016) Computer Systems: A Programmer's Perspective Macmillan Press
4 Chapman, O. (2011) Data Processing and Information Technology Pearson
5. Giblin, L. (2010) Skill with people Embassy Books
6 Taylor, M. (2017) How To Analyze People Make Profits Easy LLC
7 Kernighan P. (2015) The UNIX Programming Environment Pearson
8 Markel M. (2015) Technical Communication 11th Ed Bufallo
9 Tebtauk E. (2015) Essentials of Technical Communication 3rd Ed Embassy Books

Module Code:	**321/25/M05/1**
Module Title:	**ELECTRICAL ENGINEERING TECHNOLOGY 1**
ZNQF Level:	**4**
Credits:	**15**
Duration:	**150** hours
Relationship with Qualification Standards:	Based on Unit Standard of Qualification Standard for electrical artisans
Pre-requisite modules:	none
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an Instrumentation and Control Mechanic, electrical power artisan, computer services artisan and electronic communication artisan to understand electrical and electronic principles. This includes defining basic electrical quantities, solving DC circuit theory, demonstrating knowledge of electro-magnetic, electric field, DC transients, construction and operation of single phase transformers. The advantages of basic electrical principles includes providing a foundation for the study of the different electrical courses. Access to this module is open to all target groups including the unemployed youths, women and men wishing to establish or improve SMEs in electrical engineering.
List of Learning Outcomes:	**LO1: Units associated with basic electrical quantities** **LO2: Describe electric circuits** **LO3:Batteries and alternative sources of energy** **LO4 Series and parallel networks** **LO5 : Capacitors and capacitance** **LO6: Magnetism** **LO7: Electrical measuring instruments and measurements**
Assessment Criteria:	1.1Demonstrate knowledge of basic electrical units. 1.2 State the electrical quantities 1.3 Summary of terms, units and symbols

Content:	**Units associated with basic electrical quantities** • SI units • current • Charge • Force • Work • Power • Electrical potential and e.m.f. • Resistance and conductance • Electrical power and energy
Assessment Tasks:	10. Written assessment on the skills and knowledge required to understand the units associated with basic electrical.
Conditions/Context of assessment	11. Written assessment can be conducted in a classroom environment.
Learning Outcome 02	**Describe electric circuits**
Assessment Criteria	2.1 Demonstrate knowledge of SI units of charge, force, work and power.
	2.2 Differentiate between resistance and conductance.
	2.3 Differentiate between electrical power and energy.
	An introduction to electric circuits Electrical/electronic system block diagrams Standard symbols for electrical components Electric current and quantity of electricity Potential difference and resistance Basic electrical measuring instruments • Linear and non-linear devices Ohm's law Multiples and sub-multiples

	Conductors and insulators Electrical power and energy • Main effects of electric current Fuses Resistance variation Resistance and resistivity Temperature coefficient of resistance • Resistor colour coding and ohmic values
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand an introduction to electric circuits units.
Conditions/Context of assessment	12. Written assessment can be conducted in a classroom environment.
Learning Outcome 03	**Batteries and alternative sources of energy**
Assessment Criteria:	3.1Demonstrate knowledge of batteries and alternative sources of energy. 3.2 Differentiate between Primary and Secondary cells. 3.3 Demonstrate some knowledge associated with some terms used for batteries e.g, corrosion, cell capacity.
Content	**LO3: Batteries and alternative sources of energy** • Introduction to batteries • Some chemical effects of electricity • The simple cell • Corrosion • E.m.f. and internal resistance of a cell • Primary cells • Secondary cells • Cell capacity • Safe disposal of batteries • Fuel cells • Alternative and renewable energy sources

Assessment Tasks	1. Written and/or oral assessment on the skills and knowledge required on batteries and alternative sources of energy using content above.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment.
Learning Outcome 04	**Series and Parallel networks**
Assessment Criteria	4.1 Demonstrate knowledge of series circuits.
	4.2 Demonstrate knowledge of parallel circuits.
	4.3 Differentiate between relative and absolute voltages.
	4.4 Practically wire lamps in series and in parallel.
Content	LO4: Series and parallel networks • Series circuits • Potential divider • Parallel networks • Current division • Relative and absolute voltages • Wiring lamps in series and in parallel
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand series and parallel circuits.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 05	**Capacitors and capacitance**
Assessment Criteria:	5.1Demonstrate knowledge of electrostatics 5.2 Calculate capacitance 5.3 Solve capacitor networks

Content:	LO5: Capacitors and capacitance • Introduction to capacitors • Electrostatic field • Electric field strength • Capacitance • Capacitors • Electric flux density • Permittivity • The parallel plate capacitor • Capacitors connected in parallel and series • Dielectric strength • Energy stored in capacitors • Practical types of capacitor • Discharging capacitors
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand the fundamentals concepts of capacitors and capacitance.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 06	**Magnetism**
Assessment Criteria:	6.1Demonstrate knowledge of electromagnetism 6.2 Demonstrate knowledge of electromagnetic induction
Content:	6.1 Electromagnetism • Magnetic field due to an electric current • Electromagnets • Force on a current-carrying conductor • Principle of operation of a simple d.c. motor

	• Principle of operation of a moving-coil instrument • Force on a charge 6.2 Electromagnetic induction • Introduction to electromagnetic induction • Laws of electromagnetic induction • Rotation of a loop in a magnetic field • Inductance • Inductors • Energy stored • Inductance of a coil • Mutual inductance
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand magnetism as outlined in the assessment criteria and content above.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 07	**Electrical measuring instruments and measurements**
Assessment Criteria:	7.1 describe construction, principle of operation and application of measuring instruments 7.2 State advantages and disadvantages of moving coil and moving iron instruments
Content:	**LO7: Electrical measuring instruments and measurements** • Introduction • Analogue instruments • Moving-iron instrument • The moving-coil rectifier instrument • Comparison of moving-coil, moving-iron and moving-coil rectifier instruments

Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand the basic fundamentals of electrical measuring instruments and measurements as outlined in the assessment criteria and content above.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.

ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOPIC	WEIGHTING %
1	UNITS ASSOCIATED WITH BASIC ELECTRICAL QUANTITIES	10
2	AN INTRODUCTION TO ELECTRIC CIRCUITS	10
3	BATTERIES AND ALTERNATIVE SOURCES OF ENERGY	10
4	SERIES AND PARALLEL NETWORKS	20
5	CAPACITORS AND CAPACITANCE	20
6	MAGNETISM	20
7	ELECTRICAL MEASURING INSTRUMENTS AND MEASUREMENTS	10
	TOTAL	**100**

Approach to Teaching and Learning:

15. Observation of adult learning principles.
16. Both institution-based and work-based learning to facilitate the integration of theory and practice.
17. Face-to-face education and learning.
18. Problem-based learning.
19. Online/distance education and learning.
20. Blended/hybrid education and learning.
21. Use of social media.

Approach to Assessment:

12. Weighting of practical and theory assessment: 70% theory and 30% practical.

13. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50%.
14. Oral assessment to be conducted by a panel of two or more assessors.
15. RPL assessment.
16. Portfolio of evidence.
17. Assessment of work conducted by both individual learners and teams of learners.

Resources:

11. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

12. Facilities, Tools, Equipment and Materials

- Electrical tool box
- Digital multi meter
- Ladder
- Laptop
- Soldering kit
- PPE
- Drilling machine
- Insulation tape
- Mutton cloth
- Detergents
- Cables
- Cable enclosures
- Electrical accessories

13. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

14. Reference Materials (recommended textbooks, recommended readings)

J.O Bird (1997) Electrical circuit theory and technology Butterworth, Heinnman, Oxford
Boylestad Introduction to circuit analysis
S.A Knight (1994) Electrical and Electronic Principles 2
Ashfaq Hussan Electrical Engineering principles Butterworth, Heinnman, Oxford

Module Code:	**321/25/M05/2**
Module Title:	**ELECTRICAL ENGINEERING TECHNOLOGY 2**
ZNQF Level:	**4**
Credits:	**15**
Duration:	**150** hours
Relationship with Qualification Standards:	Based on Unit Standard of Qualification Standard for electrical artisans
Pre-requisite modules:	none
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an Instrumentation and Control Mechanic, electrical power artisan, computer services artisan and electronic communication artisan to understand electrical and electronic principles. This includes an understanding of D.C. theory, alternating voltages and currents concepts, single-phase series and parallel AC circuits, D.C. transients, operational amplifiers and digital electronics fundamentals. The advantages of basic electrical principles includes providing a foundation for the study of the different electrical courses. Access to this module is open to all target groups including the unemployed youths, women and men wishing to establish or improve SMEs in electrical engineering.
List of Learning Outcomes:	**LO1: D.C. circuit theory** **LO2: Alternating voltages and currents** **LO3:Single-phase series a.c. circuits** **LO4 Single-phase parallel a.c. circuits** **LO5 : D.C. transients**
Learning Outcome 01	**Describe D.C. circuit theory**
Assessment Criteria:	1.1Demonstrate knowledge of D.C. circuit theory. 1.2 State the different theorems 1.3 Perform simple calculations using the theorems.

<table>
<tr><td>Content:</td><td>LO1 D.C. circuit theory
Introduction
Kirchhoff's laws
The superposition theorem
General d.c. circuit theory
Thévenin's theorem
Constant-current source
Norton's theorem
Thévenin and Norton
equivalent networks
Maximum power transfer theorem</td></tr>
<tr><td>Assessment Tasks:</td><td>11. Written assessment on the skills and knowledge required to understand D.C. circuit theory.</td></tr>
<tr><td>Conditions/Context of assessment</td><td>13. Written assessment can be conducted in a classroom environment.</td></tr>
<tr><td>Learning Outcome 02</td><td>Outline alternating voltages and currents</td></tr>
<tr><td rowspan="4">Assessment Criteria</td><td>2.1 Demonstrate knowledge of Alternating Voltages and Currents.</td></tr>
<tr><td>2.2 Demonstrate an understanding of the equation of a sinusoidal.</td></tr>
<tr><td>2.3 Draw waveforms of rectification and smoothed rectification.</td></tr>
<tr><td>LO2: Alternating voltages and currents
Introduction
The a.c. generator
Waveforms
A.c. values
The equation of a sinusoidal</td></tr>
</table>

	• waveform Combination of waveforms • Rectification • Smoothing of the rectified • output waveform
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand Alternating voltages and currents.
Conditions/Context of assessment	14. Written assessment can be conducted in a classroom environment.
Learning Outcome 03	**Perform calculations on single-phase series a.c. circuits**
Assessment Criteria:	3.1Demonstrate knowledge of single phase a.c. circuits. 3.2 Demonstrate knowledge of pure resistive, inductive and capacitive a.c. circuits. 3.3 Demonstrate some knowledge of series resonance and Q-factor.
Content	**LO3: Single-phase series a.c. circuits** • Purely resistive a.c. circuit • Purely inductive a.c. circuit • Purely capacitive a.c. circuit • R–L series a.c. circuit • R–C series a.c. circuit • R–L–C series a.c. circuit • Series resonance • Q-factor
Assessment Tasks	1. Written and/or oral assessment on the skills and knowledge required on single phase a.c. series circuits using content above.
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment.
Learning Outcome 04	**Perform calculations on single-phase parallel a.c. circuits**

Assessment Criteria	4.1 Demonstrate knowledge of R-L parallel a.c. circuits. 4.2 Demonstrate knowledge of R-C parallel a.c. circuits. 4.3 Demonstrate knowledge of L-R-C parallel a.c. circuits. 4.4 Demonstrate knowledge of parallel resonance and Q-factor.
Content	Perform calculations on single-phase parallel a.c. circuits • R–L parallel a.c. circuit • R–C parallel a.c. circuit • L–C parallel circuit • LR–C parallel a.c. circuit • Parallel resonance and Q-factor
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand single-phase parallel a.c, circuits.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 05	**Perform calculations on D.C. transients**
Assessment Criteria:	5.1Demonstrate knowledge of D.C. transients. 5.2 Draw transient curves. 5.3 Demonstrate knowledge of time constants.
Content:	LO5: D.C. transients • Introduction • Charging a capacitor • Time constant for a C–R circuit • Transient curves for a C–R circuit • Discharging a capacitor • Current growth in an L–R circuit • Time constant for an L–R circuit

	• Transient curves for an L–R circuit • Current decay in an L–R circuit • Switching inductive circuits •
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand D.C. transients.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.

ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOPIC	WEIGHTING %
1	D.C. CIRCUIT THEORY	20
2	ALTERNATING VOLTAGES AND CURRENTS	20
3	SINGLE-PHASE SERIES A.C. CIRCUITS	20
4	SINGLE-PHASE PARALLEL A.C. CIRCUITS	20
5	D.C. TRANSIENTS	20
	TOTAL	**100**

Approach to Teaching and Learning:

22. Observation of adult learning principles.
23. Both institution-based and work-based learning to facilitate the integration of theory and practice.
24. Face-to-face education and learning.
25. Problem-based learning.
26. Online/distance education and learning.
27. Blended/hybrid education and learning.
28. Use of social media.

Approach to Assessment:

18. Weighting of practical and theory assessment: 70% theory and 30% practical.

19. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50%.
20. Oral assessment to be conducted by a panel of two or more assessors.
21. RPL assessment.
22. Portfolio of evidence.
23. Assessment of work conducted by both individual learners and teams of learners.

Resources:

15. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

16. Facilities, Tools, Equipment and Materials

- Electrical tool box
- Digital multi meter
- Ladder
- Laptop
- Soldering kit
- PPE
- Drilling machine
- Insulation tape
- Mutton cloth
- Detergents
- Cables
- Cable enclosures
- Electrical accessories

17. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

18. Reference Materials (recommended textbooks, recommended readings)

J.O Bird (1997) Electrical circuit theory and technology Butterworth,Heinnman,Oxford
Boylestad Introduction to circuit analysis
S.A Knight (1994) Electrical and Electronic Principles 2
Ashfaq Hussan Electrical Engineering principles Butterworth, Heinnman, Oxford

Module Code:	**321/25/M06/1**
Module Title:	**ENGINEERING MATHEMATICS 1**
ZNQF Level:	**4**
Credits:	**10**
Duration:	**100 hours**
Relationship with Qualification Standards:	Based on Unit Standard of Qualification Standard for electrical artisans
Pre-requisite modules:	none
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an Instrumentation and Control Mechanic, electrical power artisan, computer services artisan and electronic communication artisan to understand engineering mathematical principles. This includes to identify, formulate, and solve typical electrical mathematical engineering problems. Access to this module is open to all target groups including the unemployed youths, women and men wishing to establish or improve SMEs in electrical engineering.
List of Learning Outcomes:	L01 Perform basic arithmetic, algebra calculations and illustrate the relationship of algebraic equations and associated graphs. L02 Define, evaluate and solve indices, logarithms , surds and exponential functions L03 Define trigonometric functions, describe their properties and use them in calculating some basic electrical engineering relationships. L04 Define position and rotating vectors and illustrate their applications in electrical engineering. L05 Define the base of a number and perform basic calculations in various number systems.
Learning Outcome 01	**PERFORM BASIC ARITHMETIC AND ALGEBRA CALCULATIONS**
Assessment Criteria:	1.1 Demonstrate the knowledge of sketching and plotting of graph 1.2 Demonstrate the knowledge of formulating and solving linear

	equations 1.3 Demonstrate the knowledge of solving linear simultaneous and quadratic equations by different methods 1.4 With the aid of graphs differentiate between direct and inversely proportionality
Content:	L01.1 Plot a graph of a linear function and determine the gradient and intercept. L01.2 Determine the equation of a straight line given: - Two points on a graph. - Gradient and one point on a line. L01.3 Solve algebraically simple equations. L01.4. Solve algebraically linear simultaneous equations up to two unknowns by: - Elimination - Substitution - Determinant - Matrix L01.5 Solve simple quadratic equations in one unknown by: - Factorization - Quadratic formula - Completing the square L01.6 State the difference between direct and inverse proportionality and solve related problems including joint and partial variation. L01.7 Sketch graphs to illustrate direct and inverse proportionality.
Assessment Tasks:	12. Written assessment on the skills and knowledge required to understand the basic arithmetic and algebra calculations as outlined in the assessment criteria and content above.
Conditions/Context of assessment	15. Written assessment can be conducted in a classroom environment.

Learning Outcome 02	**DEFINE, EVALUATE AND SOLVE INDICES, LOGARITHMS , SURDS AND EXPONENTIAL FUNCTIONS**
Assessment Criteria	2.1 Evaluate and simplify expressions involving real number indices
	2.2 Define surds and rationalize denominators with surds.
	2.3 express numbers in standard form
	2.4 Define ,Evaluate and simplify expressions involving logarithms
	2.5 Define exponential functions and sketch graphs of exponential functions
	2.6 determine experimental laws by the use of straight line graphs
Content	L02.1 Define indices and state the laws. L02.2 Evaluate and simplify expressions involving real number indices. L02.3 Define surds and rationalize denominators with surds. L02.4 Express numbers in standard form. L02.5 Define logarithms and state the laws of logarithms. L02.6 Evaluate and simplify expressions involving logarithms. L02.7 Define exponential functions and sketch graphs of exponential functions. L02.8 Define the natural number and establish it as a base for natural logarithms. L02.9 Sketch the graphs of e^{x} and e^{-x}. L02.10 Solve logarithmic and exponential equations. L02.11 Determine experimental laws by application of straight line graphs in the form: $y = ka^{x} + b$
Assessment Tasks	1. Written and/or oral assessment on the skills and knowledge required to define, solve and evaluate indices, exponential functions, logarithms and surds as outlined in the assessment criteria and content above.

Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment.
Learning Outcome 03	**DEFINE TRIGONOMETRIC FUNCTIONS, DESCRIBE THEIR PROPERTIES AND USE THEM IN CALCULATING SOME BASIC ELECTRICAL ENGINEERING RELATIONSHIPS.**
Assessment Criteria	3.1 define and sketch graphs of trigonometry ratios
	3.2 convert radians to degrees and vice versa
	3.3 determine amplitude, frequency, period , angular frequency and the phase angle of a sine wave equation
	3.4 state and apply the compound and double angle formulae
	3.5 apply trigonometry to ac theory
Content	L03.1 Define the six trigonometrical ratios of an angle. L03.2 Sketch graphs of sine, cosine and tangent. L03.3 Convert radians to degrees and vice versa. L03.4 Convert θ-axis to time-axis in waveforms leading to $Y = Asin\ (wt \pm \propto)$ L03.5 Determine frequency and periodic time of waveforms and establish the relationship between angular frequency (ω), frequency (f) and period (T) of a function. L03.6 Derive and apply the fundamental trigonometrical identities: - $sin^2x + cos^2x = 1$ - $1 + tan^2x = sec^2x$ - $1 + cot^2x = cosec^2x$ L03.7 State and apply: - Compound angle formulae. - Double angle formulae. L03.8 Basic application of trigonometry to A. C. theory
Assessment Tasks	1. Written and/or oral assessment on the skills and knowledge required to calculate engineering relationships and apply trigonometric functions and their properties as outlined in the assessment criteria and content above.

Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment.
Learning Outcome 04	**DEFINE POSITION AND ROTATING VECTORS , AND ILLUSTRATE THEIR APPLICATIONS IN ELECTRICAL ENGINEERING**
Assessment Criteria:	4.1 distinguish between scalar and vector quantities 4. 2 determine scalar product of two vectors 4.3 add and subtract vectors 4.4 Determine the resultant of a number of phasors
Content:	L07.1 Distinguish among free vector, position vector, unit vector and component vector. L07.2 Distinguish between scalar and vector quantities giving examples. L07.4 Determine the scalar product of two vectors. L07.3 Add and subtract vectors by: - Triangular rule - Parallelogram rule - Polygon rule L07.5 Construct a phasor from a sine wave. L07.6 Determine the resultant of a number of phasors
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand vectors as outlined in the assessment criteria and content above.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 05	**DEFINE THE BASE OF A NUMBER AND PERFORM BASIC CALCULATIONS IN VARIOUS NUMBER SYSTEMS.**

Assessment Criteria:	5.1 Define binary, octal, decimal and hexadecimal number systems 5.2 convert form base to the other
Content:	L05.1 Define binary, octal, decimal and hexadecimal number systems L05.2 Convert from binary to decimal and vice versa L05.3 Convert from octal to decimal and vice versa L05.4 Convert from hexadecimal to decimal and vice versa L05.5Convert from binary to decimal and vice versa via octal L05.6 Convert from binary to decimal and vice versa via hexadecimal
Assessment Tasks:	13. Written assessment on the skills and knowledge required to understand number system as outlined in the assessment criteria and content above.
Conditions/Context of assessment	16. Written assessment can be conducted in a classroom environment.

Approach to Teaching and Learning:

29. Observation of adult learning principles.
30. Both institution-based and work-based learning to facilitate the integration of theory and practice.
31. Face-to-face education and learning.
32. Problem-based learning.
33. Online/distance education and learning.
34. Blended/hybrid education and learning.
35. Use of social media.

Approach to Assessment:

24. Weighting of practical and theory assessment: 70% theory and 30% practical.
25. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50%.
26. Oral assessment to be conducted by a panel of two or more assessors.
27. RPL assessment.
28. Portfolio of evidence.
29. Assessment of work conducted by both individual learners and teams of learners.

Resources:

19. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

20. Facilities, Tools, Equipment and Materials

Scientific Calculator.

Mathematical Set
Ruler
Sharp pencil

3.0 ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOTAL	WEIGHTING%
1	ALGEBRA	25
2	INDICES AND LOGARITHMS	25
3	TRIGONOMETRY AND CIRCULAR MEASURE	30
4	VECTORS	10
5	NUMBER SYSTEMS	10
	TOTAL	**100%**

4 Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

5 Reference Materials (recommended textbooks, recommended readings)

J.O Bird Basic engineering mathematics 4th to 7th edition Ed Elsevier
Stroud, K.A. and Booth, D.J. (2013) *Engineering Mathematics* Prentice Hall

Stroud, K.A. (2017) *Advanced Engineering Mathematics* McMillan UK

Davison R. 2017) *Engineering Mathematics* Pearson Education Limited

Module Code:	**321/25/M06/2**
Module Title:	**ENGINEERING MATHEMATICS 2**
ZNQF Level:	**4**
Credits:	**10**
Duration:	**100** hours
Relationship with Qualification Standards:	Based on Unit Standard of Qualification Standard for electrical artisans
Pre-requisite modules:	Engineering Mathematics 1
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an Instrumentation and Control Mechanic, electrical power artisan, computer services artisan and electronic communication artisan to understand engineering mathematical principles. This includes to identify, formulate, and solve typical electrical mathematical engineering problems. Access to this module is open to all target groups including the unemployed youths, women and men wishing to establish or improve SMEs in electrical engineering.
List of Learning Outcomes:	L01 Define a function and its derivative and use the derivatives to find extremes and rates of change, applying the techniques to problems in electrical engineering. L02 Define definite and indefinite integrals and perform basic integration with applications, determination of areas under curves. L03 apply complex number arithmetic to periods phenomenon L04 explain motion in terms of time, displacement, velocity and acceleration in a straight line as well as a circle and perform basic relevant calculations. L05 define angular velocity and constant angular acceleration, relate angular to linear motion. Apply equations of angular motion
Learning Outcome 01	**DEFINE A FUNCTION AND ITS DERIVATIVE AND USE THE**

	DERIVATIVE AND RATES OF CHANGE, APPLYING TECHNIQUES TO PROBLEMS IN ELECTRICAL ENGINEERING
Assessment Criteria:	1.1 demonstrate the knowledge of differentiation from first principles 1.2 demonstrate the knowledge of differentiation by the general rule 1.3 differentiate a function of a function, a product and a quotient 1.4 apply differential calculus to rates of change of electrical quantities
Content:	L01.1 Define $\frac{lim}{\Delta x \to 0} \frac{\Delta y}{\Delta x}$ and show that it is the gradient function of $y = f(x)$. - Apply the gradient function to obtain gradients of functions at particular points of the function. L01.2 Differentiate from first principles functions with powers of x up to x^3, sin x, cos x , $1n\ x$ and e^x. L01.3 Differentiate by the general rule, a function of the form $y = ax^n + bx^{n-1} + cx^{n-2} +$ -------------- L01.4 Differentiate: - Function of a function - Product Quotient L01.5 Apply differentiation in rates of change of quantities in the electrical engineering trade.
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand calculus as outlined in the assessment criteria and content above.
Conditions/Context of assessment	17. Written assessment can be conducted in a classroom environment.
Learning Outcome 02	**DEFINE DEFINITE AND INDEFINITE INTEGRALS AND PERFORM BASIC INTEGRATION WITH APPLICATIONS,**

	DETERMINATION OF AREAS UNDER THE CURVE.
Assessment Criteria:	2.1 demonstrate the knowledge of integrating by the general rule 2.2 integrate by substitution 2.3 determine the area under a curve by integration 2.4 determine mean and root mean square values of a sine wave using integration
Content:	**LO2.1** State indefinite integrals of functions such as $ax^n, \frac{1}{x}, \sin nx, \cos x$ *and* e^x. L02.2 Integrate by substitution functions such as: - $\frac{1}{(1+x)^n}$ where n= 1 or n= 2 - $sin(ax)$ - $sinxcosx$ - sin^2xcosx -sin^2x - e^{ax} - $lnax$ L02.3 Determine the area under a curve. L02.4 Determine the mean and *root mean square* value of a sine wave.
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand calculus as outlined in the assessment criteria and content above.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 03	**APPLY COMPLEX NUMBER ARITHMETIC TO PERIODS PHENOMENON**
Assessment Criteria:	3.1 evaluate the powers of the imaginary number up to j^5 3.2 Add, subtract, multiply and divide complex numbers (include powers and roots). 3.3 Represent complex numbers on an Argand diagram. 3.4 interconvert between Cartesian and polar forms 3.5 solve complex number equations

Content:	L03.1 Define a complex number. L03.2 Evaluate the powers of the imaging number up to j^5 L03.3 Add, subtract, multiply and divide complex numbers in Cartesian form (include powers and roots). L03.4 Represent complex numbers on an Argand diagram. L03.5 Interconvert between Cartesian and polar form. L03.6 Divide, multiply, complex numbers in polar form. L03.7 solve complex numbers equations
Assessment Tasks:	14. Written assessment on the skills and knowledge required to understand complex numbers as outlined in the assessment criteria and content above.
Conditions/Context of assessment	18. Written assessment can be conducted in a classroom environment.
Learning Outcome 04	**EXPLAIN MOTION IN TERMS OF TIME, DISPLACEMENT, VELOCITY AND ACCELERATION IN A STRAIGHT LINE. PERFORM BASIC RELEVANT CALCULATIONS**
Assessment Criteria:	4.1 demonstrate the knowledge of velocity/time graphs for solving problems associated with linear motion 4.2 apply linear motion equations to solve linear motion problems
Content:	L04.1 Define: -Displacement -Velocity -Acceleration L04.2 Construct the velocity/time graphs and apply them to solve problems associated with linear motion. L04.3 Apply the following equations in solving problems associated with linear motion:

	$v = u + at$ $s = ut + \frac{1}{2}\, at^2$ $v^2 - u^2 = 2as$ L04.4 Adapt equations in LO4.3 for vertical motion under gravity.
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand displacement, velocity and acceleration as outlined in the assessment criteria and content above.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.
Learning Outcome 05	**DEFINE ANGULAR VELOCITY AND ANGULAR ACCELERATION, RELATE ANGULAR AND LINEAR MOTION. APPLY EQUATIONS OF ANGULAR MOTION**
Assessment Criteria:	5.1 State and apply Newton's 3 laws of motion 5.2 convert angular velocity to speed in revs per minute 5.3 relate angular motion to linear motion 5.4 apply angular motion equations to angular motion problems
Content:	L05.1 State and apply Newton's three Laws of motion L05.2 Define angular velocity and constant angular acceleration. L05.3 Convert angular velocity to speed in revs per minute. L05.4 Connect frequency and period with angular velocity. L05.5 Relate angular motion to linear motion. L05.6 Apply the following equations of angular motion: $ù = ù_0 + át$ $\quad$ $è = w_0 t + \frac{1}{2}\, át^2$ $ù^2 - ù_0 = \frac{1}{2} áè$
Assessment Tasks:	1. Written assessment on the skills and knowledge required to understand circular motion as outlined in the assessment criteria and content above.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment.

Approach to Teaching and Learning:

36. Observation of adult learning principles.
37. Both institution-based and work-based learning to facilitate the integration of theory and practice.
38. Face-to-face education and learning.
39. Problem-based learning.
40. Online/distance education and learning.
41. Blended/hybrid education and learning.
42. Use of social media.

Approach to Assessment:

30. Weighting of practical and theory assessment: 70% theory and 30% practical.
31. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50%.
32. Oral assessment to be conducted by a panel of two or more assessors.
33. RPL assessment.
34. Portfolio of evidence.
35. Assessment of work conducted by both individual learners and teams of learners.

Resources:

1 Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

2 Facilities, Tools, Equipment and Materials

Scientific Calculator.

Mathematical Set

Ruler

Sharp pencil

3 **ASSESSMENT SPECIFICATION GRID**

TOPIC NO.	TOTAL	WEIGHTING%
1	DIFFERENTIAL CALCULUS	20
2	INTEGRAL CALCULUS	20
3	COMPLEX NUMBERS	20
4	LINEAR MOTION	20
5	ANGULAR MOTION	20
	TOTAL	**100%**

4 Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

5 Reference Materials (recommended textbooks, recommended readings)

J.O Bird Basic engineering mathematics 4th to 7th edition Ed Elsevier
Stroud, K.A. and Booth, D.J. (2013) *Engineering Mathematics* Prentice Hall

Stroud, K.A. (2017) Advanced *Engineering Mathematics* McMillan UK

Davison R. (2017) *Engineering Mathematics* Pearson Education Limited

Module Code:	**321/25/M08**
Module Title:	**PROGRAMMABLE LOGIC CONTROLLERS**
ZNQF Level:	**4**
Credits:	**15**
Duration:	**150**hours
Relationship with Qualification Standards:	**Programmable Logic Controllers** Qualification Standard for Instrumentation and Control Artisan
Purpose of Module:	This module describes the skills, knowledge and attributes required by an Instrumentation and Control Systems artisan to configure, program and wire Programmable Logic Controllers. Access to this module is open to all target groups including unemployed/employed youths, women and men wishing to establish or improve their knowledge and skills in Automation.
List of Learning Outcomes:	**LO1: Checking equipment operation** **LO2: Program writing, testing and simulating** **LO3: Devices configuration and program uploading** **LO4: PLC Maintenance** **LO5: Documentation**
Learning Outcome 01	**Checking equipment operation**
Assessment Criteria:	1.1 Carryout equipment survey 1.2 Present equipment wiring diagram
Content:	• Carryout equipment survey. ✓ Identify different cables ✓ Identify different cable connectors ✓ Identify different inputs and outputs ✓ Identify different communication cables and protocols ✓ Identify the right power supplies • Produce equipment wiring diagram

	✓ Identify different device/component symbols used with INSTRUMENTATION as per given schematic diagram ✓ Demonstrate how to read different wiring diagrams ✓ Produce wiring diagrams ✓ Wire the devices
Assessment Tasks:	15. Show knowledge of identifying equipment 16. Show knowledge of how to read wiring diagrams
Learning Outcome 02	**Program writing, testing and simulating**
Assessment Criteria	16.2 knowledge of programming languages for use is demonstrated 16.3 Correct procedure for writing a program is followed 16.4 Program simulation 16.5 Program debugging 16.6 Program test run is carried out 16.7 Backups of program created
Content	• Knowledge of programming languages ✓ List the programming languages used with PLCs • Program writing • Demonstrate 1.inputs addressing 2.outputs addressing 3.system memory addressing 4.timers addressing 5.counters addressing • Demonstrate knowledge of Latch and Unlatch • Demonstrate knowledge of the RESET function • Correct sequence of building a program is followed • Knowledge of converting from one language to another is displayed • Knowledge of using Logic gates is displayed • Program simulation, debug and run ✓ Correct procedure of simulating a program is followed ✓ Correct procedure of debugging is followed ✓ Online mode activated ✓ Test run mode activated • Backups of program created ✓ A backup of the program is created/saved in a flash drive
Assessment Tasks	1. Show the knowledge of programming languages 2. Write a program from a given situation 3. Simulate the program

	4. Show debugging skills 5. Show knowledge of backing up programs
Learning Outcome 03	**Devices configuration and program uploading**
Assessment Criteria	1.1 Configure devices 1.2 Upload program
Content	• Configure devices ✓ Equipment is set ✓ The programmer connected to the PLC ✓ Devices made/set to communicate with each other • Upload program ✓ Program uploaded into the PLC ✓ The plc set to run mode
Assessment Tasks	1. Setting the equipment 2. Making the programmer and the PLC communicate with each other 3. Uploading the program into the PLC 4. Run the whole setup
Learning Outcome 04	**PLC Maintenance**
Assessment Criteria	4.1 Program backing up 4.2 PLC protection 4.3 Checking connections
Content	• Program backup - Demonstrate how to use different types of backup storage devices - Create a backup of the uploaded program • PLC protection - Check-up environmental factors, eg humidity, temperature - Identify corrosive and conductive contaminants - Demonstrate how to clear debris and dust • Checking connections - Check loose connections - Inspect I\O devices for proper adjustments
Assessment Tasks	1. Show knowledge of creating a backup 2. Show how to clear debris and dust 3. Show skills of checking loose connections
Learning Outcome 05	**Documentation**
Assessment Criteria	4.1 Create documents
Content	• Create documents ✓ Produce documents of all done work ✓ Work proposal is drafted Work proposal is presented

Assessment Tasks	1 Show knowledge of preparing necessary documents

ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOTAL	WEIGHTING%
1	**CHECKING EQUIPMENT OPERATION**	20
2	**PROGRAM WRITING, TESTING AND SIMULATING**	20
3	**DEVICES CONFIGURATION AND PROGRAM UPLOADING**	20
4	**PLC MAINTENANCE**	20
5	**DOCUMENTATION**	20
	TOTAL	**100%**

Resources:

21. Facilities, Tools, Equipment and Materials

- Electrical tool box
- PPE
- Insulation tape
- Cables/wires
- Input devices
- Output devices
- PLCs
- Programmer
- Communication cable
- Power supply
- Connectors

Approach to Teaching and Learning:

43. Observation of adult learning principles.
44. Both institution-based and work-based learning to facilitate the integration of theory and practice.
45. Face-to-face education and learning.
46. Problem-based learning.
47. Online/distance education and learning.
48. Blended/hybrid education and learning.
49. Use of social media.

Approach to Assessment:

36. Weighting of practical and theory assessment: 70% practical and 30% theory.

37. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50% work-based assessment.
38. Oral assessment to be conducted by a panel of two or more assessors.
39. RPL assessment.*
40. Portfolio of evidence.
41. Assessment of work conducted by both individual learners and team of learners.

Resources:

22. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualifications and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

The lecturer should have a minimum of National Diploma in Electrical Power Engineering

23. Facilities, Tools, Equipment and Materials

- Electrical tool box
- Ladder
- PPE
- Drilling machine
- Welding machine
- Megger
- Insulation tape
- Mutton cloth
- Detergents
- PPC
- Cables
- Cable enclosures
- Electrical accessories

24. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

25. Reference Materials (recommended textbooks, recommended readings)

14. Petruzella F.D. (2017) *Programmable Logic Controllers* McGraw Hill Publishing Rockis G. (2014)
15. Electrical Motor Controls for Integrated Systems American Technical Alexander C.K. (2017)
16. Fundamentals of Electrical Circuits McGraw Hill Publishing Tyler, D. W. (2012)
17. Electrical & Electronic Applications 2 Butterworth & Co. Ltd. Lewis, M. L. (2014)
18. Electrical Installation Technology Hutchinson & Co. Ltd. London W. Bolton (2015)
19. Programmable Logic Controllers, 4th Edition McGraw Hill Publishing Lewis M.L. (2016)
20. Electrical Installation Technology Hutchinson & Co. Ltd. Mark H. (1995)

21. Solar Electric System for Africa Common Wealth Science Council Bimbhra P.S. (2008)
22. Electrical Machinery Khanna IEE Regulations 15 or 16th Edition Z Wiring Regulations and code of Practice (latest edition)

Module Code:	**321/25/M09**
Module Title:	**OVERHEAD AND UNDERGROUND LINE CONSTUCTION.**
ZNQF Level:	**4**
Credits:	**15**
Duration:	**150** hours
Relationship with Qualification Standards:	Based on Unit Standard of Qualification Standard for electrical artisans
Pre-requisite modules:	**SAFETY, HEALTH, ENVIRONMENTAL AND QUALITY MANAGEMENT** **POWER GENERATION, TRANSMISSION , DISTRIBUTION**
Purpose of Module:	This unit will enable an individual to design, source for resources an install, and carry out maintenance on overhead and underground distribution network that meets the IEE, SAZ, and CAS, wiring Regulations and Electricity Supply Regulations.
List of Learning Outcomes:	Upon successful completion of this program the apprentice should be able to perform the following outcomes. ***1.*** **Observe safety regulations in the Trade.** 2. **Outline line construction theory** 3. **Construct Overhead lines.** 4. **Lay Underground lines**
Learning Outcome 01	**Observe safety regulations in the Trade**
Assessment Criteria:	1.1 Use industry standard practices for climbing, lifting, rigging and hoisting in this trade. 1.2 Describe personal protective grounding.
Content:	1. Describe manual lifting procedures. 2. Describe types rigging hardware *(wire ropes, jacks, bolts, and turnbuckles etc).* 3. Select equipment for rigging loads. 4. Describe hoisting and load moving procedures. 5. Identify PPE for climbing, lifting and load moving equipment.

	6. Explain the need of personal protective grounding/bonding. 7. Identify the electrical and mechanical requirements of a personal protective ground. 9. Outline the procedure for installing and removing personal protective grounds.
Assessment Tasks:	Written/oral assessment on the knowledge required to understand safety regulations associated with power line and underground line construction
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment. 2. Oral assessment can be conducted in the workshop or work environment.
Learning Outcome 02	**Line construction theory.**
Assessment Criteria:	4.1 Explain the requirements and methods of sagging for overhead conductors 4.2 Explain the requirements for the identification, mapping and switching of new underground installations. 4.3 Explain insulation testing.
Content:	1. Explain the effects on a line if the sagging is either too tight or too loose *(conductor Arrangement, Ground clearance, wire crossing clearance, sag table*). 2. Choose the appropriate sag chart, given the necessary line and conductor information (initial or final). 3. Explain in detail the line and sight method of sagging. 4. Explain methods of determining sag on an existing energized line. 5. Explain the effect on sag when changing span length. 6. Explain ruling span and actual (sagging) span. 7. Prepare a switching program to energize a new section of an underground system given a single line diagram. 8. State the reasons for insulation testing.

	9. Explain the term high potential testing and the hazards involved.
Assessment Tasks:	Written assessment on the knowledge required for line construction theory.
Conditions/Context of assessment	Written assessment can be conducted in a classroom environment.
Learning Outcome 03	**Construct Overhead line.**
Assessment Criteria	2.1 Explain power line structures and their handling. 2.2 Explain pole installation and climbing. 2.3 Explain the characteristics of insulators used in power line construction. 2.4 Explain the application of anchors and guys used in power line construction 2.5 Describe the type of conductors used in power line construction. 2.6 Explain methods of stringing and recovering power lines. 2.7 Explain how to splice, connect and terminate overhead conductors. 2.8 Explain rigging and hoisting practices.
Content	Overhead Support structures 1. Describe the types of power line structures (PCC, RCC, Wooden, Rail,) 2. Describe the treatments used for power line poles. 3. Define the differences between classes of wood poles. 4. Identify the information found on pole stamps. 5. Describe basic framing and attachments to poles (*attachment of cross arms*). 6. Describe the procedures for erecting poles.

	7. Describe the forces exerted on power line structures. 8. Describe methods of setting poles. 9. Describe the factors that affect how poles are faced in various situations. 10. Describe problems caused when poles are improperly backfilled and tamped. 11. Explain the hazards involved in climbing poles. Overhead Line Insulators 12. List the common types of insulator materials used on power systems (*i.e pin, disc, suspension strain etc*). 13. Describe the different type of insulators and their applications. 14. Describe the following terms associated with insulators - B.I.L. (basic impulse level) rating. - Flashover and leakage current. - Dielectric strength of insulating materials. 15. Identify typical causes of insulator failure. **Anchors and Guys** 16. Describe the types and holding capacities of power line anchors. 17. Describe the proper installation of typical anchor types. 18. Describe the proper placement of anchors and state reasons for anchor failure (us*e of Stay Rod, brackets*). Overhead line Conductors 19. Describe the advantages and disadvantages of various types of material used for line conductors 20. Recognise conductor sizes. 21. Explain the relationship between the size and ampacity of conductors. 22. Describe common aerial conductor types. 23. Describe common methods of handling and storage of reels of conductor. 24. Describe common methods of stringing and recovering power line conductors.

	25. Describe the fundamental relationship between tension and sag. 26. Describe the methods used for conductor tie-in. 271. Describe the use of manual and power driven presses. 28. Describe how to check for proper compression. 29. Describe the preparation of conductors for splicing and dead ending. 30. Identify the proper conductor splicing sleeves and press dies using reference charts *include:* o *making joints using single strands and multi strands conductors.* o *Soldering Joints.* o *Use of Crimping tool, wire stripper, different.* o *Use of clamps* o *types of lugs.* 31. Identify the proper insulated sleeve for low voltage conductor splicing
Assessment Tasks	Written and/or practical assessment on the skills and knowledge required for overhead line construction.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment. 2. Practical assessment on conducting the construction of overhead power lines assessment in the workshop field.
Learning Outcome 04	**Lay underground cables**
Assessment Criteria	3.1 Explain underground distribution systems. 3.2 Describe the various types of cables. 3.3 Explain methods used to terminate and splice underground cables. 3.5 Explain methods of laying cables. 3.6 Describe the various types of cable enclosures. 3.4 Explain testing and fault locating.
Content	1. Describe the advantages and disadvantages of underground distribution systems. 2. Identify associated components of underground systems.

	3. Describe the construction, advantages and disadvantages, and application of the following common types of cables under the following headings: - *Power cables: PVC steel wire armoured, mineral insulated,* - *Wave-conal, Consac, XPLE (cross linked poly-ethylene),* - *Armoured with thermosetting,* - *Installation cables: single-phase single core PVC insulated and sheathed,* - *XPLE, mineral insulated metal sheathed.* 3. Use I. E. E. and or SAZ regulations to select minimum cable sizes under the following headings: - *Maximum demand and diversity.* - *Cable size selection calculations up to checking/verifying voltage drop.* 4. Describe direct burial and ductwork installation of underground cable. 5. Describe the construction, advantages and disadvantages, and application of the following common types of cable enclosures under the following headings: - *Conduit* - *Trunking* - *Ducting* - *Cable Tray.* 6. Use I. E. E. and or SAZ regulations to select minimum cable enclosure sizes. 7. State the relevant I.E.E and or SAZ Regulations on cable enclosure installation. 8. Describe commonly used primary and secondary cables. 9. Describe the components and purpose of primary and secondary underground cables. 10. Describe the components and purpose of a high voltage termination. 11. Describe methods used to terminate and splice primary and secondary cables. *(Cable Jointing, cable joining techniques, types of cable joining kits and its specifications).* 12. Explain the purpose of single line diagrams used for troubleshooting. 13. Interpret the operation of fault indicators. 14. Describe cable testing and fault locating methods.
Assessment Tasks	Written and/or practical assessment on the skills and knowledge required for underground line construction.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment. 2. Practical assessment on conducting the underground cable installation, termination and/splicing.

ASSESSMENT SPECIFICATION GRID

TOPIC NO.	TOPIC	WEIGHTING %
1	SAFETY REGULATIONS	10
2	LINE CONSTRUCTION THEORY	20
3	OVERHEAD SUPPORT STRUCTURES	20
4	OVERHEAD LINE INSULATORS	10
5	OVERHEAD LINE CONDUCTORS	10
6	UNDERGROUND CABLES	10
7	UNDERGROUND CABLE TERMINATION AND SPLICING.	10
8	UNDERGROUND LINE TESTING & FAULT FINDING.	10
	TOTAL	**100**

Resources:

- Basic Electrician tool kit.
- Angor bit.
- Come alone.
- Ladder.
- Safety belts.
- Pick and shovel.
- Slashers and axes.
- Chain saw.
- Drilling machine.
- Generator.
- Multimeter.
- Treated poles.
- D-irons.
- Insulators.
- Stay wire kit.
- Overhead cables.
- insulation resistance tester
- jack hammer and compressor
- shovels
- picks
- wheel barrow

- Blow lamp, 0.5 ltr
- Conductor (AAA, ACSR, AACSR)
- Megger 1000 V, 1500 V
- Pulley Block
- Earth Tester
- Safety belt with provision for keeping tools
- Ropes & Rigging As required
- Fist Aid Box
- Brackets
- Pulley block
- Crimping Tool(Light Duty)
- Crimping Tool(Heavy Duty)

MODULE **PROJECT (ONE PROJECT TO BE NATIONALLY SET)**

CODE: **321/25/M10**

DURATION: **40 HOURS SUPERVISED TIME**

1.0 AIM

The aim of the module is to provide the student with knowledge and skills of the

Installation, Maintenance and Repair of electrical plant and equipment observing SHEQ procedures.

2.0. LEARNING OUTCOMES

At the end of the module the student should be able to:

2.1 demonstrate proper workshop health and safety standards.

2.2 demonstrate the proper selection, use and care of all tools and apparatus common to electrical craft tradesmen.

2.3 identify and select all materials and hardware common to electrical installation work.

2.4 transfer information from technical drawings and specifications to an installation.

2.5 construct and test circuits and apparatus common to electrical craft, consistent with all relevant IEE Regulations.

2.6 supervise subordinates.

DESIGN LEGTH

40 hours supervised time.

3.0 PROJECT CONDITIONS

3.1 One project will be administered in all the centres.

3.2 Project should be submitted as a completed portfolio

3.3 Project should be a simulation of real plant/installation.

3.4 Field trip to be done for a specific project.

3.5 Students to defend their projects to interview panel where necessary.

4.0 PROJECT LEARNING OUTCOMES

4.1 identify and list the tools, equipment and materials used.

4.2 interpret and apply relevant wiring rules and regulations as SAZ and IEE.

- Calculate the size of cables and protective devices for a particular circuit/installation or load.

- Calculate the number of luminaries and power points for a particular Circuit/load/installation

4.3 determine cable routes.

4.4 list, in logical sequence, the procedures to be taken when fault finding and/or repairing electrical equipment and systems.

4.5 design a planned maintenance program based on the project.

4.6 draw electrical/electronic circuits using relevant symbols, e.g. BS3939.

4.7 test and commission electrical installation.

4.8 present a logical write-up based on the requirements as specified on the question paper.

5.0 ASSESSMENT SPECIFICATION GRID

The examiners are advised to ensure that the project questions based on the objectives are in line with the % mark allocation below. The total marks may be up to 1000.

Objective	% mark allocation
identify and list the tools, equipment and materials used.	5
interpret and apply relevant wiring rules and regulations as SAZ and IEE. - Calculate the size of cables and protective devices for a particular circuit/installation or load. - Calculate the number of luminaries and power points for a particular Circuit/load/installation	 10 10
determine cable routes.	10

list, in logical sequence, the procedures to be taken when fault finding and/or repairing electrical equipment and systems	5
design a planned maintenance program based on the project.	10
draw electrical/electronic circuits using relevant symbols, e.g. BS3939.	10
test and commission electrical installation.	10
present a logical write-up based on the requirements as specified on the question paper.	10
Project Defending	15
Presentation, neatness and originality	5

Module Code:	**402/25/M01**
Module Title:	**ENTREPRENEURSHIP SKILLS DEVELOPMENT**
ZNQF Level:	**4**
Credits:	**5**
Duration:	**50 HOURS**
Relationship with Qualification Standards:	Based on Unit Standard **TBA ENTREPRENEURSHIP SKILLS DEVELOPMENT OF UNIT STANDARD FOR AN ENTREPRENEUR**
Pre-requisite modules:	NON
Purpose of Module:	This module describes the skills, knowledge and attitudes required by an entrepreneur to acquire leadership, business and time management, creative thinking and problem-solving in a job role and industries. This module will ensure that the entrepreneur will formulate a business plan, register a company and operate a business. The advantages of entrepreneurship skills development are that growth and development are constant, beneficial network is developed and work life autonomy is possible. Access to this module is open to all youth, man and woman who want to own a business.
List of Learning Outcomes:	**LO1: Formulate a business** **LO2: Register a company** **LO3: Operate a business**
Learning Outcome 01	**Formulate a business**
Assessment Criteria:	1.1 Formulate a business idea 1.2Produce business plan 1.3 Research on business market 1.4 Compile a financial plan 1.5 Position a product/service 1.6 Envelope survival strategies 1.7 Establish a business environment 1.8 Mobilise financial resources
Content:	**1.1.Formulate a business idea** 1.1.1Define an entrepreneur 1.1.2 Discuss the various concepts of entrepreneurship

	1.1.3 Analyse the various forms of business ownership **1.2. Produce business plan** 1.2.1 Define a business plan 1.2.2Produce an executive summary of your business 1.2.3Describe the business 1.2.4 Provide the organisational structure of the business 1.2.5Describe product/services 1.2.6 Provide market analysis 1.2.7 Give marketing strategies 1.2.8 Provide a financial plan **1.3 Research on business market** 1.3.1 Define business market 1.3.2 Study market trends 1.3.3 Analyse market segmentation 1.3.4 Analyse competitors in the market **1.4 Compile a financial plan** 1.4.1Plan for staffing and employees 1.42 Forecast on profit and loss 1.4.3 Analysis of cashflow 1.4.5 Prepare a balance sheet **1.5 Position products/services** 1.5.1 Define positioning of products and services 1.5.2 Describe the types of product and services positioning 1.5.3 Discuss the importance of product/service positioning **1.6 Envelope survival strategies** 1.6.1 Define survival strategies 1.6.2Describe the types of survival strategies 1.6.3Discuss the importance of survival in business **1.7 Establish a business environment** 1.7.1Conduct SWOT analysis 1.7.2Discuss price and position products/ services 1.7.3Conduct viable promotions

	1.8 Mobilise Financial resources 1.8.1Provide a detailed account of how to bring revenue and funding to get started 1.8.2Balancing financial statement
Assessment Tasks:	17. Written and/or oral assessment on the skills and knowledge required to formulate a business as outlined in the assessment criteria and content above. 18. Practical assessment on the formulation of a business plan
Conditions/Context of assessment	19. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 20. The practical assessment will be conducted in the workplace or simulated work environment in the training institution. 21. The context of assessment should include the facilities, tools, equipment and materials as per entrepreneur's occupation.
Learning Outcome 02	**Register a company**
Assessment Criteria	2.1Prepare company documents 2.2 Process business registration 2.3Secure a place of business operation 2.4Compile rules and regulations
Content	**2.1Prepare company documents** 2.1.1 Identify business documents 2.1.2 Explain the purpose of books of accounts (cashbooks,ledger,etc) 2.1.3 Explain the importance of business documents **2.2 Process business registration** 2.2.1Define company registration 2.2.2Identify the types of companies that can be registered 2.2.3 Describe the requirements needed to register different companies 2.2.4 Discuss the procedures for company registration 2.2.5 Describe the documents that are received after company registration **2.3 Secure a place of business operation** 2.3.1 Identify factors that influence an entrepreneur in securing a place of business operation 2.3.2 Discuss the macro and micro environmental factors affecting entrepreneurship

	2.3.3 Define SMEs(Small and Medium Enterprises) 2.3.4 Discuss the roles of SMEs 2.**4Compile rules and regulations** 2.4.1 Define rules and regulations in business 2.4.2 Compile guiding rules and regulations in business 2.4.3 Explain the importance of rules and regulations in business
Assessment Tasks	1. Written and/or oral assessment on the skills and knowledge required to registering a company as outlined in the assessment criteria and content above. 2. Practical assessment on the registering of a business plan
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the workplace or simulated work environment in the training institution. 3. The context of assessment should include the facilities, tools, equipment and materials as per entrepreneur's occupation
Learning Outcome 03	**Operate a business**
Assessment Criteria	3.1Manage a business according to organisation policy 3.2Allocate resources according to line of business 3.3 Cost products in line with procedues 3.4 Price products according to company procedures 3.5 Update and maintain records 3.6 Control stock in line with organisation requirements 3.7 Formulate market plans 3.8 Manage risks in line with organisation requirements 3.9 Adopt growth strategies 3.10 Observe business and give social responsibility 3.11 Practise customer care 3.12 Motivate employees in line with organisational requirements
Content	**3.1Manage a business according to organisation policy** 3.1.1Define business management 3.1.2 Explain the roles of management in a business 3.1. Discuss the importance of computers as a business management tool 3**.2 Allocate resources according to line of business** 3,2.1 Define resource allocation 3.2.2 Explain the importance of properly allocating resources (human, capital, material)

	3.3 Cost products in line with procedures 3.3.1 Define various costing terms 3.3.2 Explain the importance of costing to a business 3.3.3 Describe the costing processes of a business 3.3.4 Calculate using the basic cost - pricing and profit methods in relation to products/ services **3.4 Price products in line with business policy** 3.4.1 Define various pricing terms 3.4.2 Explain the importance of pricing to a business 3.4.3 Analyse the pricing processes of a business 3.4.4 Calculate prices of products 3.4.5 Describe pricing strategies **3.5 Update and maintain records** 3.5.1Define record keeping in business 3.5.2 Identify source business documents 3.5.3 Explain the importance of record keeping 3.5.4 Describe the purposes of books of accounts **3.6 Control stock in line with organisation requirements** 3.6.1Define stock control in business 3.6.2Describe the importance of stock control 3.6.3 Outline effective stock control procedures **3.7 Formulate market plans** 3.7.1Define marketing 3.7.2 Devise a marketing plan for a business 3.7.3 Explain the Ps of marketing 3.7.4 Discuss the marketing mix strategies **3.8 Manage risks in line with organisation requirements** 3.8.1 Define risk management 3.8.2 Discuss the importance of risk covers in entrepreneurship 3.8.3Explain the principles of risk management to a business 3.8.4 Analyse the steps involved in risk management process 3.8.4Identify the various risk management strategies in business **3.9 Adopt growth strategies** 3.9.1 Define business growth strategies 3.9.2 Explain the four business growth strategies

	3.10 Observe business ethics and give social responsibility 3.10.1 Define business ethics and social responsibility 3.10.2 Explain the imporatance of business ethics to entrepreneurs 3.10.3 Outline social responsibility principles 3.10.4 Explain the importance of social responsibility to the entreptreneur 3.10.5 Illustrate acts of social responsibility by an entrepreneur in a community **3.11 Practise customer care** 3.11.1 Define customer care 3.11.2 Discuss ten tips of customer care 3.11.3 Explain benefits of customer care **3.12 Motivate employees in line with organisational requirements** 3.12.1 Define motivation 3.12.2 Outline theories of staff motivation in business 3.12.3 Discuss the importance of motivation
Assessment Tasks	1. Written and/or oral assessment on the skills and knowledge required to operate a business as outlined in the assessment criteria and content above. 2. Practical assessment on operating a business
Conditions/Context of assessment	1. Written and/or oral assessment can be conducted in a classroom environment. Oral assessment can also be conducted by the assessor during the performance of the practical assessment by the trainees. 2. The practical assessment will be conducted in the workplace or simulated work environment in the training institution. 3. The context of assessment should include the facilities, tools, equipment and materials as per entrepreneur's occupation

Approach to Teaching and Learning:

50. Observation of adult learning principles.
51. Both institution-based and work-based learning to facilitate the integration of theory and practice.
52. Face-to-face education and learning.
53. Problem-based learning.
54. Online/distance education and learning.
55. Blended/hybrid education and learning.
56. Use of social media.

Approach to Assessment:

42. Weighting of practical and theory assessment: 60% theory and 40% practical.
43. Weighting of institution-based and work-based assessment: 50% institution-based assessment and 50%.
44. Oral assessment to be conducted by a panel of two or more assessors.
45. Portfolio of evidence.
46. Assessment of work conducted by both individual learners and teams of learners.

Resources:

1 Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone ZNQF accredited training programmes and should have qualification and experience recognised by the Zimbabwe National Qualifications Authority (ZNQA).

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26. Facilities, Tools, Equipment and MaterialsFacilities, Tools, Equipment and Materials

- Computer
- Communication equipment
- Data storage devices
- Television
- DVD Recorder/playerGeneric which are relevant to the type of business

27. Learning Resources

Relevant training manual (learners' guide) and facilitators' guide

28. Reference Materials (recommended textbooks, recommended readings)

Alderman,P.,J.,(2011) Entrepreneurial Finance,Pearson Education LTD, London

Appleby R (1994) Modern Business Administration

Barringer,B.,R., &Ireland,D.,R.,(2006) Entrepreneurship:Successfully Launching New Ventures, Pearson Education

Bridge, S., O'Neill , K. & Martin, F., (2009) Understanding Entrerprise: Entrepreneurship &Small Business, Third Edition, Palgrave Macmillan, London

Burns, P. & Dewhurst, J., (eds)91990)Small Business and Entrepreneurship, Macmillan Education LTD, Hampshire

City and Guilds, (2012) Hospitality supervision & Leadership, Heinemann, Essex,

Deakins, D., & Freel, M., (2012) Entrepreneurship and Small Firms, McGraw-Hill,Berkshire

Hisrich, R. D. & Peters, M. P. (2016) Entrepreneurship, Tatq McGraw Hill New Delphi Entre

Holt, D.T., (2017) Entrepreneurship, Prentice Hall, London

Jarskoy, H. & Stevenson, D., (2014) International Labour Organisation Start Your Business. ILO, Harare

Justin Smith (2000) Business Management Trainer's Guide

Kotler Philip & Armstrong G (2001) Principles of Marketing

Kuratiko, D., F., (2008) Introduction to Entrepreneurship, Cengage Learning , Hampshire

Lee, C.,L., &Melicher,W.,(2012) Entrepreneurial Finance,4th Edition, Cengage Learning, South Western

Marcourse, I.(2016) Business Studies @nd Ed Hodder Arnold, London

McGuckin Frances (1988) Business for Beginners (A simple step by step Guide to Start Your New Business)

Mullins L (1999) Management and Organisational Behaviour 5th edition

Need Harm, D, & Dransfield, R.(2000) Advanced Business and Dexel, Oxford

Rae, D., (2007) Entrepreneurship, From opportunity to action, Palgrave Macmillan, New York

Rwegema , V., U., Entrepreneurship: theory in practice, 2nd edition, Oxford University Press, Cape Town

Stokes, D., Wilson, N. & Mador, M.,(2010) Entrepreneurshp, Cengage Learning EMEA, Hampshire

Stoner, J., A. F., Freeman, R., E. & Gilbert, D, R., J. R. (2017) Management 6th Edition, Prentice Hall International Englenwood Cliffs, New Jersey.

Van Der Wagen & Davies, C.(1998) Supervision and Leadership, Hospitality Press Pvt Ltd Elsternwick Victoria

Zimmerer T.W, Scarborough M Norman – Essentials of Entrepreneurship and Small

UNIT 1

Unit Code	**402/22/M01**
Unit Title:	**ENTREPRENEURSHIP SKILLS DEVELOPMENT**

Level of Unit: **Generic**

Credits: **8**

Occupation: **ENTREPRENEUR**

Date of Promulgation: **TBA**

Review Date: **TBA AIM OF THE UNIT STANDARD**

This unit enables an individual to acquire skills and knowledge in leadership, business and time management, creative thinking and problem-solving in a job role and industries.

ELEMENT AND PERFORMANCE CRITERIA

Element 1.1	**Formulate a Business**

Performance Criteria:

1.1.1 Business idea formulated according to requirements
1.1.2 Business plan produced
1.1.3 Business market researched in line with policy
1.1.4 Financial plan compiled
1.1.5 Product or service positioned in line with specifications
1.1.6 Survival strategies enveloped
1.1.7 Business environment established according to requirements
1.1.8 Financial resources Mobilised

Element 1.2	**Register a company**

Performance Criteria:

1.2.1 Company documents prepared in line with procedures
1.2.2 Business registration processed according to policies
1.2.3 Place of business operation secured
1.2.4 Rules and regulations compiled according to business requirements

Element 1.3	**Operate a business**

Performance Criteria:

1.3.1 Business managed according to organisation policies
1.3.2 Resources allocated according to line of business
1.3.3 Products costed in line with procedures
1.3.4 Products priced according to company procedures
1.3.5 Records updated and maintained
1.3.6 Stock controlled in line with organisation requirements
1.3.7 Marketing plan formulated
1.3.8 Risks managed in line with organisation requirements
1.3.9 Growth strategies adopted
1.3.10 Business ethics observed and social responsibility given
1.3.11 Customer care practised
1.3.12 Employees motivated in line with organisational requirements

COMPETENCIES REQUIRED IN READINESS FOR ASSESSMENT:

Accounting skills
Record keeping
Customer care skills
Management skills (decision making, planning, organising)
Technological awareness
Marketing skills
Business conduct
Legal awareness
Mobilisation skills
Self-Supervision
Patriotism
Environmental awareness (PESTEL)

GENERIC SKILLS:
Practical skills
Calculations Skills
Creativity
Sense of initiative
Ability to Marshall Resources

Technological knowledge
Communication
Planning
Organization
Controlling
Human relation skills
Interpersonal skills
Analytical skills

RANGE STATEMENT:

Tools and Equipment
Generic which are relevant to the type of business
Materials
Generic which are relevant to the type of business

Duration: 50 hours

ASSESSMENT AND CERTIFICATION:
In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.

Accredited assessors will conduct assessment. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence of their achievements.

Module Code:	**401/24/M01**
Module Title:	**NATIONAL STUDIES**
ZNQF Level:	Generic
Credits:	5
Duration:	50 hours
Relationship with Qualification Standards:	**Based on Unit Standard TBA** **NATIONAL STUDIES UNIT STANDARD FOR PATRIOTIC CITIZEN**
Pre-requisite modules:	NON
Purpose of Module:	This module describes the skills, knowledge and attitudes required by a patriotic citizen to develop values that make them proud to be Zimbabweans. This includes maintaining a Zimbabwean culture, preserving Zimbabwean History, assembling components of colonial effects, analysing post-independence socio-economic and political developments, assembling components of legal and parliamentary affairs, carrying out a feasibility study on peace, conflict and resolution as well as participating in civic responsibilities. This is important in producing an informed and responsible citizen prepared to defend and develop the country. Access to this module is open to all target groups, which include the unemployed youth, men and women willing to develop their country.
List of Learning Outcomes:	**LO1: Maintain a Zimbabwean culture** **LO2: Preserve Zimbabwean History** **LO3: Assemble components of colonial effects** **LO4: Analyse post-independence socio-economic and political developments** **LO5: Carry out a feasibility study on peace, conflict and resolution** **LO6: Participate in civic responsibilities** **LO7 Assemble components of legal and parliamentary affairs**
Learning Outcome 01	**Maintain a Zimbabwean culture**

Assessment Criteria:	**1.1. Preserve cultural heritage** **1.2. Conserve cultural artefacts** **1.3. Demonstrate knowledge of Zimbabwean culture** **1.4. Capture records of maintaining natural resources of Zimbabwe** **1.5. Preserve indigenous knowledge systems**
Content:	**1.1.Preserve cultural heritage** 1.1.1. Definition of cultural heritage 1.1.2. Types of cultural heritage 1.1.3. Importance of cultural heritage 1.1.4.Indigenous methods of preserving and conserving cultural heritage 1.1.5.Modern ways of preserving and conserving cultural heritage 1.1.6.Role of national and international organisations in protecting cultural heritage **1.2. Conserve cultural artefacts** 1.2.1. Identification of cultural artefacts 1.2.2. Threats to cultural artefacts 1.2.3. Importance of cultural artefacts 1.2.4. Ways of protecting cultural artefacts **1.3. Demonstrate knowledge of Zimbabwean culture** 1.3.1. Components of Zimbabwean culture 1.3.2. Significance of components of the Zimbabwean Culture 1.3.3. Threats to various components of the Zimbabwean Culture 1.3.4. Ways of upholding the Zimbabwean Culture **1.4. Capture records of maintaining natural resources of Zimbabwe** **1.4.1.** Types of natural resources 1.4.2. Importance of natural resources 1.4.3. Indigenous and modern methods of protecting natural Resources

	1.4.4. National and international statutes that protect national Resources **1.5. Preserve indigenous knowledge systems** 1.5.1. Definition of indigenous knowledge systems 1.5.2. Components of indigenous knowledge systems 1.5.3. Meanings and significance of indigenous knowledge systems 1.5.4. Insights gained from indigenous knowledge systems
Assessment Tasks:	19. Written assessment on the skills and knowledge required maintain a Zimbabwean Culture as highlighted above. 20. Practical based assignment on ways of preserving cultural heritage sites within their communities.
Conditions/Context of assessment	22. Written assessment can be conducted in a classroom environment. 23. The practical based assignment assessment will be conducted based on observations in their communities
Learning Outcome 02	**Preserve Zimbabwean History**
Assessment Criteria	**2.1 Identify pre-colonial states** **2.2 Analyse precolonial political structure** **2.3 Record achievements of precolonial history** **2.4 Record colonial history** **2.5 Record role of Christian missionaries** **2.6 Record occupation of Zimbabwe** **2.7 Trace causes of first /second Chimurenga**
Content	**2.1 Identify pre-colonial states** 2.1.1 Defining term pre-colonial 2.1.2. Identifying precolonial states 2.1.3 Pre- colonial socio-economic organisation 2.1.4. Causes of decline of pre-colonial states

	2.1.5. Influence of pre-colonial civilisation on contemporary society **2.2 Analyse precolonial political structure** 2.2.1 System of governance of pre-colonial states 2.2.2 Features of the pre-colonial system 2.2.3. Influence of precolonial governance on contemporary society **2.3 Record achievements of precolonial history** **2.3.1** Impact of precolonial achievements and political development **2.4 Record colonial history** 2.4.1 Partition and colonisation of Africa 2.4.2 Berlin conference 2.4.3 Causes/ reasons for the colonisation/occupation of Zimbabwe 2.4.4 Colonisation steps/processes in Zimbabwe **2.5. Record role of Christian missionaries** 2.5.1 Socio-economic and political impact of Christian missionaries in Zimbabwe **2.6. Record occupation of Zimbabwe** **2.6.1** Colonial Administration from 1894 to 1923 1.2.6.2 Socio-economic and political impact of colonisation in Zimbabwe **2.7 Trace causes of first /Second Chimurenga** 2.7.1 Causes and results of the Anglo-Ndebele war 1.7.2 Causes and results of the 1^{st} Chimurenga/Umvukela 2.7.3. African reaction-to socio-economic and political colonial administration 2.7.4. Causes and results of the 2^{nd} Chimurenga

	2.7.5. Socio-economic and political impact of the 1st and 2nd Chimurenga 2.7.6. Prosecution of the war of liberation 2.7.6 Social and political impact of heroes/heroines
Assessment Tasks:	1. Written or oral assessment on the skills and knowledge required to assess the understanding of Zimbabwean History. 2. Practical activities based on observations within and outside the institution that demonstrate understanding of Zimbabwean history.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment or practical activities conducted within or outside the institution. 2. The practical based assignment/activities will be conducted based on participation/observations in their communities
Learning Outcome 03	**Assemble components of colonial effects**
Assessment Criteria	3.1 Demarcate administrative boundaries 3.2. Exploit natural resources (minerals, wildlife, land, water Vegetation, etc.) 3.3. Change traditional religion 3.4. Introduce foreign food crops and livestock 3.5. Change forms of trade 3.6. Change education systems 3.7. Introduce new legal systems 3.8. Introduce Capitalistic relations 3.9. Violate Human rights 3.10. Analyse results of colonisation
Content	**3.1 Demarcate administrative boundaries** 3.1.1. Factors that led to demarcation of boundaries 3.1.2. Distribution of land and uses

	3.1.3. Consequences of establishing administrative boundaries **3.2. Exploit natural resources (minerals, wildlife, land, water Vegetation etc.)** 3.2.1. Geographical distribution of available resources 3.2.2. Measures enacted to exploit the resources 3.2.3. Consequences of exploiting the resources (Social, political, economic) **3.3. Change traditional religion** 3.3.1. The nature of African traditional religion prior to colonisation 3.3.2. The role of religion in the African societies 3.3.3. The introduction of foreign religion 3.3.4. The effect of foreign religion on African societies 3.3.5. The place of African Traditional religion in contemporary society **3.4. Introduce foreign food crops and livestock** 3.4.1. Nature and significance of African food crops and livestock 3.4.2. Types of foreign crops introduced 3.4.3. Consequences of the foreign crops and livestock on African Societies 3.4.4. The sustainability of traditional versus foreign crops and livestock in contemporary Zimbabwean society **3.5. Change forms of trade** 3.5.1. Nature and benefits of trade prior to colonisation 3.5.2. Nature of trade during colonisation 3.5.3. Effects of trade during colonial era on African societies. 3.5.4. Influence of trade patterns to contemporary society

	3.6. Change education systems 3.6.1. Nature and purpose of Traditional African Education system 3.6.2. Nature and purpose of Colonial education 3.6.3. Consequences of Colonial education on African Societies 3.6.4. Influence of colonial education to contemporary society **3.7. Introduce new legal systems** 3.7.1. Nature of African legal system prior to colonisation. 3.7.2. Nature of colonial legislation (social, political and economic) 3.7.3. Purpose of colonial legal system 3.7.5. Consequences of colonial legal system to colonial and contemporary African societies **3.8. Introduce Capitalistic relations** 3.8.1. Nature of African relations before colonisation 3.8.2. Introduction of capitalist relations 3.8.3. Effects of capitalist relations during the colonial era and the contemporary society **3.9. Violate Human rights** 3.9.1. Definition of human rights 3.9.2. Nature of human rights violations in the colonial era 3.9.3. Response to human rights violations during the colonial era **3.10. Analyse results of colonisation** 3.10.1. Social effects of colonisation on African Societies 3.10.2. Economic effects of colonisation on African Societies 3.10.3. Political effects of colonisation on African societies 3.10.4. Benefits and non-benefits of colonisation

Assessment Tasks	1. Written assessment on the skills and knowledge required to assess the consequences of colonisation on the African Societies. 2. Practical based assignment on observable socio-politico and economic effects of colonisation within their communities.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment. 2. The practical based assignment assessment will be conducted based on observations in their communities
Learning outcome O4	**Analyse post-independence socio-economic and political developments**
Assessment Criteria	**4.1 Analyse socio-economic, political developments** **4.2Formulate Policies** **4.3 Adopt measures to address colonial injustices**
Content	**4.1 Examine socio-economic and political developments** 4.1.1 Social-economic and political post-independence developments 4.1.2 Critique of post-independent development **4.2 Formulate Policies** 4.2.1 Legislation that addressed colonial injustices 4.2.2 Impact of post-independent legislation 4.2.3 Comparison of colonial and post-independence legislation **4.3 Adopt measures to address colonial injustices** 4.3.1 Socio-economic and political measures to address colonial injustices 4.3.2 Impact of measures to address colonial injustices 4.3.3 Colonial vestiges 4.3.4 Strategies to address colonial vestiges
Assessment Tasks	1. Written assessment on the skills and knowledge required to assess the achievements and challenges of post-independent in Zimbabwe. 2. Practical based assignment on observable socio-economic and political developments in their communities.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment. 2. The practical based assignment assessment will be conducted based on observations in their communities

Learning Outcome 05	**Carry out a feasibility study on peace, conflict and resolution**
Assessment Criteria	**5.1. Demonstrate Conflict and resolution styles** **5.2. Demonstrate 3Cs between Zimbabwe and the global community** **5.3. Analyse Strategies for sustaining peace** **5.4. Analyse the influence of multi-national companies in developing countries** **5.5. Examine the benefits of International capital to developing countries**
Content	**5.1. Demonstrate Conflict and resolution styles** 5.1.1 Defining conflict and conflict resolution 5.1.2 Identifying and explaining conflict resolution styles 5.1.3 Impact of conflict resolution to socio-economic development 5.1.5 Traditional African conflict resolution methods. **5.2. Demonstrate 3Cs between Zimbabwe and the global community** 5.2.1 Defining terms Conflict, competition and co-operation 5.2.2 Impact of 3Cs to economic development 5.2.3 Approaches/Theories to International Relations 5.2.4 Global power balance 5.2.5 Zimbabwean foreign policy 5.2.6 Zimbabwean regional and international interventions **5.3. Analyse Strategies for sustainable peace** 5.3.1 Defining peace and sustainable peace 5.3.2 Importance of peace to socio-economic and political Development 5.3.3. Impact of sanctions on development

	5.3.4 Strategies for sustainable peace 5.3.5 Role of NGOs in the development of sustainable peace 5.3.6 Role of media in promoting and maintaining peace **5.4. Analyse the influence of multi-national companies in developing countries** 5.4.1 Role of multi-national companies in developing countries 5.4.2 International capital and imperialism 5.4.3 The IMF/WB Institutions 5.4.4 Impact of Non-Governmental Organisations in developing countries **5.5 Examine the benefits of International capital to developing countries** 5.5.1 Characteristics of finance capital 5.5.2 International economic relations 5.5.3 Features of Globalisation 5.5.4 Benefits and non-benefits of globalisation
Assessment Tasks	1. Written assessment on the skills and knowledge required to assess the importance of understanding the importance of peace, conflict and resolution on socio-economic developments in Zimbabwe. 2. Practical based assignment on observable socio-economic and political developments in their communities.
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment. 2. The practical based assignment assessment will be conducted based on observations in their communities
Learning Outcome 06	**Participate in civic responsibilities**
Assessment Criteria	**6.1 Undertake Civic responsibilities** **6.2 Observe participation in Disaster Management** **6.3 Adopt Citizen duties**

Content	**6.1 Undertake Civic responsibilities** 6.1.1. Definition of civic responsibilities 6.1.2. Civic responsibility activities 6.1.3. Justification for civic responsibilities **6.2. Observe participation in Disaster Management** 6.2.1. Definition of disaster management 6.2.2. Justification for participation in disaster management 6.2.3. Sustainable disaster management practices **6.3 Adopt Citizen duties** 6.3.1. Definition of terms: citizen and citizen duties 6.3.2. Identification and explanation of citizen duties(Socio-economic & political) 6.3.3. Citizen rights 6.3.4. Importance of citizen duties
Assessment Tasks	1. Written or oral assessment on the skills and knowledge required to assess the understanding of citizen duties and responsibilities. 2. Practical activities within and outside the institution that demonstrate civic duties and responsibilities by community participation
Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment or practical activities conducted within or outside the institution. 2. The practical based assignment/activities will be conducted based on participation/observations in their communities
Learning outcome 07:	**Assemble components of legal and parliamentary affairs**
Assessment Criteria	**7.1 Identify and explain origins of law** **7.2 Observe constitutional provisions** **7.3 Identify and explain arms of the state** **7.4 Explain Law making process**

Content	7.1 **Identify and explain the origins of law** 7.1.1 Definition of legal terms 7.1.2 Purpose of the law to the community 7.1.3 Classification of the law 7.1.4 Sources of law in Zimbabwe **7.2 Observe constitutional provisions** 7.2.1 Justification of a Zimbabwean constitution 7.2.2 Constitutional Rights as enshrined in the Zimbabwean constitution 7.2.3 Benefits of constitutional rights to the community 7.2.3 Enforcement of rights 7.2.4 Role played by stakeholders in upholding constitutional rights (NGO, Civil Societies and victim friendly units) 7.2.5. Impediments to exercising human rights 7.2.6. Role of constitution in the community **7.3 Observe arms of the state** 7.3.1 Identification of the three arms of state 7.3.2 Duties and functions of the three arms of the state 7.3.3 Importance of separation of powers to Zimbabwe 7.4 **Explain Law making process** 7.4.1 Steps in the Law making 7.4.2 Role of community in law making process
Assessment Tasks	1. Written or oral assessment on the skills and knowledge required to assess the understanding of legal and parliamentary affairs. 2. Practical activities within and outside the institution that demonstrate the importance of participating in legal and parliamentary activities. 3.

Conditions/Context of assessment	1. Written assessment can be conducted in a classroom environment or practical activities conducted within or outside the institution. **2.** The practical based assignment/activities will be conducted based on participation/observations in their communities.

Approach to Teaching and Learning:

57. Observation of adult learning principles; both institution-based and work-based learning to facilitate the integration of theory and practice.
58. Face-to-face education and learning.
59. Problem-based learning.
60. Online/distance education and learning.
61. Blended/hybrid education and learning.
62. Use of social media.

Approach to Assessment:

47. Weighting of institution-based and examination -based assessment: 60% institution-based assessment and 40% examination.
48. Portfolio of evidence.

Resources:

29. Qualifications and experience of Trainers, Assessors and Moderators

All trainers, assessors and moderators should have undergone a Bachelor's Degree in History or equivalent.

30. Facilities, Tools, Equipment and Materials

- Computer
- Communication equipment
- Data storage devices
- Television
- DVD Recorder/player
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31. Learning Resources

- Artefacts
- Resource persons
- Museums and heritage sites
- Videos and audio materials

32. Reference Materials (recommended textbooks, recommended readings)

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Shaw, W.H., 2003. 'They Stole Our Land': debating the expropriation of white farms in Zimbabwe. *The Journal of Modern African Studies*, *41*(1), pp.75-89.

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UNIT 1

Unit Code	401/25/M01
Unit Title:	**National Studies**

Level of Unit: **Generic**

Credits: **5**

Occupation: Patriotic Citizen

Date of Promulgation: **TBA**

Review Date: **TBA**

AIM OF THE UNIT STANDARD

This unit helps people to develop values that make them proud to be Zimbabweans.

ELEMENT AND PERFORMANCE CRITERIA

Element 1.1	**Maintain a Zimbabwean culture**

Performance Criteria:

1.1.9 Cultural heritage preserved
1.1.10 Cultural artefacts conserved
1.1.11 Knowledge of Zimbabwe culture demonstrated
1.1.12 Records of maintaining natural resources of Zimbabwe captured
1.1.13 Indigenous knowledge systems preserved

Element 1.2	**Preserve Zimbabwean History**

Performance Criteria:

1.2.1 Pre-colonial states identified
1.2.2 Precolonial political structure analysed
1.2.3 Achievements of precolonial history recorded
1.2.4 Colonial history recorded
1.2.5 Role of Christian missionaries recorded
1.2.6 Occupation of Zimbabwe recorded
1.2.7 Causes of First /Second Chimurenga traced

Element 1.3	**Assemble components of colonial effects**

Performance Criteria:

1.3.1 New administrative boundaries demarcated

1.3.2 Natural resources exploited (minerals, wildlife, land, water, vegetation etc)
1.3.3 Traditional religion changed
1.3.4 Foreign food crops and livestock introduced
1.3.5 Education systems changed
1.3.6 Capitalistic relations introduced
1.3.7 New legal systems introduced
1.3.8. Forms of trade changed
1.3.9 Human rights violated
1.3.10 Results of colonisation analysed

Element 1.4	**Analyse post-independence socio-economic and political developments**

Performance Criteria:

1.4.1 Socio-economic and political developments examined
1.4.2 Policies formulated
1.4.3 Measures to address colonial injustices adopted

Element 1.5	**Carry out a feasibility study on peace, conflict and resolution**

Performance Criteria:

1.5.1 Conflict and resolution styles demonstrated
1.5.2 3Cs between Zimbabwe and the global community demonstrated
1.5.3 Strategies for sustaining peace analysed
15.4 Influence of multi-national companies in developing countries analysed
1.5.5 Benefits of International capital to developing countries examined.

Element 1.6	**Participate in civic responsibilities**

Performance Criteria:

1.6.1 Civic responsibilities undertaken
1.6.2 Participation in disaster management observed
1.6.3 Citizen duties adopted

Element 1.7	**Assemble components of legal and parliamentary affairs**

Performance Criteria:

1.7.1 Origins of law identified and explained
1.7.2 Constitutional provisions observed

1.7.3 Arms of the state identified and explained
1.7.4 Law making process explained

COMPETENCIES REQUIRED IN READINESS FOR ASSESSMENT:

Record keeping skills

Customer care skills
Management skills (decision making, planning, organising)
Technological awareness
Problem-solving skills
Interpersonal skills
Legal awareness
Mobilisation skills
Upholding norms, values and social aspects of Zimbabwean culture.
Patriotism

Environmental awareness skills
Legal awareness
Critical thinking skills
Research skills
Problem-solving skills
Maintaining Zimbabwean culture
Social responsible
Abreast with global current events
Tool handling skills

GENERIC SKILLS:
Patriotic
Practical skills
Tolerance skills
Technological knowledge
Communication
Positive regard
Good attitude
Good morals
Acceptance of others
Servant hood
Committed cadre to National Agenda
Quest for more knowledge
Social skills

Planning
Organisation
Controlling
Human relation skills
Interpersonal skills
Critical thinking skills
Analytical skills

TOOLS AND EQUIPMENT:

Generic which are relevant to the type of business.

MATERIALS:

Generic which are relevant to the type of business.

Duration: **8**0 hours

ASSESSMENT AND CERTIFICATION:

In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.

Accredited assessors will conduct assessment. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence

MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT

HIGHER EDUCATION EXAMINATION COUNCIL

(HEXCO)

QUALIFICATION STANDARD

FOR AN

ELECTRICIAN

SECTOR: ELECTRICAL & ELECTRONICS ENGINEERING

QUALIFICATION FOR A ELECTRICIAN

QUALIFICATION CODE: TBA

LEVEL: NATIONAL CERTIFICATE

DATE OF PROMULGATION: TBA

DEFINITION OF TERMS

Assessment	A process of collecting evidence of a learner's work to measure and make judgements about the achievement or non-achievement of the specified National Qualifications Framework standards or qualifications.
Certification	Awarding of approved documentary evidence of a qualification.
Competences required in readiness for assessment	Critical relevant knowledge, skills and attitudes a learner requires in order to achieve specified outcomes before assessment.
Credit	The value assigned to a unit completed or a value assigned to a unit standard which reflects the relative time and effort required to complete the outcomes.
Date of promulgation	Date when standard and qualification have been approved, registered and gazetted.
Duration	The minimum notional hours required by a learner to attain all the competences in a unit standard.
Element	The smallest component of a unit with a meaningful outcome.
Generic skills	Universal skills which apply to more than one occupation.
Level descriptor	A specific indicator of competence level on the ZQF.
Occupation	A group of related economically beneficial work activities performed by a person.
Performance criteria	A statement of competence or achievement against which the attainment of outcomes is measured.
Qualification	Formal award of recognition of the achievement of the required competency and/or capability level of the Zimbabwe Qualifications Framework as may be determined by the relevant bodies registered for such purpose by the Authority.
Range statement	The context or conditions within which a competence is performed and assessed that include tools, equipment, materials and duration.
Review Date	Date of revision of qualification standard as and when necessary but not later than three years from date of issue.

Sector	A section of the economy in which operators produce or provide similar products or services.
Standard	Registered statement of desired education and training outcomes and their assessment criteria.
Unit	The smallest combination of work activities capable of being a full-time economically beneficial occupation.
Unit Standard	Registered statement(s) of desired education and training outcomes, their associated assessment criteria together with administrative information as specified.
ZQF	National qualifications framework approved by the minister for registration of national standards and qualifications.

UNIT TITLES

NO.	UNIT	CREDITS
1	Plant Maintenance	40
2	Plant and Equipment Installation	38
3	Electrical Equipment Repair	48
4	Safety, Health, Environmental and Quality Management	40
5	Power Distribution Lines Construction	30
6	Electrical Circuit Designing	36
7	PLC Manipulation	20

SUMMARY OF STANDARD

UNIT NO.	UNIT TITLE	CREDITS	ELEMENTS
1	Plant Maintenance	40	Monitor running Plant 1.2 Design maintenance plan 1.3 Request required resource 1.4 Carry out planned maintenance 1.5 Carry out breakdown maintenance 1.6 Test and re-commission
2	Plant and Equipment Installation	38	2.1 Conduct installation survey 2.2 Design installation plan 2.3 Produce quantity of required resources 2.4 Conduct installation 2.5 Inspect installation 2.6 Inspect, test and commission
3	Electrical Equipment Repair	48	3.1 Ensure Safety 3.2 Troubleshoot faults 3.3 Rectify faults
4	Safety, Health, Environmental and Quality Management	40	4.1 Use appropriate personal protective equipment 4.2 Adhere to work permits 4.3 Adhere to local and international SHE standards 4.4 Use equipment safely
5	Power Distribution Lines Construction	30	5.1 Carryout ground survey 5.2 Clear line route 5.3 Determine labour and material requirements 5.4 Erect poles and draw lines
6	Electrical Circuit Designing	36	6.1 Obtain circuit application 6.2 Implement the designed circuit 6.3 Modify circuit diagram
7	PLC Manipulation	20	7.1 Check equipment operation parameters 7.2 Write a program 7.3 configure devices 7.4 Test and simulate program 7.5 Up load program 7.6 Produce documentation

UNIT 1

Unit Title 1	PLANT MAINTENANCE
Unit Code	321/25/CR/MO

ZQF Level: **NATIONAL CERTIFICATE**

Credits: **40**

Occupation: **Electrician**

Date of Promulgation: **TBA**

Review Date: **TBA**

Aim of the unit standard

This unit will enable an individual to design a maintenance plan, source for resources and carry out maintenance of all electrical equipment in a working environment.

ELEMENTS AND PERFORMANCE CRITERIA

Element 1.1	Monitor running plant

Performance Criteria:

1.1.1 Safety, health, environmental and quality procedures are adhered to.
1.1.2 Operation checklist is completed
1.1.3 Operation breakdowns are attended to
1.1.4 Frequency and nature of operation breakdowns are documented

Element 1.2	Design maintenance plan

Performance Criteria:

1.2.1 Plant equipment and machinery are categorised
1.2.2 Different maintenance schedules are noted
1.2.3 Breakdown history is collated
1.2.4 Maintenance plan is drafted
1.2.5 Maintenance activities are scheduled

Element 1.3	**Request required resources**

Performance Criteria:

Necessary resources are noted
1.3.2 Resource parameters are identified
1.3.3 Relevant resources are quantified
1.3.4 Requested resources are verified
1.3.5 Requisition form is completed

Element 1.4	**Carry out planned maintenance**

Performance Criteria:

1.4.1 Maintenance work order and permit are obtained and analysed
1.4.2 SHEQ is observed
1.4.3 Tasks are carried out according to the maintenance work order
1.4.4 Tasks are carried out according to the maintenance work order
1.4.5 Operation and performance are tested and commissioned
1.4.6 Maintenance work order and work permit are closed

Element 1.5	**Carry out breakdown maintenance**

Performance Criteria:

1.5.1 Maintenance work order and permit are obtained and analysed
1.4.2 SHEQ is observed
1.4.3 Machine history is analysed
1.4.4 Breakdown machine is diagnosed according to laid down procedure
1.4.5 Repaired machine is inspected and tested
1.4.6 Breakdown work order and work permit are closed

Element 1.6	**Test and re-commission**

Performance Criteria:

1.6.1 Manufacturer's operating procedures are adhered to
1.6.2 Operation inspection is carried out according to checklist
1.6.3 Visual inspection is carried out
1.6.4 Corrective action is carried out where necessary
1.6.5 Proper handover procedure is carried out

Competences required for assessment
Planning
Knowledge of costing materials and labour
Repairing techniques
Knowledge of Faulty diagnosing techniques
Testing procedure
Safety and health practises
Technoprenuer
Operating standards

Generic Skills:
Literacy
Communication
Sober minded
Team work
Organisation

Range Statement:

Tools and Equipment
Basic Electrician tool kit
Tachometer
Stethoscope
Laptop
Ladder
Bearing puller
Chain block
Megger
Infrared camera
Lux meter
Earth leakage tester

Materials
Cables
Insulation tape
PVC conduit
Electrical accessories

Duration: **400** hours

ASSESSMENT AND CERTIFICATION:

In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.

Assessment will be conducted by accredited assessors. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence of their achievements.

UNIT 2

Unit Title:2	PLANT AND EQUIPMENT INSTALLATION
Unit Code	321/25/CR/MO

ZQF Level: NATIONAL CERTIFICATE

Credits: 38

Occupation: Electrician

Date of Promulgation: TBA

Review Date: TBA

Aim of the unit standard

This unit will enable an individual to install electrical circuits and equipment in domestic and industrial situations

ELEMENTS AND PERFORMANCE CRITERIA

Element 2.1	Conduct Installation survey

Performance Criteria:

2.1.1 Power supply adequacy is assessed
2.1.2 Environmental factors that affect installation are established
2.1.3 SHEQ issues are established
2.1.4 Installation requirements are established
2.1.5 Survey report is produced

Element 2.2	Design installation plan

Performance Criteria:

Installation standards and regulations are established
Site plan/floor plan is obtained
Installation positions / cable roots/DBs / panels/equipment/ power permits/ luminaries are indicated on the plan
Installation wiring diagrams are produced
Installation overlays are produced according to wiring standards and regulations

Element 2.3	Produce quantities of required resources

Performance Criteria:

2.3.1 Installation materials are determined
2.3.2 Installation human resource requirements is established
2.3.3 Project critical path is established

Element 2.4	Conduct Installation

Performance Criteria:

2.4.1 SHEQ is observed
2.4.2 Installations are prepared
2.4.3 Installation positions are marked according to standard
2.4.4 Cable enclosures are installed
2.4.5 Distribution boards / panels are mounted
2.4.6 Cables are drawn in
2.4.7 Switchgear/ fitting/ equipment/ accessories are connected
2.4.8 Relevant standards and regulations are adhered to

Element 2.5	Inspect Installation

Performance Criteria:

2.5.1 Visual inspection is conducted
2.5.2 Termination / connections are verified
2.5.3 Fire / labels/ correct colour coding is checked
2.5.4 Circuit segregation is verified
2.5.5 Installation checklist is completed

Element 2.6	Inspect, test and commission

Performance Criteria:

2.6.1 Continuity is tested
2.6.2 Polarity test is carried out
2.6.3 Insulation resistance test is carried out
2.6.4 Earth loop impedance test is carried out
2.6.5 Supply voltage is verified
2.6.6 Phase rotation is verified

2.6.7 Installation test checklist is completed
2.6.8 Installation test run is conducted
2.6.9 Pre-commission test is done
2.6.10 Production test run s performed
2.6.11 Installation documentation and hand over is done

Competences required for assessment

Cables selecting
Earthing systems
Continuity testing techniques
Recommended resistance
Coupling methods
Cable enclosures
SHEQ
Health and safety practices
TEEE regulations, SAZ
Circuit segregation

Common Essential Skills:

Literacy
Numeracy
Computer literacy
Communication
Planning

Range Statement:

Tools and Equipment
Basic Electrician tool kit
Fish tape
Spring bender
Crimping tool
Chasing hammer
Heat gun
Drilling machine
Ladder
Safety belts
Chain block
Personal protective equipment
Grinder
Masonry disks and bits
Scaffolding
Installation tester

Materials
PVC conduit and accessories
Electrical cables
Electrical accessories

Duration: **380** hours

ASSESSMENT AND CERTIFICATION:

In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.

Assessment will be conducted by accredited assessors. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence of their achievements.

UNIT 3

Unit Title:3	**ELECTRICAL EQUIPMENT REPAIR**
Unit Code	**321/25/CR/MO**

ZQF Level: **NATIONAL CERTIFICATE**

Credits: **48**

Occupation: **Electrician**

Date of Promulgation: **TBA**

Review Date: **TBA**

Aim of the unit standard

This unit will enable an individual to troubleshoot and repair electrical appliances and equipment

ELEMENTS AND PERFORMANCE CRITERIA

Element 3.1	**Gather Operation History**

Performance Criteria:

3.1.1 Work order/ job card is obtained
3.1.2 SHEQ is observed
3.1.3 Operation /operator's report is analysed
3.1.4 Machine / equipment history is established

Element 3.2	**Diagnose faults**

Performance Criteria:

3.2.1 Visual inspection is carried out
3.2.2 Necessary tests is carried out using appropriate tools and equipment
3.2.3 Faulty components/ devices/ equipment are identified

Element 3.3	**Rectify faults**

Performance Criteria:

3.3.1 Root Cause analysis is carried out
3.3.2 Appropriate remedial action is carried if necessary
3.3.3 Request for required material if necessary following laid down procedure
3.3.4 Faulty components are fixed or replaced where possible as per diagnosis

Element 3.4	**Test equipment**

Performance Criteria:

3.4.1 Visual inspection is done
3.4.2 Appropriate tests is carried following laid down procedure
3.4.3 Bench testing is carried out where necessary
3.4.4 In situ testing is carried out
3.4.5 Handover is done following laid down procedure
3.4.6 Work order is closed

Competences required for assessment

Common faults
Diagnosing techniques
Disassembling and assembling components
Circuit interpretation and designing techniques out
Acquaintance with IEE regulations
Safety and health practises (SHEQ)
Handling different devices

Common Essential Skills:
Literacy
Numeracy
Computer literacy
Communication
Planning

Range Statement:

Tools and Equipment
Basic electrician tool kit
Crimping tool
Heat gun

Drilling machine
Ladder
Chain block
Personal protective equipment

Materials
PVC conduit and accessories
Electrical cables
Electrical accessories

Duration: **480** hours

ASSESSMENT AND CERTIFICATION:

In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.

Assessment will be conducted by accredited assessors. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence of their achievements.

UNIT 4

Unit Title 4	SAFETY, HEALTH, ENVIRONMENTAL AND QUALITY MANAGEMENT
Unit Code	321/25/CR/MO

ZQF Level: **NATIONAL CERTIFICATE**

Credits: **40**

Occupation: **Electrician**

Date of Promulgation: **TBA**

Review Date: **TBA**

Aim of the unit standard

This unit will enable an individual to maintain safe working environment and exercise safety precautions.

ELEMENTS AND PERFORMANCE CRITERIA

Element 4.1	Conduct Hazard Identification risk Assessment (HIRA)

Performance Criteria:

4.1.1 Hazards and risks associated with a task are identified
4.1.2 Relevant regulations and permits are adhered to
4.1.3 Possible mitigatory measures are instigated.

Element 4.2	Guarantee SHEQ at workplace

Performance Criteria:

4.2.1 Appropriate PPE is issued/ worn.
4.2.2 PPE is inspected for suitability
4.2.3 Training needs are identified
4.2.4 Appropriate training is done
4.2.5 Adherence to SHEQ regulations is monitored

Element 4.3	Perform Housekeeping

Performance Criteria:

4.3.1 Materials to be sorted are classified
4.3.2 The 5-S (sorting, setting, shining, standardise, sustain) principle is applied as per procedure
4.3.3 Appropriate waste disposal procedures are carried out.

Element 4.4	Adhere to emergency preparedness procedures

Performance Criteria:

4.4.1 Pre- induction is carried out
4.4.2 Workplace dangers and appropriate mitigatory measures are identified
4.4.3 Emergency alarms are adhered to

Competences required for assessment

First Aid techniques
Personal protective equipment
Firefighting methods
Local and international SHEQ standards

Generic Skills:

Literacy
Communication
Planning
Team work

Range Statement:

Tools and Equipment

First Aid Kit
Personal Protective Equipment

Materials

Tablets
Bandages

Duration: **400** hours

ASSESSMENT AND CERTIFICATION:

In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.

Assessment will be conducted by accredited assessors. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence of their achievements.

UNIT 5

Unit Title 5	POWER DISTRIBUTION LINES CONSTRUCTION
Unit Code	321/25/CR/MO

ZQF Level: NATIONAL CERTIFICATE

Credits: 30

Occupation: Electrician

Date of Promulgation: TBA

Review Date: TBA

Aim/purpose of the unit standard

This unit will enable an individual to construct transmission and distribution power lines

ELEMENTS AND PERFORMANCE CRITERIA

Element 5.1	Carryout line ground work survey

Performance Criteria:

5.1.1 Route soil structure is assessed
5.1.2 Topography landscape is surveyed
5.1.3 Line route is determined

Element 5.2	Determine resource requirements

Performance Criteria:

5.2.1 Power line materials and equipment are determined are determined
5.2.3 Human resources requirements established
5.2.3` Power line construction critical path is established

Element 5.3	**Monitor line route clearance**

Performance Criteria:

5.3.1 SHEQ is observed
5.3.2 Clearance resources are mobilised
5.3.3 Route obstacles removal is monitored

Element 5.4	**Construct power Distribution lines**

Performance Criteria:

5.4.1 SHEQ is observed
5.4.2 Hole marking out is carried out according to standards
5.4.3 Hole digging out is monitored
5.4.4 Power line poles are dressed with accessories
5.4.5 Power line poles are positioned and plumbed
5.4.6 Power line stayers are tensioned
5.4.7 Power line struters are positioned
5.4.8 Line conductors are drawn and tied

Element 5.5	**Install necessary equipment**

Performance Criteria:

5.5.1 SHEQ is observed
5.5.2 Necessary equipment site structure is prepared
5.5.3 Necessary equipment mounting and connections are done
5.5.4 Physical and electrical protection is provided

Element 5.6	**Inspect constructed line**

Performance Criteria:

5.6.1 Visual inspection is carried out
5.6.2 Necessary tests are carried out using appropriate tools and equipment
5.6.3 Protection systems are visually inspected
5.5.4 Physical and electrical protection is provided

Competences required for assessment
Topography

Labour costing techniques
Treatment methods
Cable selection methods
Weather patterns
Cable drawing techniques

Common Essential Skills:
Literacy
Numeracy
Computer literacy
Communication
Planning

Range Statement:

Tools and Equipment
Basic Electrician tool kit
Angor bit
Come alone
Ladder
Safety belts
Pick and shovel
Slashers and axes
Chain saw
Drilling machine
Generator

Materials
Treated poles
D-irons
Insulators
Stay wire kit
Overhead cables

Duration: 300 hours

ASSESSMENT AND CERTIFICATION:

In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.

Assessment will be conducted by accredited assessors. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence of their achievements.

UNIT 6

Unit Title 6	**ELECTRICAL CIRCUIT DESIGNING**
Unit Code	**321/25/CR/MO**

ZQF Level: **NATIONAL CERTIFICATE**

Credits: **36**

Occupation: **Electrician**

Date of Promulgation: **TBA`**

Review Date: **TBA**

Aim/purpose of the unit standard

This unit will enable an individual to design, interpret and wire circuit diagrams.

ELEMENTS AND PERFORMANCE CRITERIA

Element 6.1	**Draft circuit application**

Performance Criteria:

6.1.1 Oral, written and circuit applications are obtained
6.1.2 Control/ power circuits are drafted
6.1.3 Designed draft circuits are simulated.

Element 6.2	**Produce circuit diagrams**

Performance Criteria:

6.2.1 Standard drawing software/ CAD are applied
6.2.2 Drawing instruments/ tools/ paper are gathered
6.2.3 Drafted circuits are drawn to standards
6.2.4 Drawn circuits are sent for approval
6.2.5 Approved circuits are filed/ shelved

Element 6.3	**Modify circuit diagram**

Performance Criteria:

6.3.1 Existing copy of circuit diagram is retrieved
6.3.2 Copy of circuit diagram is analysed
6.3.3 Changes are incorporated
6.3.4 Modified circuit diagram is simulated
6.3.5 Modified circuit diagrams are adopted

Competences required for assessment
Knowledge of circuit designing and interpretation
Knowledge of cable sizes
Wiring techniques
Knowledge of computer aided designing techniques

Generic Skills:
Literacy
Numeracy
Computer literacy
Communication
Planning

Range Statement:
Tools and Equipment
Basic Electrician tool kit
Drawing pencils
Drawing kit
Computer and printer

Materials
Bond paper
Cables
Wiring accessories
Software

Duration: **360 hours**

ASSESSMENT AND CERTIFICATION:

In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.
Assessment will be conducted by accredited assessors. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence of their achievements.

UNIT 7

Unit Code	**321/25/CR/MO**
Unit Title:	**PLC MANIPULATION**

Level of Unit: **NATIONAL CERTIFICATE**

Credits: **20**

Occupation: **Electrician**

Date of Promulgation: **TBA**

Review Date: **TBA**

Aim of the Unit Standard

This unit will enable an individual to produce and modify components to specified standards

ELEMENT AND PERFORMANCE CRITERIA

Element 7.1	**Conduct feasibility study**

Performance Criteria:

7.1.1 Stakeholders are consulted
7.1.2 Data is analysed
7.1.3 Program proposal is drafted
7.1.4 Program proposal presented

Element 7.2	**Write a program**

Performance Criteria:

7.2.1 Knowledge of programming languages for use is demonstrated
7.2.2 Correct procedure for writing a program is followed
7.2.3 Programming equipment compatibility is checked
7.2.4 Program is simulated
7.2.5 Program debuying is carried out
7.2.6 Program test run is carried out
7.2.7 Program coding is carried out
7.2.8 Backups of program created

Element 7.3	**Upload program**

Performance Criteria:

7.3.1 Knowledge to upload the program is demonstrated
7.3/2 Programmer is connected to intelligent device
7.3.3 Safety measures are adhered to
7.3.4 A constant power supply is maintained throughout the uploading process
7.3.5 Intelligent device is set to run mode

Competencies Required in Readiness for Assessment:
Information skills
Knowledge of programming languages

Generic Skills:

Planning
Numeracy
Coordination
Problem solving
Communication
Counselling
Organising
Team sprit
Tool handling

Range Statement:

TOOLS AND EQUIPMENT

Programmer
Intelligent
Printer

MATERIALS
Computer consumables
Stationery

DURATION **150** Hours

ASSESSMENT AND CERTIFICATION

In order to gain credits for this unit standard, a candidate must be assessed and demonstrate competency in all the elements and performance criteria of this unit standard.

Assessment will be conducted by accredited assessors. The results of the assessment will be submitted to ZIMEQA. A candidate can apply to ZIMEQA for documentary evidence of their achievements.

	MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT SKILLS PROFICIENCY SCHEDULE	**CODE**

INDUSTRY: ELECTRICAL
TRADE/ OCCUPATION: ELECTRICIAN
CLASS/ LEVEL: ONE

DUTY A: MAINTAIN SAFETY, HEALTH , ENVIRONMENT AND QUALITY 10

Pre-requisites: **CLASS TWO** **Approval Date:** **Review Date:**

TASK	**STEPS**	**PROFICIENCY INDICATORS**	RELATED KNOWLEDGE	WORKPLACE ESSENTIAL SKILLS
A1: Create safe working environment	✓ Check size, settings and operation of protective devices in accordance with set standards ✓ Inspect protective equipment for wear and damage ✓ Install guards on appropriate positions ✓ Check earthing of equipment ✓ Check condition of equipment supports ✓ Identify possible dangers or risks ✓ Select safety equipment ensuring conformity to standards	➢ Risk assessment has been carried out. ➢ Proper selection and inspection of safety gear and safety equipment according to SHEQ standards. ➢ Safety gear worn correctly. ➢ Safety guards and safety switches are installed and functioning according to IEE regulations. Protective equipment properly used. ➢ Test equipment earth continuity to the supply earth. ➢ Risk assessment carried out. ➢ Selected safety equipment conforms to SHEQ standards. ➢ Safety equipment in good	▪ Safety and health practices. ▪ Testing equipment and procedures. ▪ Knowledge of Institute of Engineers (IEE) Regulations. ▪ Knowledge of personal protective equipment. ▪ Knowledge of fire-fighting methods. ▪ Electrical technology. ▪ First aid.	<u>*Reading text*</u> ○ Instructions ○ Report writing ○ Manuals ○ Planning. ○ Communication skills. ○ Team work. ○ First aid techniques.

	✓ Check condition of safety equipment ✓ Recommend repair or disposal of faulty safety equipment ✓ Operate safety equipment properly. ✓ Obtain instructions ✓ Make dead ✓ Isolate power ✓ Lock out circuit ✓ Check circuit for dead and earth ✓ Install barricades and post danger warning signs and safety notices ✓ Issue safety document / permit to work	condition according to SHEQ standards. ➢ Instructions obtained. ➢ Switch gear isolated, locked, earthed and tagged. ➢ Danger warning signs are posted and barricades installed. ➢ Permit is available.		
A2: Use appropriate PPE	✓ Identify possible dangers or risks and raise awareness ✓ Select and inspect appropriate safety gear according to standards ✓ Select and inspect appropriate safety equipment according to standards ✓ Wear safety gear correctly ✓ Ensure safety guards and safety switches are installed and are functioning	➢ Risk assessment has been carried out. ➢ Proper selection and inspection of safety gear and safety equipment according to SHEQ standards. ➢ Safety gear worn correctly. ➢ Safety guards and safety switches are installed and functioning according to IEE regulations. Protective equipment properly used. ➢ Test equipment earth continuity to the supply earth.		

	✓ Use protective equipment correctly ✓ Ensure effective earthing			
A3: Perform Housekeeping	✓ Clean work area after use ✓ Clear gangway of obstacles ✓ Clean and store tools and equipment after use ✓ Dispose waste materials according to regulations	➢ Work area cleaned after use ➢ Gangway cleared of obstacles ➢ Tools and equipment cleaned and stored appropriately after use ➢ Waste material disposed accordingly		
A4: Conduct safety awareness campaign	✓ Install danger warning signs ✓ Barricade working area ✓ Advise stakeholders of dangers associated with working area ✓ Install anti-climb/ anti-intrusion devices	➢ Danger warning signs installed ➢ Working area barricaded ➢ Stakeholders educated ➢ Anti-climb/ anti-intrusion devices installed		
A5: Perform basic first aid	✓ Switch off source of supply ✓ Remove victim from source of danger ✓ Render first aid ✓ Send/call for help ✓ Monitor victim	➢ Source of power switched off ➢ Victim removed from source of danger ➢ First aid administered ➢ Help called/sent for ➢ Victim monitored		

TOOLS AND EQUIPMENT NECESSARY TO COMPLETE THIS DUTY:

- Basic Electrician tool kit.
- First aid kit.
- Megger.
- Multimeter.

HEALTH, SAFETY AND ENVIRONMENTAL ISSUES RELATED TO THIS DUTY:

- Personal protective clothing
- Use of proper insulated tools
- Use of calibrated instruments incompliance to ISO standards
- Free gang ways in the workshop and plant
- Floors free of slippery conditions
- Safety insignia post at all relevant areas

SPECIFIC WORKER TRAITS REQUIRED COMPLETING THIS DUTY:

- Team work
- Sober minded
- Communication
- Honest
- Punctual
- Target orientated.

	MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT SKILLS PROFICIENCY SCHEDULE	CODE

INDUSTRY: ELECTRICAL — TRADE/ OCCUPATION: ELECTRICIAN — CLASS/ LEVEL: ONE

DUTY B: INSTALL PLANT AND EQUIPMENT 20

Pre-requisites: **CLASS TWO** **Approval Date:** **Review Date:**

TASK	STEPS	PROFICIENCY INDICATORS	RELATED KNOWLEDGE	WORKPLACE ESSENTIAL SKILLS
B1: Assess area of installation	✓ Assess Environment of installation ✓ Locate power source ✓ Ascertain compatibility of power supply ✓ Determine the route of supply cables ✓ Establish SHEQ issues ✓ Produce installation assessment report	➢ Area clear of obstacles and environmental hazards likely to disturb installation. ➢ Use of closed power supply. ➢ Power supply compatible with equipment. ➢ Optimum cable route identified. ➢ SHEQ issues established ➢ Installation assessment report produced	• Safety and health practices. • Testing of equipment procedures. • Institute of Electrical Engineers (IEE) Regulations. • Cable selection. • Types of cable enclosures. • Methods of coupling. • Electrical technology. • Knowledge of earthing systems.	*Numeracy* ○ Estimation skills ○ Measuring skills ○ Calculations *Reading text* ○ Instructions ○ Drawings ○ Specifications ○ Manuals *Document use* ○ Interpret drawings and sketches.
B2: Verify Installation compliance	✓ Assess SHEQ ✓ Establish installation standards and regulations ✓ Verify supply source parameters ✓ Verify switchgear and cables sizes	➢ SHEQ issues assessed ➢ Installation standards and regulations established ➢ Supply source parameters verified ➢ Switchgear and cable sizes verified		

	✓ Analyse installation drawings/ diagram	➢ Installation drawings/diagram analysed		○ Write bills of quantity. ○ Read material codes *Planning.* ○ Scheduling of work.
B3: Produce required resources	✓ Analyse installation drawing/ diagram ✓ Determine required materials ✓ Determine required human resources ✓ Establish required tools and equipment ✓ Estimate time frame of the project	➢ Installation drawing/ diagram analysed ➢ Required resources established ➢ Required human resources established ➢ Required tools and equipment established ➢ Timeframe estimated		
B4: Carryout installation	✓ Select appropriate tools and equipment ✓ Observe SHEQ ✓ Prepare installation area/site ✓ Mark installation positions according to the diagram/ specifications ✓ Mount and secure electrical equipment and machinery ✓ Install cable enclosures ✓ Mount panels, distribution boards and marshaling kiosks ✓ Draw in/ lay cables ✓ Terminate cables and connect accessories	➢ Appropriate tools and equipment selected ➢ SHEQ observed ➢ Installation area/site prepared ➢ Installation positions marked according to diagram ➢ Electrical equipment and machinery mounted ➢ Cable enclosures installed ➢ Panel, distribution boards and marshaling kiosks mounted ➢ Cables drawn in/ laid. ➢ Cables terminated and accessories connected.		
B5: Inspect installation	✓ Verify terminations ✓ Check for loose connections ✓ Verify circuit segregation ✓ Check for labels and colour coding	➢ Terminations verified ➢ Loose connections identified and corrected ➢ Circuit segregation verified ➢ Labels and colour coding		

	✓ Verify installation specifications and regulation compliance	checked ➢ Installation specifications verified		
B6: Test and commission installation	✓ Perform continuity test on all circuits ✓ Perform insulation resistance test on circuits ✓ Perform short / open circuit test ✓ Conduct earth logs impedance ✓ Carryout polarity test ✓ Verify phase rotation ✓ Conduct phasing of supply transformer ✓ Measure contact resistance ✓ Carryout circuit breaker timing test ✓ Complete installation test checklist ✓ Energise completed installation ✓ Perform installation test run ✓ Prepare commissioning report	➢ Continuity test conducted ➢ Insulation resistance test conducted ➢ Circuit tests conducted ➢ Earth logs impedance conducted ➢ Polarity tests conducted ➢ Phase rotation verified ➢ Phasing of supply transformer conducted ➢ Contact resistance established ➢ CIRCUIT BREAKER timing test conducted ➢ Installation checklist completed ➢ Installation energised ➢ Installation test run conducted ➢ Commissioning report compiled		

TOOLS AND EQUIPMENT NECESSARY TO COMPLETE THIS DUTY:

- ➢ Basic Electrician tool kit.
- ➢ Fish tape
- ➢ Laptop
- ➢ Ladder.
- ➢ Spring bender

- Chain block.
- Megger.
- Multimeter.
- Crimping tool.
- Chasing hammer.
- Soldering station.
- Drilling machine.
- Safety belts.

HEALTH, SAFETY AND ENVIRONMENTAL ISSUES RELATED TO THIS DUTY:

- Personal protective clothing
- Use of proper insulated tools
- Use of calibrated instruments incompliance to ISO standards
- Free gang ways in the workshop and plant
- Floors free of slippery conditions
- Safety insignia post at all relevant areas

SPECIFIC WORKER TRAITS REQUIRED COMPLETING THIS DUTY:

- Team work
- Sober minded
- Communication
- Honest
- Punctual
- Target orientated.

	MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT SKILLS PROFICIENCY SCHEDULE	CODE

INDUSTRY	TRADE/ OCCUPATION	CLASS/ LEVEL
ELECTRICAL	ELECTRICIAN	ONE

DUTY C: MAINTAIN PLANT & EQUIPMENT 20

Pre-requisites: **CLASS TWO** **Approval Date:** **Review Date:**

TASK	STEPS	PROFICIENCY INDICATORS	RELATED KNOWLEDGE	WORKPLACE ESSENTIAL SKILLS
C1: Design plant maintenance schedule	✓ Prepare asset / equipment register ✓ Prepare maintenance schedule according to equipment specifications ✓ Prepare maintenance check list as per machine ✓ Prepare material and special equipment requirements lists ✓ Allocate man hours according to task	➢ Asset/equipment register is completed. ➢ Maintenance schedule prepared according to equipment specifications. ➢ Maintenance check list prepared as per machine. ➢ Material and special equipment list compiled. ➢ Manhours fairly allocated according to task.	▪ Safety and health practices. ▪ Testing of equipment procedures. ▪ Institute of Electrical Engineers (IEE) Regulations. ▪ Bearing and seals technology. ▪ Knowledge of cleaning solvents. ▪ Electrical technology. ▪ Fault diagnosis techniques.	<u>*Numeracy*</u> ○ Estimation skills ○ Measuring skills ○ Calculations <u>*Reading text*</u> ○ Instructions ○ Drawings ○ Specifications ○ Manuals <u>*Document use*</u> ○ Interpret drawings and sketches. ○ Write bills of
C2: Monitor running plant	✓ Attend morning briefing ✓ Adhere to SHEQ procedures ✓ Complete operation checklist ✓ Document nature and frequency of breakdown	➢ Meeting attended/ minutes produced ➢ SHEQ procedures adhered to ➢ Operation checklist completed ➢ Nature and frequency of breakdown documented		

C3: Request required resources	✓ Identify nature of problem ✓ List requirements ✓ Raise a requisition of requirements ✓ Seek requisition requirements authorization ✓ Collect required resources ✓ Verify requested resources	➢ Nature of problem established ➢ Requirements list compiled ➢ Requisition raised ➢ Authority for requisition obtained ➢ Required resources obtained ➢ Requested resources verified		quantity. ○ Read material codes *Planning.* ○ Scheduling of work.
C4: Conduct planned maintenance	✓ Consult planned maintenance schedule ✓ Obtain work permit ✓ Isolate work area/ ✓ Test for dead on work area ✓ Earth/ barricade/ lockout work area ✓ Carryout maintenance work as scheduled ✓ Close work permit/ work order	➢ Maintenance schedule analysed ➢ Work permit obtained ➢ Work area isolated ➢ Work area tested for dead ➢ Work area earthed/ barricaded/ locked ➢ Maintenance work carried out as scheduled ➢ Work permit/ work order closed		
C5: Carryout breakdown maintenance	✓ Analyse operator's job card ✓ Investigate root cause of breakdown ✓ Obtain work permit/ work order ✓ Observe SHEQ ✓ Analyse machine/ equipment history ✓ Isolate and lock area ✓ Test for dead on work area ✓ Earth/ barricade work area ✓ Carryout necessary repairs ✓ Close work permit/ order	➢ Operator's job card analysed ➢ Breakdown root cause established ➢ Work permit/ work order obtained ➢ SHEQ regulations adhered to ➢ Machine history outlined ➢ Area isolated/ locked ➢ Test for dead conducted ➢ Work area barricaded ➢ Repairs conducted ➢ Work permit/ order closed		

C6: Inspect plant	✓ Verify terminations ✓ Check for loose connections ✓ Verify circuit segregation ✓ Check for labels and colour coding ✓ Verify installation specifications and regulation compliance	➢ Terminations verified ➢ Loose connections identified and corrected ➢ Circuit segregation verified ➢ Labels and colour coding checked ➢ Installation specifications verified		
C7: Test and commission plant	✓ Perform continuity test on all circuits ✓ Perform insulation resistance test on circuits ✓ Perform short / open circuit test ✓ Conduct earth logs impedance ✓ Carryout polarity test ✓ Verify phase rotation ✓ Conduct phasing of supply transformer ✓ Measure contact resistance ✓ Carryout circuit breaker timing test ✓ Complete installation test checklist ✓ Energise completed installation ✓ Perform installation test run ✓ Prepare commissioning report	➢ Continuity test conducted ➢ Insulation resistance test conducted ➢ Circuit tests conducted ➢ Earth logs impedance conducted ➢ Polarity tests conducted ➢ Phase rotation verified ➢ Phasing of supply transformer conducted ➢ Contact resistance established ➢ CIRCUIT BREAKER timing test conducted ➢ Installation checklist completed ➢ Installation energised ➢ Installation test run conducted ➢ Commissioning report compiled		

TOOLS AND EQUIPMENT NECESSARY TO COMPLETE THIS DUTY:

- Basic Electrician tool kit.
- Tachometer.
- Laptop

- Ladder.
- Bearing puller.
- Chain block.
- Megger.
- Multimeter.

HEALTH, SAFETY AND ENVIRONMENTAL ISSUES RELATED TO THIS DUTY:

- Personal protective clothing
- Use of proper insulated tools
- Use of calibrated instruments incompliance to ISO standards
- Free gang ways in the workshop and plant
- Floors free of slippery conditions
- Safety insignia post at all relevant areas

SPECIFIC WORKER TRAITS REQUIRED COMPLETING THIS DUTY:

- Team work
- Sober minded
- Communication
- Honest
- **Punctual**
- **Target orientated.**

	MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT SKILLS PROFICIENCY SCHEDULE	**CODE**

INDUSTRY ELECTRICAL | TRADE/ OCCUPATION ELECTRICIAN | CLASS/ LEVEL ONE

DUTY D: CONSTRUCT ELECTRICAL POWER DISTRIBUTION LINES 10

Pre-requisites: CLASS TWO **Approval Date:** **Review Date:**

TASK	STEPS	PROFICIENCY INDICATORS	RELATED KNOWLEDGE	WORKPLACE ESSENTIAL SKILLS
D1: Assess line construction area	✓ Check soil properties ✓ Consider existing infrastructure, fauna and flora ✓ Consider SHEQ ✓ Measure length of proposed line	➢ Soil properties checked ➢ Existing infrastructure, fauna and flora considered ➢ SHEQ regulations adhered to ➢ Length of proposed line determined	▪ Safety and health practices. ▪ Testing equipment and procedures. ▪ Knowledge of Institute of Engineers (IEE) Regulations. ▪ Knowledge of topography. ▪ Knowledge of weather patterns. ▪ Electrical technology. ▪ Cable selection methods.	*Numeracy* ○ Estimation skills ○ Measuring skills ○ Calculations. *Reading text* ○ Instructions ○ Drawings ○ Specifications ○ Manuals *Document use* *Planning.*
D2: Determine required resources	✓ Compute line hardware and accessories ✓ Determine required human resources ✓ Select required tools and equipment ✓ Select appropriate PPE	➢ Line hardware and accessories computed ➢ Required human resources determined ➢ Appropriate tools and equipment selected ➢ Appropriate PPE selected		
D:3 Monitor line route clearance	✓ Observe SHEQ ✓ Ensure availability of resources ✓ Supervise obstacles removal	➢ SHEQ adhered to ➢ Availability of resources ensured ➢ Removal of obstacles supervised		

D4: Erect poles and string conductors	✓ Peg Line according to ZESA engineering instructions ✓ Dig or drill holes according to ZESA engineering Instructions ✓ Dress poles with necessary accessories ✓ Position and plumb poles ✓ Fit and tension stay wire according to electrical engineering standards ✓ String Line conductors to engineering standards ✓ Test and commission	➢ Holes aligned and drilled according to engineering instructions. ➢ Poles are dressed and plumbed. ➢ Stay wires are tensioned according to engineering standards. ➢ Line conductors are strung according engineering standards.		
D5: Inspect construct line	✓ Verify installation specification and regulation compliance ✓ Check for loose connections ✓ Check for damaged hardware	➢ Installation specification and regulation compliance verified ➢ Loose connections identified ➢ Damaged hardware identified		

TOOLS AND EQUIPMENT NECESSARY TO COMPLETE THIS DUTY:

- Basic Electrician tool kit.
- Angor bit.
- Come alone.
- Ladder.
- Safety belts.
- Pick and shovel.
- Slashers and axes.
- Chain saw.
- Drilling machine.
- Generator.
- Multimeter.
- Treated poles.

- D-irons.
- Insulators.
- Stay wire kit.
- Overhead cables.

HEALTH, SAFETY AND ENVIRONMENTAL ISSUES RELATED TO THIS DUTY:

- Personal protective clothing
- Use of proper insulated tools
- Use of calibrated instruments incompliance to ISO standards
- Free gang ways in the workshop and plant
- Floors free of slippery conditions
- Safety insignia post at all relevant areas

SPECIFIC WORKER TRAITS REQUIRED COMPLETING THIS DUTY:

- Team work
- Sober minded
- Communication
- Honest
- Punctual
- Target orientated.

	MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT SKILLS PROFICIENCY SCHEDULE	**CODE**

INDUSTRY	TRADE/ OCCUPATION	CLASS/ LEVEL
ELECTRICAL	ELECTRICIAN	ONE

DUTY E: REPAIR ELECTRICAL COMPONENTS & EQUIPMENT 15

Pre-requisites: CLASS TWO Approval Date: Review Date:

TASK	STEPS	PROFICIENCY INDICATORS	RELATED KNOWLEDGE	WORKPLACE ESSENTIAL SKILLS
E1: **Ensure Safety**	✓ Obtain work permit to carry out repairs ✓ Carry out risk assessment ✓ Isolate power and isolate switch gear ✓ Using the appropriate instruments verify whether the correct circuit has been isolated ✓ Post danger warning tags at point of isolation	➢ Work permit obtained. ➢ Risk assessment carried out as per check list. ➢ Equipment isolated, tagged, locked out and earthed.	▪ Safety and health practices. ▪ Testing of equipment procedures. ▪ Institute of Electrical Engineers (IEE) Regulations. ▪ Bearing and seals technology. ▪ Knowledge of cleaning solvents. ▪ Electrical technology ▪ Fault diagnosis techniques. ▪ Knowledge of	*Numeracy* ○ Estimation skills ○ Measuring skills ○ Calculations *Reading text* ○ Instructions ○ Drawings ○ Specifications ○ Manuals *Document use* ○ Interpret drawings and sketches. ○ Write bills of quantity. ○ Read material
E 2: Diagnose Fault	✓ Obtain equipment history from Operator, job card and equipment display ✓ Check for recurring problems/faults ✓ Visually inspect equipment for	➢ Equipment history has been obtained. ➢ Faults identified. ➢ Causes of faults identified. ➢ Use of correct equipment to identify the faults.		

	mechanical damages, loose components, terminals and signs of overheating on windings or circuit boards ✓ Smell burnt insulation of cables or components ✓ Identify the special tools required to diagnose fault ✓ Test for short circuits, continuity and insulation resistance with reference to electrical drawings	➢	disassembling and assembling components. ▪ Circuit interpretation and designing techniques.	codes *<u>Planning.</u>* ○ Scheduling of work.
E 3: Rectify Fault	✓ Identify the materials, spares and special tools required to rectify fault ✓ Repair or replace defective components ✓ Test necessary parameters in the circuit ✓ Restore power and energize equipment ✓ Measure applicable variables, eg. current, voltage, speed, temperature torque and power	➢ Description of fault. ➢ Defective components and spares have been replaced or repaired. ➢ Correct component and equipment ratings are maintained. ➢ Parameters are measured against set values. ➢		
E4: Inspect repaired equipment	✓ Verify terminations ✓ Check for loose connections ✓ Verify circuit segregation ✓ Check for labels and colour coding ✓ Verify installation specifications and regulation compliance	➢ Terminations verified ➢ Loose connections identified and corrected ➢ Circuit segregation verified ➢ Labels and colour coding checked ➢ Installation specifications verified		

E5: Test and commission repaired equipment	✓ Perform continuity test on all circuits ✓ Perform insulation resistance test on circuits ✓ Perform short / open circuit test ✓ Conduct earth logs impedance ✓ Carryout polarity test ✓ Verify phase rotation ✓ Conduct phasing of supply transformer ✓ Measure contact resistance ✓ Carryout circuit breaker timing test ✓ Complete installation test checklist ✓ Energise completed installation ✓ Perform installation test run ✓ Prepare commissioning report	➢ Continuity test conducted ➢ Insulation resistance test conducted ➢ Circuit tests conducted ➢ Earth logs impedance conducted ➢ Polarity tests conducted ➢ Phase rotation verified ➢ Phasing of supply transformer conducted ➢ Contact resistance established ➢ Circuit Breaker timing test conducted ➢ Installation checklist completed ➢ Installation energised ➢ Installation test run conducted ➢ Commissioning report compiled		

TOOLS AND EQUIPMENT NECESSARY TO COMPLETE THIS DUTY:

- Basic Electrician tool kit.
- Crimping tool.
- Laptop
- Ladder.
- Bearing puller.
- Chain block.
- Megger.
- Multimeter.
- Soldering station.
- Drilling machine.

HEALTH, SAFETY AND ENVIRONMENTAL ISSUES RELATED TO THIS DUTY:

- Personal protective clothing
- Use of proper insulated tools
- Use of calibrated instruments incompliance to ISO standards
- Free gang ways in the workshop and plant
- Floors free of slippery conditions
- Safety insignia post at all relevant areas

SPECIFIC WORKER TRAITS REQUIRED COMPLETING THIS DUTY:

- Team work
- Sober minded
- Communication
- Honest
- Punctual
- Target orientated.

	MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT **SKILLS PROFICIENCY SCHEDULE**	**CODE**

INDUSTRY ELECTRICAL | TRADE/ OCCUPATION ELECTRICIAN | CLASS/ LEVEL ONE

DUTY F: DESIGN ELECTRICAL CIRCUIT	**10**

Pre-requisites: **CLASS TWO** **Approval Date:** **Review Date:**

TASK	STEPS	PROFICIENCY INDICATORS	RELATED KNOWLEDGE	WORKPLACE ESSENTIAL SKILLS
F1: Obtain oral, written circuit application	✓ Obtain drawing instruments and software ✓ Read instructions and record key conditions to be achieved ✓ Compare process or requirements with machine specification ✓ Compile parameters of each component on the machine or equipment ✓ Analyze instructions / application formulate correct sequence of operation ✓ Adhere to safety procedures at all times	➢ Key conditions listed. ➢ Comparison of machine specifications and process requirement are made. ➢ Parameters for each component on the machine are listed. ➢ Correct sequence of operation is formulated. ➢ Appropriate SHEQ standards are observed.	▪ Safety and health practices. ▪ Testing equipment and procedures. ▪ Knowledge of Institute of Engineers (IEE) Regulations. ▪ Knowledge of Circuit designing and interpretation. ▪ Knowledge of cable sizes. ▪ Wiring techniques. ▪ Knowledge of	*Numeracy* ○ Estimation skills ○ Measuring skills ○ Calculations *Reading text* ○ Instructions ○ Drawings ○ Specifications ○ Manuals *Document use* ○ Interpret drawings and sketches. ○ Write bills of quantity. ○ Read material codes
F 2: Draw wiring	✓ Use BS 3939 standards to draw a rough draft and incorporate all	➢ A properly labeled working diagram is drawn according to		

diagram	necessary protection features ✓ Label components correctly ✓ Simulate the draft using the suitable software or wire and test on the work bench ✓ Make changes draft were necessary ✓ Draw final diagram to BS 3939 standards ✓ Insert key or legend for wiring diagram ✓ Seek design approval ✓ File approved circuits	BS3939 standards.	computer aided designing (CAD) techniques. ▪ Electrical technology	*Planning.* ○ Scheduling of work. *Computer literacy* ○ Use of application software
F 3: Determine size and quantities of control gear accessories	✓ Perform power calculations to determine full load current of the circuit from given information ✓ Select properly rated power components from calculations ✓ Calculate cable size with reference to IEE Regulations Cable size selection procedures ✓ Prepare a qualitative and quantitative list of materials required	➢ Selected components and cables are according to IEE regulations. ➢ Bill of quantities produced.		
F4: Produce cable layout	✓ Assess environment in which cables are to be installed (Use existing map) ✓ Consider alternative cable routes ✓ Consider underground, ducting, overhead and trunking for high	➢ Cables are installed according to IEE regulations. ➢ Use of site plan. ➢ Cable routes marked and site plan updated.		

	voltage without interfering with low and extra low voltage circuits or electronic equipment ✓ Ensure cable enclosures are adequate for nominal operation and future expansion ✓ Ensure easy access to cable enclosures ✓ Mark cable routes with tags and update site plan ○			

TOOLS AND EQUIPMENT NECESSARY TO COMPLETE THIS DUTY:

- Basic electrician tool kit.
- Drawing pencils.
- Laptop.
- Printer.
- Ladder.
- Drawing kit.
- Cables and wiring accessories.
- Megger.
- Multimeter.

HEALTH, SAFETY AND ENVIRONMENTAL ISSUES RELATED TO THIS DUTY:

- Personal protective clothing
- Use of proper insulated tools
- Use of calibrated instruments incompliance to ISO standards
- Free gang ways in the workshop and plant
- Floors free of slippery conditions
- Safety insignia post at all relevant areas

SPECIFIC WORKER TRAITS REQUIRED COMPLETING THIS DUTY:

- Team work

- Sober minded
- Communication
- Honest
- Punctual
- Target orientated.

	MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT SKILLS PROFICIENCY SCHEDULE	**CODE**

INDUSTRY ELECTRICAL | TRADE/ OCCUPATION ELECTRICIAN | CLASS/ LEVEL ONE

DUTY G : MANIPULATE PROGRAMMABLE LOGIC CIRCUIT 10

Pre-requisites: CLASS TWO **Approval Date:** **Review Date:**

TASK	STEPS	PROFICIENCY INDICATORS	RELATED KNOWLEDGE	WORKPLACE ESSENTIAL SKILLS
H1: Design PLC program	✓ Analyse situation parameters ✓ Compile parameter of each component ✓ Select suitable program structure ✓ Write program algorithm	➢ Situation parameters analysed ➢ Component parameters established ➢ Suitable program selected ➢ Program algorithm compiled	▪ Programming languages ▪ Circuit designing	*Numeracy* ○ Estimation skills ○ Measuring skills ○ Calculations *Reading text* ○ Instructions ○ Drawings ○ Specifications ○ Manuals *Document use* ○ Interpret drawings and sketches. ○ Write bills of quantity. ○ Read material
H 2: Configure PLC program	✓ Select appropriate tools and equipment ✓ Select appropriate programming language ✓ Input/ code designed program ✓ Run designed program	➢ Appropriate tools and equipment used ➢ Appropriate programming language selected ➢ Designed program coded ➢ Designed program tested		
H 3: Program PLC devices	✓ Verify designed program compatibility to ladder logics ✓ Insert logics according to designed program ✓ Input designed program into PLC CPU	➢ Designed program compatibility to ladder logistics verified ➢ Logistics inserted according to designed program ➢ Designed program input into CPU		

H4: Test and simulate program	✓ Connect relevant components to CPU ✓ Verify connections ✓ Observe SHEQ ✓ Run designed program ✓ Vary parameters ✓ Modify program ✓ Document results	➢ Relevant components connected to CPU ➢ Connections verified ➢ SHEQ adhered to ➢ Designed program run ➢ Parameters varied ➢ Results recorded		codes *Planning.* ○ Scheduling of work. *Computer literacy* ○ Use of application software

TOOLS AND EQUIPMENT NECESSARY TO COMPLETE THIS DUTY:

- Basic electrician tool kit.
- Drawing pencils.
- Laptop.
- Printer.
- Cables and wiring accessories.
- Multimeter.

HEALTH, SAFETY AND ENVIRONMENTAL ISSUES RELATED TO THIS DUTY:

- Personal protective clothing
- Use of proper insulated tools
- Use of calibrated instruments incompliance to ISO standards
- Free gang ways in the workshop and plant
- Floors free of slippery conditions
- Safety insignia post at all relevant areas

SPECIFIC WORKER TRAITS REQUIRED COMPLETING THIS DUTY:

- Team work
- Sober minded
- Communication
- Honest
- Punctual
- Target orientated

	MINISTRY OF HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT SKILLS PROFICIENCY SCHEDULE	**CODE**

INDUSTRY ELECTRICAL — TRADE/ OCCUPATION ELECTRICIAN — CLASS/ LEVEL ONE

DUTY H: TRAIN AND SUPERVISE SURBODINATES 5

Pre-requisites: **Approval Date:** **Review Date:**

TASK	STEPS	PROFICIENCY INDICATORS	RELATED KNOWLEDGE	WORKPLACE ESSENTIAL SKILLS
H1: Demonstrate use of Tools and Equipment	✓ Demonstrate safe working practices. ✓ Induct subordinates on use of tools and equipment. ✓ Explain how to interpret diagrams and manuals. ✓ Demonstrate adherence to procedures and standards	➢ Safe working practices demonstrated. ➢ Correct use of machine ➢ /equipment. ➢ Working according to specifications/ manufacturer's standards. ➢ Procedures followed correctly	▪ Tools and Machine Handling. ▪ SHEQ ▪ Construction Drawing ▪ Facilitation	○ Facilitation ○ Teamwork ○ Tool Handling Planning ○ Supervisory ○ Communication ○ Training and Development ○ Demonstration ○ Evaluation
H2: Assign Duties	✓ Identify tasks to be done ✓ Identify resources required ✓ Allocate resources ✓ Distribute tasks	➢ Tasks identified ➢ resources identified ➢ resources allocated ➢ tasks distributed	▪ Management ▪ Computer Literacy ▪ Metrology ▪ Engineering	○ Problem Solving ○ Tool Handling ○ Drawing ○ Accuracy

H3: Supervise subordinates	✓ Verify tasks according to work schedule ✓ Monitor execution of duties ✓ Conduct performance appraisal ✓ Recommend for further training/upgrade ✓ Monitor use/distribution of resources ✓ Create appropriate documentation	➢ tasks verified according to work schedule ➢ execution of duties monitored ➢ performance appraisal conducted ➢ further training/upgrade recommended ➢ use/distribution of resources monitored ➢ appropriate documentation created	Drawing ▪ Workshop Practice ▪ Workshop Technology ▪ Code of Conduct	○ Supervision ○ Evaluation ○ Estimation ○ Numeracy ○ Quality Control ○ Training and Development ○ Counseling

TOOLS AND EQUIPMENT NECESSARY TO COMPLETE THIS DUTY:

- Computer and accessories
- Stationery

HEALTH, SAFETY AND ENVIRONMENTAL ISSUES RELATED TO THIS DUTY:

- Protective Equipment (P.P.E)
- Personal Warning Signs

SPECIFIC WORKER TRAITS REQUIRED TO COMPLETE THIS DUTY:

- Hardworking
- Disciplined
- Healthy and strong
- Reliable
- Punctual
- Result oriented